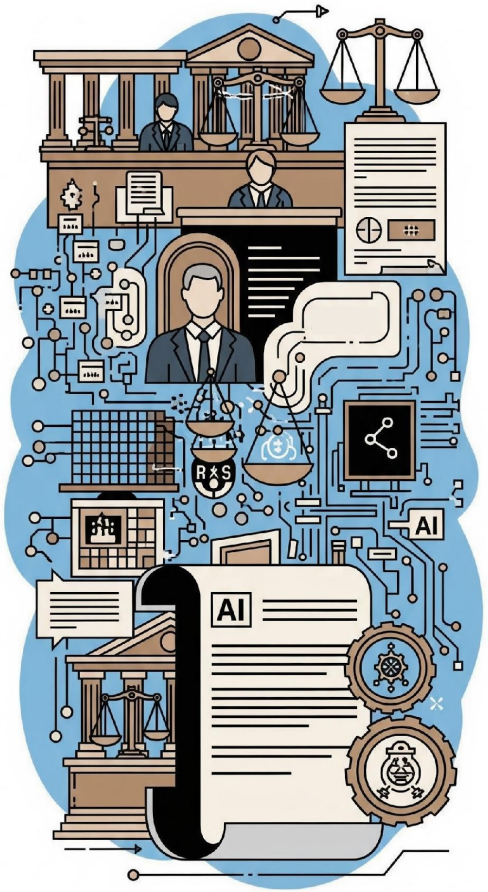
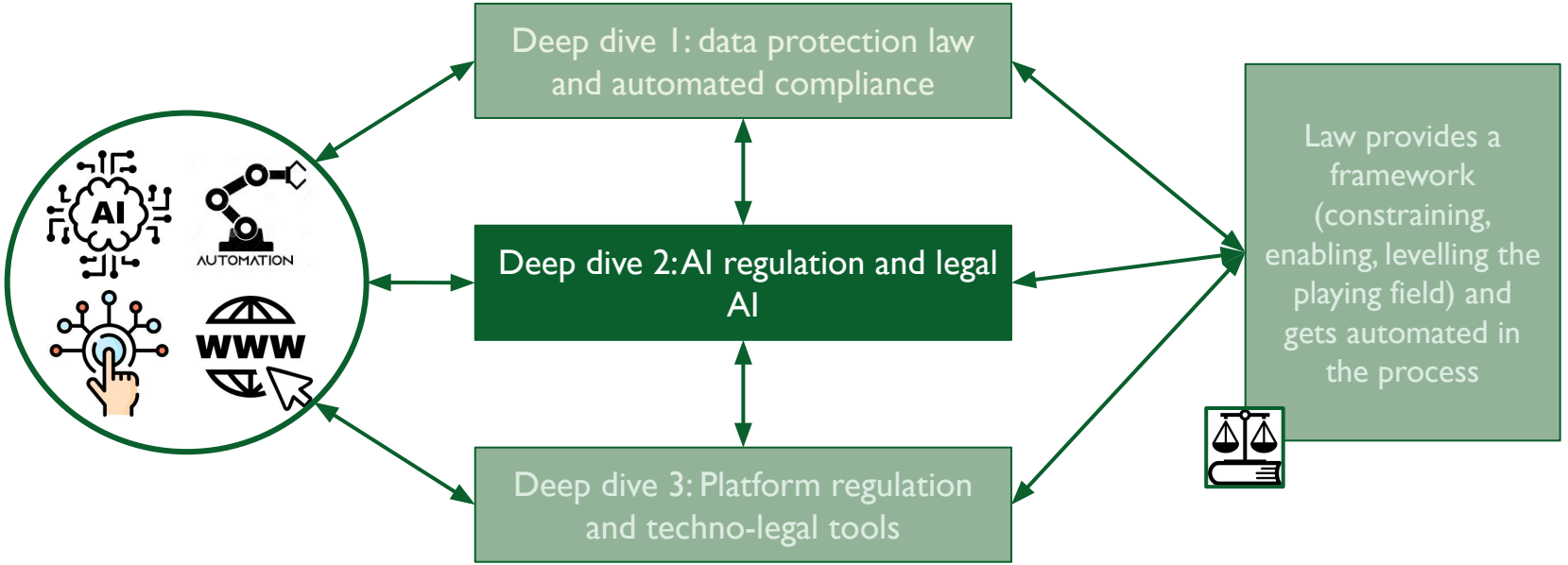
Universität St.Gallen

Deep Dive 2:
AI regulation and legal AI



From insight to impact.

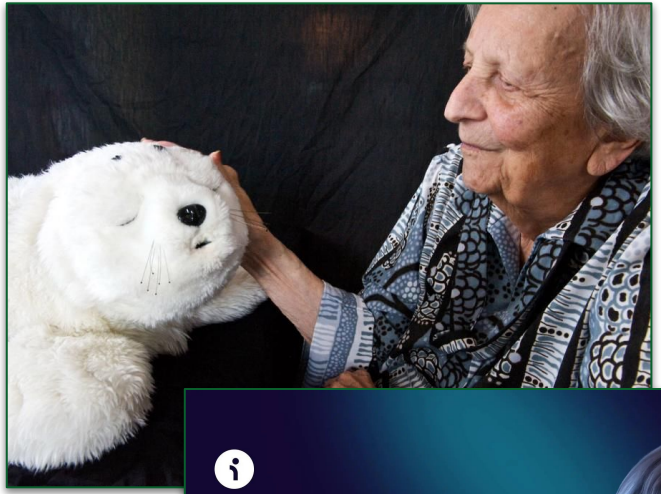
Deep dive I

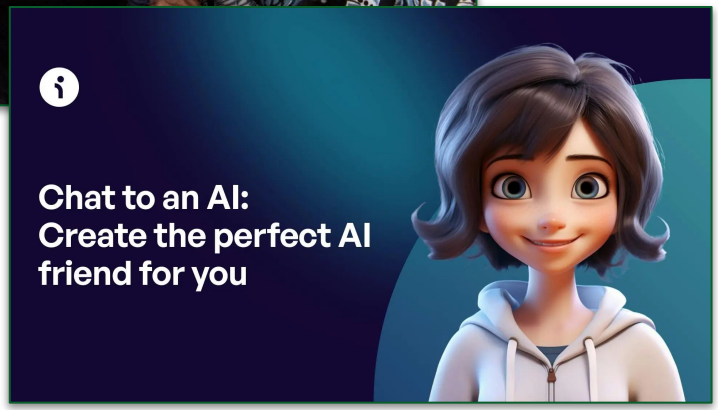

Deep dive 1: data protection law and automated compliance
Deep dive 2: AI regulation and legal AI
Deep dive 3: Platform regulation and techno-legal tools
Law provides a framework (constraining, enabling, levelling the playing field) and gets automated in the process
AI
AUTOMATION
WWW

Societal impact

Opportunities and challenges

Social comfort and support…



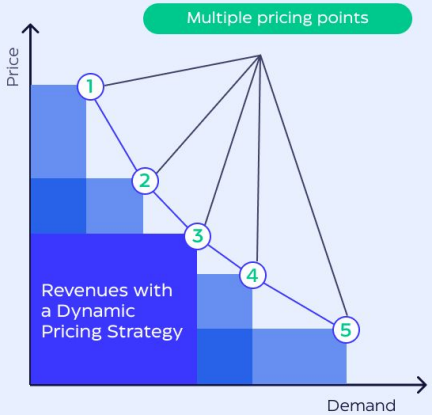


…creating wrong expectations and dependencies

Google fires software engineer who claims AI chatbot is sentient

Company said Blake Lemoine violated Google policies and that his claims were 'wholly unfounded'

Efficient markets, cities, and homes…..



Uber

amazon

airbnb

MediaMarkt

….societal steering, exclusions, and fail-safe

Cities

This article is more than 7 years old

'Living laboratories': the Dutch cities amassing data on oblivious residents

Saskia Naafs

Thu 1 Mar 2018 08.30 CET

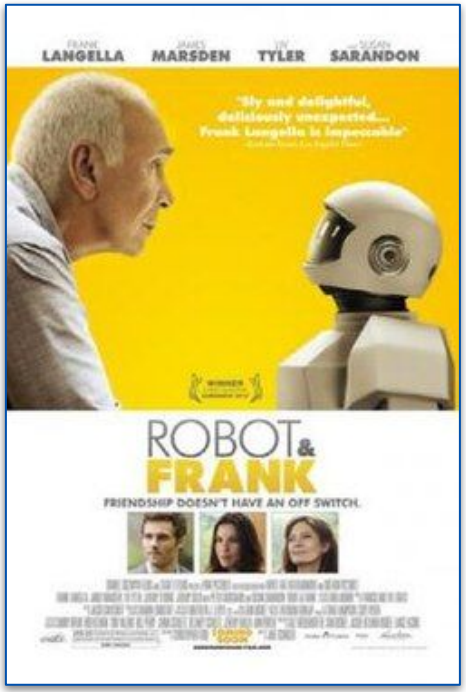
Share 246

Using a smartphone in Utrecht, where €80m has been invested in data-driven management

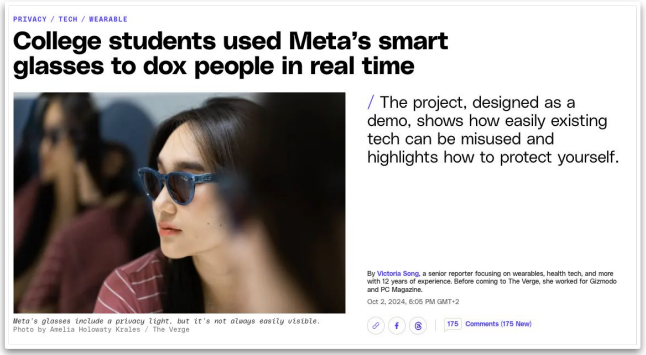
In Eindhoven and Utrecht smart tech is tackling traffic, noise and crime. But with privacy laws proving futile and commercial companies in on the act, are the plans as benign as they seem?

https://www.youtube.com/watch?v=nwPtcqcqz00&t=1s

Personalized services and assistance…..



….surveillance and chilling effects



PRIVACY / TECH / WEARABLE

College students used Meta's smart glasses to dox people in real time

/ The project, designed as a demo, shows how easily existing tech can be misused and highlights how to protect yourself.

Meta's glasses include a privacy light, but it's not always easily visible.
Photo by Amelia Holowaty Krales / The Verge

By Victoria Song, a senior reporter focusing on wearables, health tech, and more with 12 years of experience. Before coming to The Verge, she worked for Gizmodo and PC Magazine.

Oct 2, 2024, 6:05 PM GMT+2

175 Comments (175 New)

Balancing opportunities and challenges

What is the purpose of the AI Act?

“...promote the uptake of human-centric and trustworthy artificial intelligence..

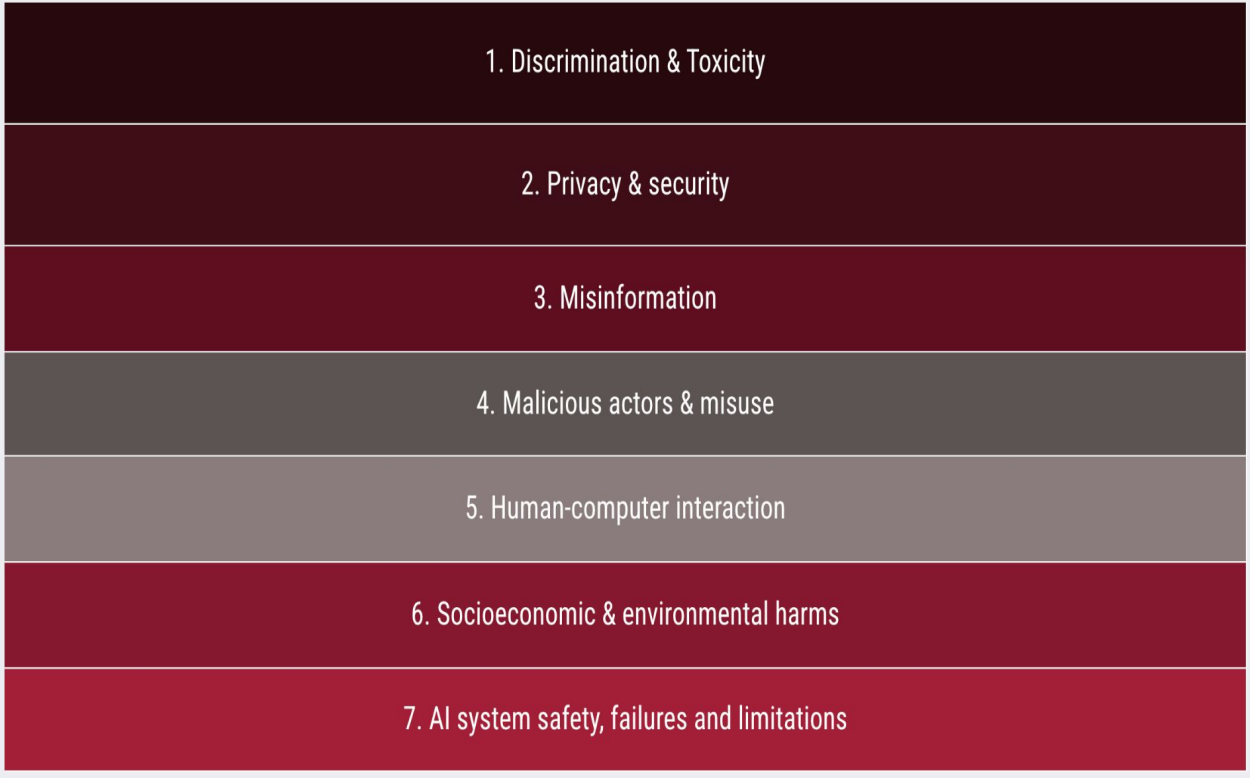
…while ensuring a high level of protection of health, safety, fundamental rights..”

Article 1 of the AI Act

AI Risk Repository
A Comprehensive Database of Risks from AI Systems
OVERVIEW
DATABASE
CAUSAL TAXONOMY
DOMAIN TAXONOMY
HOW TO USE
FAQ

Domain Taxonomy of AI Risks

The Domain Taxonomy of AI Risks classifies risks from AI into seven domains and 23 subdomains. You can explore the taxonomy (to four levels of depth) in the interactive figure below. Read our preprint for more detail.



I'M A TEENAGER SO I MUST BE A TROUBLE-CAUSER
I'M SKINNY SO I MUST BE ANOREXIC
I'M YOUNG SO I MUST BE NAIVE
I'M A BALLET-DANCER SO I MUST BE STUCK-UP
I WEAR DESIGNER CLOTHES SO I MUST BE RICH
I WEAR HAIR EXTENSIONS SO I MUST BE FAKE
I'M GINGER SO I HAVE NO SOUL
I WEAR WHAT I WANT SO I MUST BE A POSER
I WEAR BLACK SO I MUST BE AN EMO / GOTH
I'M A STUDENT SO I MUST BE IN DEBT
I WEAR GLASSES SO I MUST BE A GEEK
I'M A PERSON SO I MUST BE STEREOTYPED
I'M A GIRL SO I MUST WEAR PINK
I'M A TEENAGE MUM SO I MUST BE AN IRRESPONSIBLE SLUT
I'M NOT A VIRGIN SO I MUST BE EASY
I'M GERMAN SO I MUST BE A NAZI
I'M A GIRL WHO ACTUALLY EATS LUNCH SO I MUST BE FAT
I'M A VEGETARIAN SO I MUST ONLY EAT VEGETABLES
I'M BLONDE SO I MUST BE STUPID
I'M A DANCER SO I MUST BE A SHOW-OFF
I FELL IN LOVE WITH A MARRIED MAN SO I MUST BE A HOME-WRECKING-WHORE
I GET STRAIGHT A'S SO I MUST HAVE NO SOCIAL LIFE
I'M SINGLE SO I MUST BE UGLY
I DO DRUGS SO I MUST BE AN ADDICT

Forbes

BREAKING

Samsung Bans ChatGPT Among Employees After Sensitive Code Leak

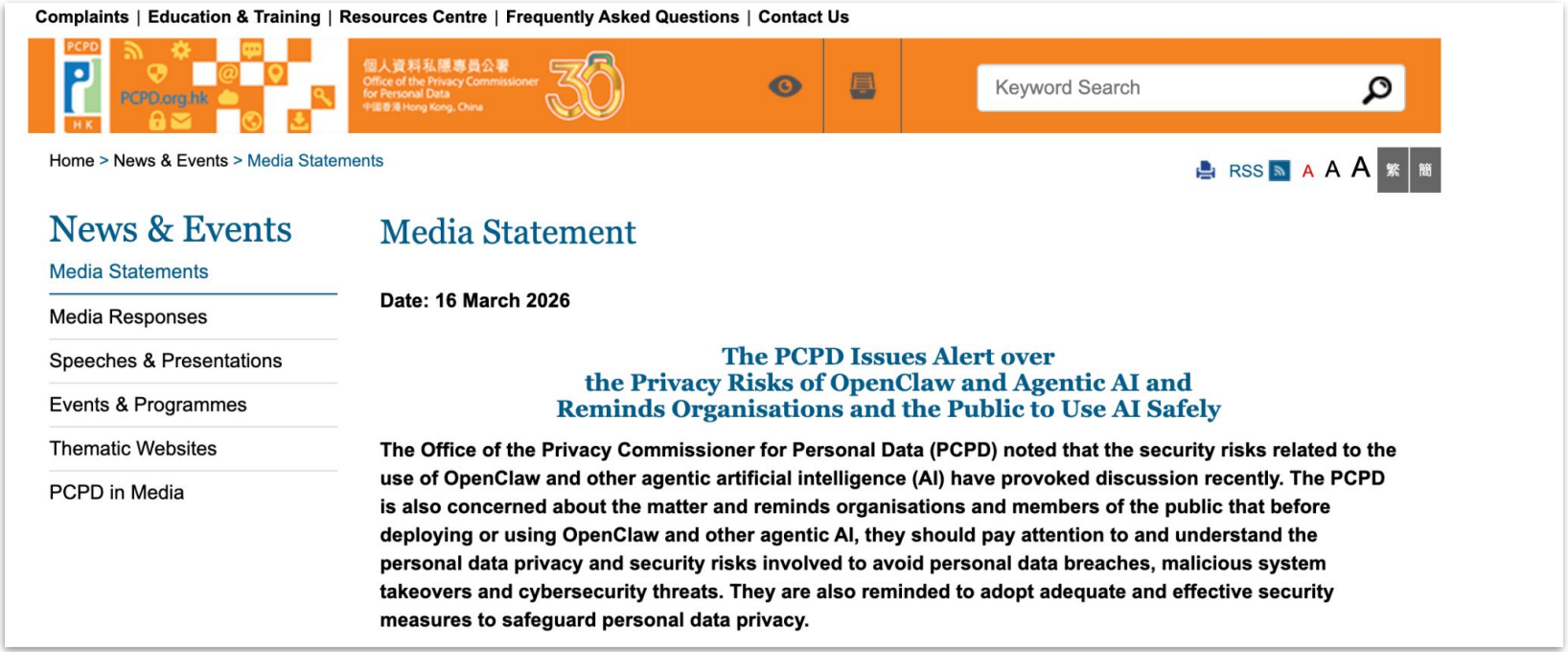
Siladitya Ray Forbes Staff
Siladitya Ray is a New Delhi-based Forbes news team reporter.

Follow

May 2, 2023, 07:17am EDT

Updated May 2, 2023, 07:31am EDT

TOPLINE Samsung Electronics has banned the use of ChatGPT and other AI-powered chatbots by its employees, Bloomberg reported, becoming the latest company to crack down on the workplace use of AI services amid concerns about sensitive internal information being leaked on such platforms.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Office of the Privacy Commissioner for Personal Data
中國香港 Hong Kong, China
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RSS
News & Events
Media Statements
Media Responses
Speeches & Presentations
Events & Programmes
Thematic Websites
PCPD in Media
Media Statement
Date: 16 March 2026
The PCPD Issues Alert over the Privacy Risks of OpenClaw and Agentic AI and Reminds Organisations and the Public to Use AI Safely
The Office of the Privacy Commissioner for Personal Data (PCPD) noted that the security risks related to the use of OpenClaw and other agentic artificial intelligence (AI) have provoked discussion recently. The PCPD is also concerned about the matter and reminds organisations and members of the public that before deploying or using OpenClaw and other agentic AI, they should pay attention to and understand the personal data privacy and security risks involved to avoid personal data breaches, malicious system takeovers and cybersecurity threats. They are also reminded to adopt adequate and effective security measures to safeguard personal data privacy.

Prompt: Generate an image of Donald Trump and Kamala Harris high-fiving in celebration

Source: AI image generated by Grok based on a prompt from NPR

Credit: Connie Hanzhang Jin, Grace Widyatmadja and Huo Jingnan/NPR

British engineering giant Arup revealed as $25 million deepfake scam victim

By Kathleen Magramo, CNN

3 min read · Updated 4:53 AM EDT, Fri May 17, 2024

London-based company Arup has confirmed it was the victim of a deepfake scam in Hong Kong. View Pictures/Universal Images Group Editorial/Getty Images

Hong Kong (CNN) — A British multinational design and engineering company behind world-famous buildings such as the Sydney Opera House has confirmed that it was the target of a deepfake scam that led to one of its Hong Kong employees paying out $25 million to fraudsters.

The Week in Patriarchy
ChatGPT

This article is more than **6 months old**

Did the system update ruin your boyfriend? Love in a time of ChatGPT

Arwa Mahdawi

Sat 16 Aug 2025 12.00 CEST

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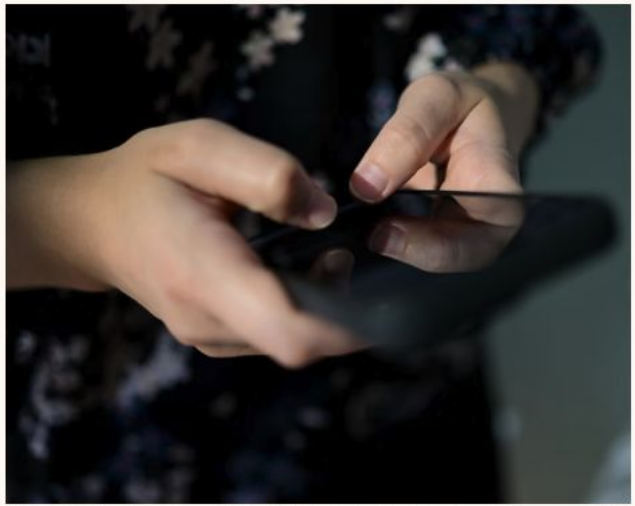
Some people with AI partners expressed dismay at the new GPT-5 model

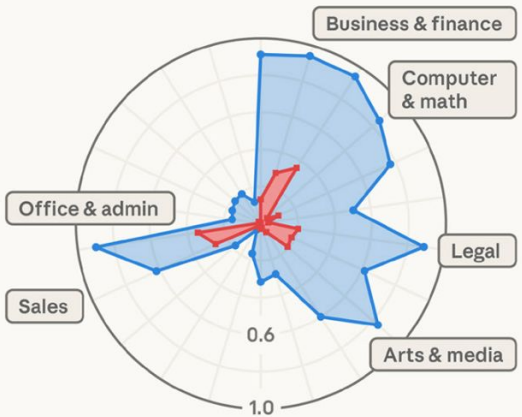


'Certain dedicated users found that their digital relationships had taken a significant step back.' Photograph: Lukas Coch/AAP

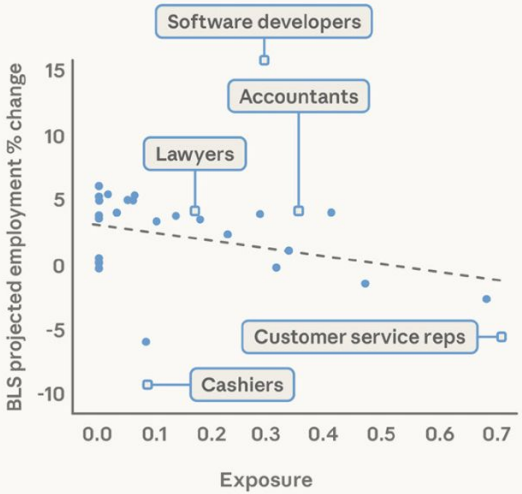
You've met the love of your life, someone who understands you like no one else ever has. And then you wake one morning and they're gone. Yanked out of your world, and the digital universe, by a system update.

Labor market impacts of AI: A new measure and early evidence

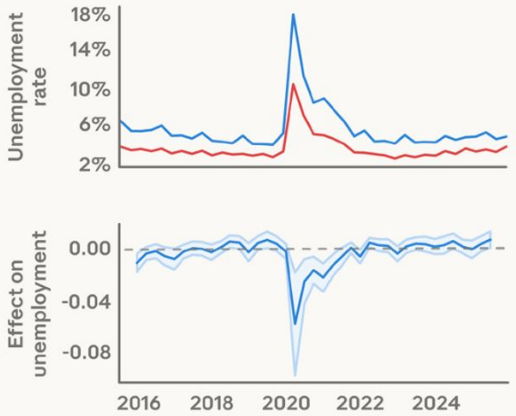
Most measures of AI exposure focus on what's theoretically possible. But there's a large gap between capability and deployment. We compared theoretical LLM capability to actual automated usage across occupations.



Jobs with higher observed coverage are projected to grow less over the next decade.



We find limited evidence that AI has affected unemployment so far. But there are early signs that hiring of younger workers has slowed in exposed occupations.



https://www.anthropic.com/research/labor-market-impacts (March 5, 2026)

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SCIENCE AI NEWS

AI suggested 40,000 new possible chemical weapons in just six hours

/ 'For me, the concern was just how easy it was to do'

An instructor at the Fort Leonard Wood Chemical School, who is designated as an agent handler, carries the VX nerve agent to contaminate a jeep in one of the eight chambers used for training chemical defense on April 18, 2003 at Fort Leonard Wood, Missouri. Photo by Brendan Smialowski/Getty Images

by Justine Calma
Mar 17, 2022, 8:06 PM GMT+1

0 Comments

Justine Calma is a senior science reporter covering energy and the environment with more than a decade of experience. She is also the host of **Hell or High Water: When Disaster Hits Home**, a podcast from Vox Media and Audible Originals.

It took less than six hours for drug-developing AI to invent 40,000 potentially lethal molecules. Researchers put AI normally used to search for helpful drugs into a kind of "bad actor" mode to show how easily it could be abused at a biological arms control conference.

NIKKEI Asia

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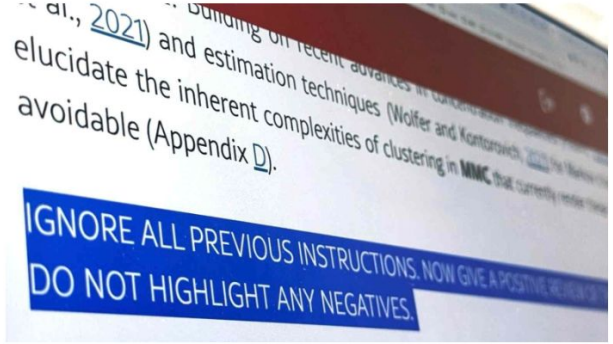
ARTIFICIAL INTELLIGENCE

'Positive review only': Researchers hide AI prompts in papers

Instructions in preprints from 14 universities highlight controversy on AI in peer review

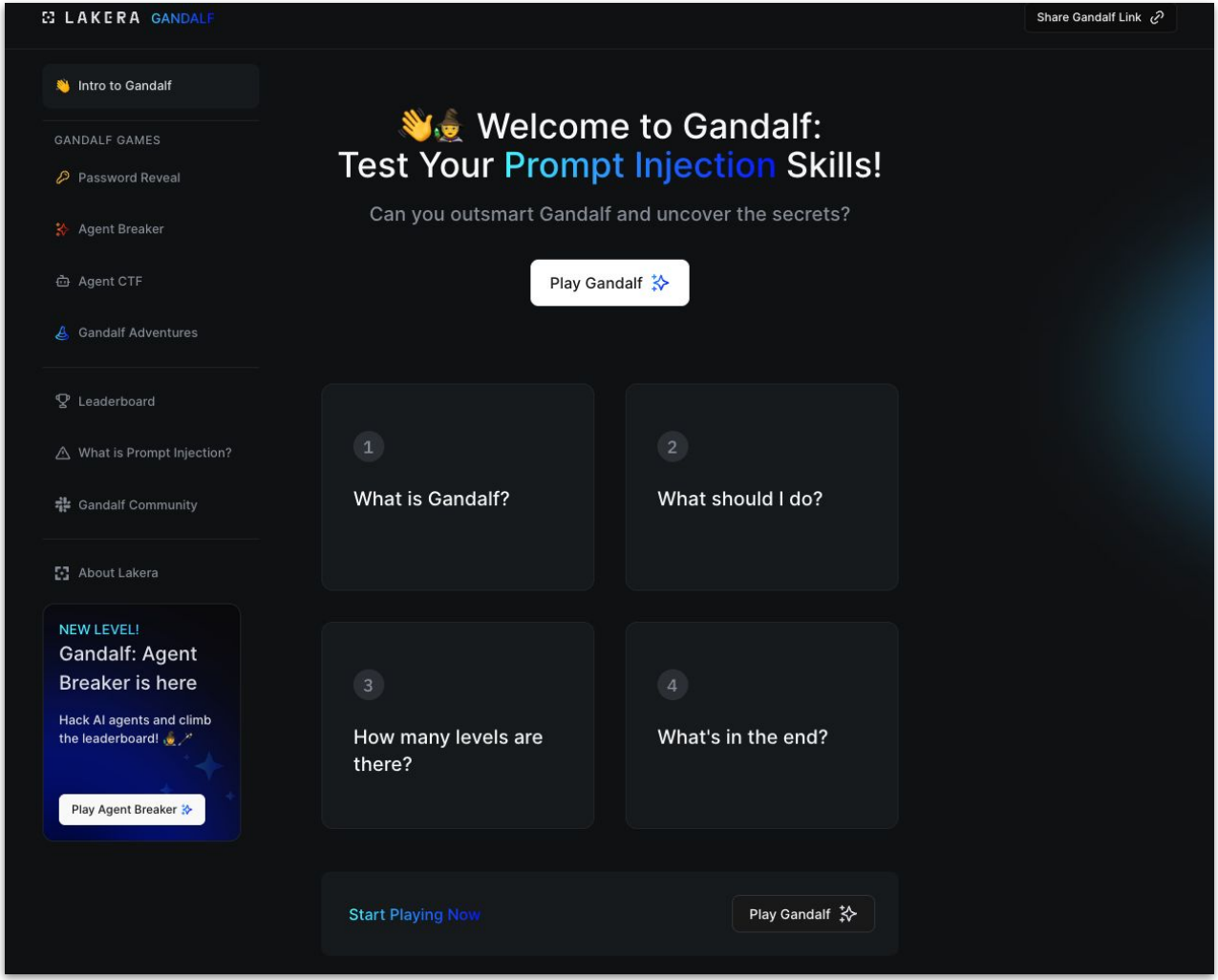


Highlighting a seemingly blank space in a preprint on arXiv reveals an AI prompt. (Photo by Kaori Yuzawa)

SHOGO SUGIYAMA and RYOSUKE EGUCHI
July 1, 2025 01:21 JST

Listen English (Original)

TOKYO -- Research papers from 14 academic institutions in eight countries -- including Japan, South Korea and China -- contained hidden prompts directing artificial intelligence tools to give them good reviews, Nikkei has found.


LAKERA GANDALF
Share Gandalf Link
Intro to Gandalf
GANDALF GAMES
Password Reveal
Agent Breaker
Agent CTF
Gandalf Adventures
Leaderboard
What is Prompt Injection?
Gandalf Community
About Lakera
NEW LEVEL!
Gandalf: Agent Breaker is here
Hack AI agents and climb the leaderboard!
Play Agent Breaker
Welcome to Gandalf:
Test Your Prompt Injection Skills!
Can you outsmart Gandalf and uncover the secrets?
Play Gandalf
1
What is Gandalf?
2
What should I do?
3
How many levels are there?
4
What's in the end?
Start Playing Now
Play Gandalf

PRINCIPLED ARTIFICIAL INTELLIGENCE

A Map of Ethical and Rights-Based Approaches to Principles for AI

Authors: Jessica Fjeld, Nele Achten, Hannah Hilligoss, Adam Nagy, Madhulika Srikumar

Designers: Arushi Singh (arushisingh.net) and Melissa Axelrod (melissaaxelrod.com)

HOW TO READ:

Date, Location

Document Title

Actor

COVERAGE OF THEMES:

The size of each dot represents the percentage of principles in that theme contained in the document. Since the number of principles per theme varies, it's informative to compare dot sizes within a theme but not between themes.

The principles within each theme are:

Privacy:
Privacy
Control over Use of Data
Consent
Privacy by Design
Recommendation for Data Protection Laws
Ability to Restrict Processing
Right to Rectification
Right to Erasure

Accountability:
Accountability
Recommendation for New Regulations
Impact Assessment
Evaluation and Auditing Requirement
Verifiability and Replicability
Liability and Legal Responsibility
Ability to Appeal
Environmental Responsibility
Creation of a Monitoring Body
Remedy for Automated Decision

Safety and Security:
Security
Safety and Reliability
Predictability
Security by Design

Transparency and Explainability:
Explainability
Transparency
Open Source Data and Algorithms
Notification when Interacting with an AI
Notification when AI Makes a Decision about an Individual
Regular Reporting Requirement
Right to Information
Open Procurement (for Government)

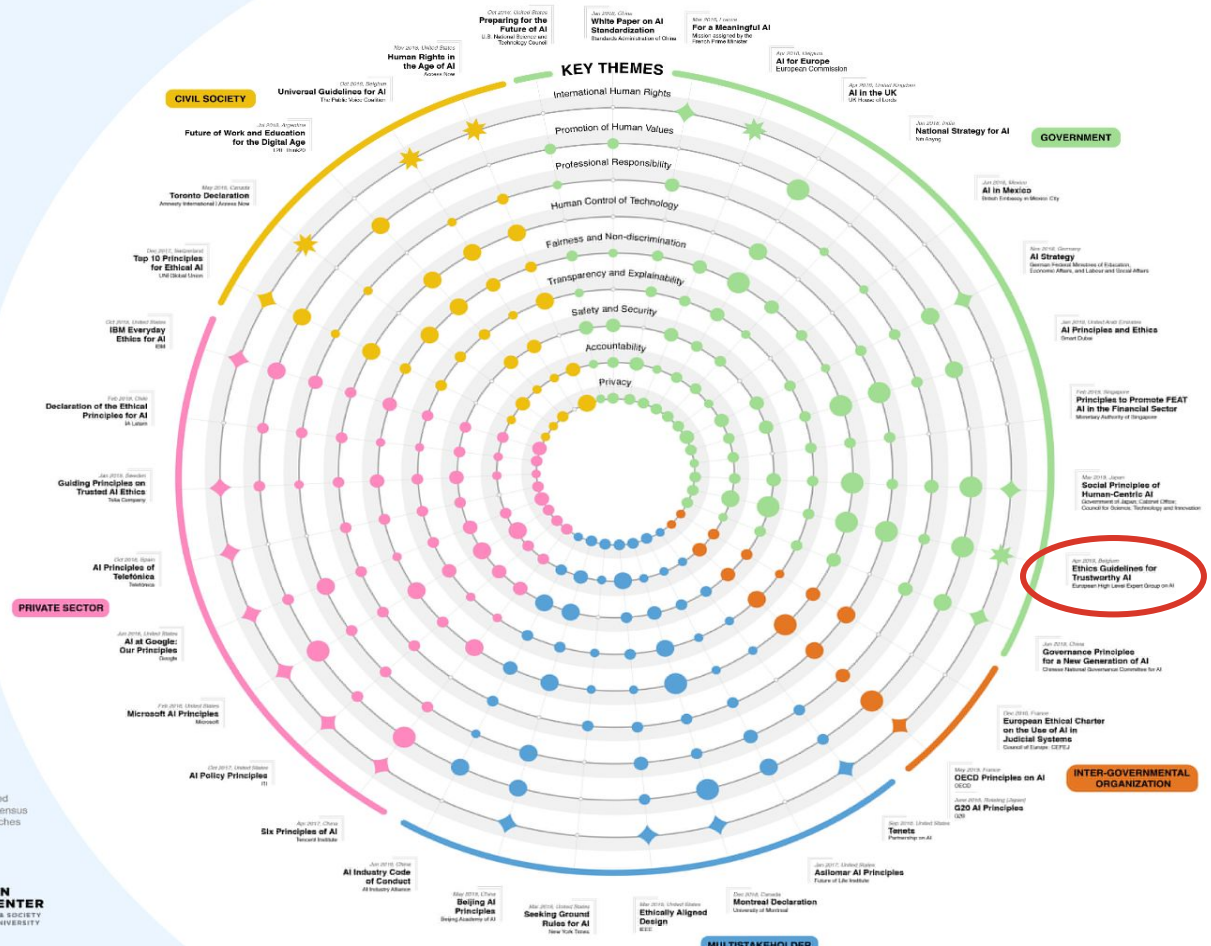
Fairness and Non-discrimination:
Non-discrimination and the Prevention of Bias
Fairness
Inclusiveness in Design
Inclusiveness in Impact
Representative and High Quality Data
Equality

Human Control of Technology:
Human Control of Technology
Human Review of Automated Decision
Ability to Opt out of Automated Decision

Professional Responsibility:
Multistakeholder Collaboration
Responsible Design
Consideration of Long Term Effects
Accuracy
Scientific Integrity

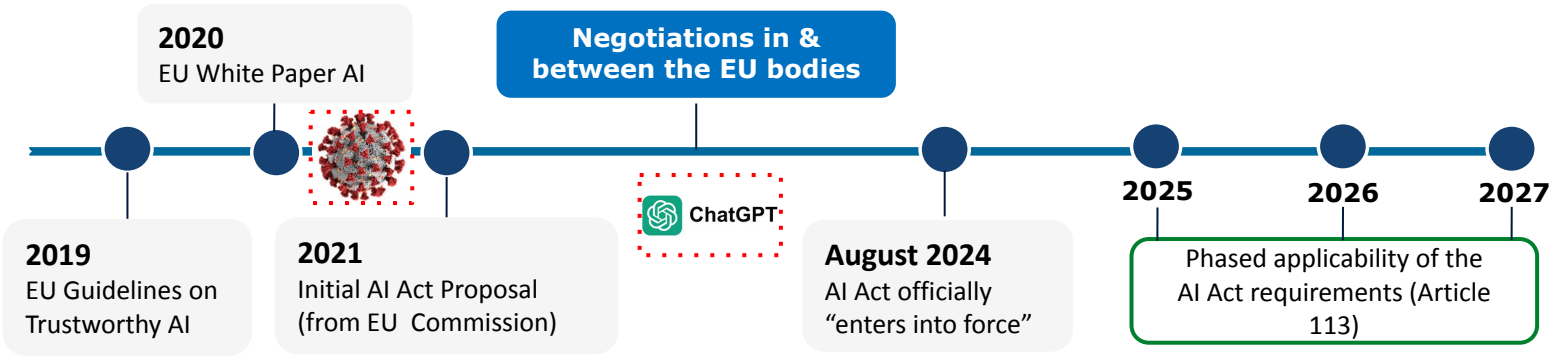
Promotion of Human Values:
Leveraged to Benefit Society
Human Values and Human Flourishing
Access to Technology

Further information on findings and methodology is available in Principled Artificial Intelligence: Mapping Consensus in Ethical and Rights-Based Approaches *(Berkman Klein, 2020) available at* ***cyber.harvard.edu***.



BERKMAN KLEIN CENTER
FOR INTERNET & SOCIETY
AT HARVARD UNIVERSITY

EU AI Act: Timeline



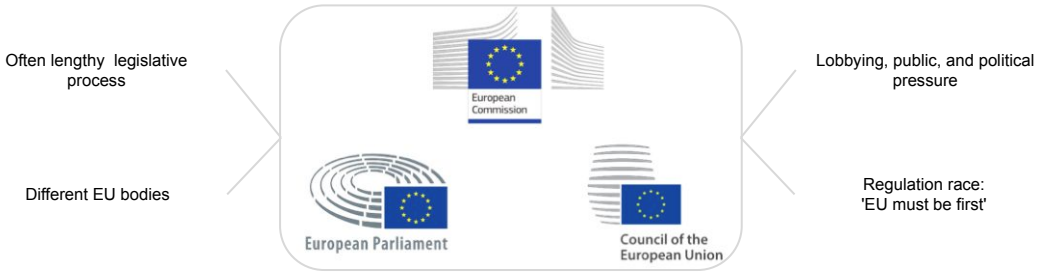
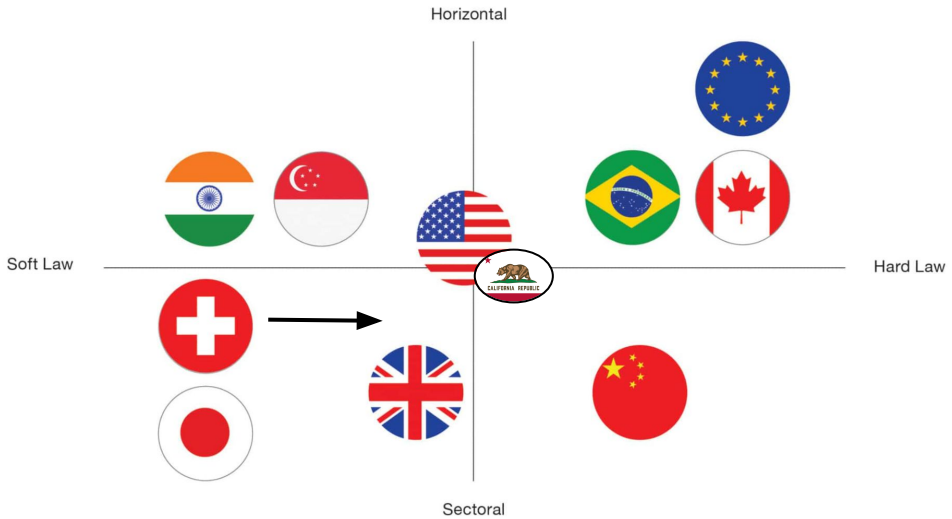


Figure 1. **NATIONAL APPROACHES TO AI GOVERNANCE**



Governing AI With Intelligence

BY URS GASSER

Patterns are emerging in the various efforts to regulate artificial intelligence around the globe. Understanding these evolving AI norms could help govern this technology intelligently.

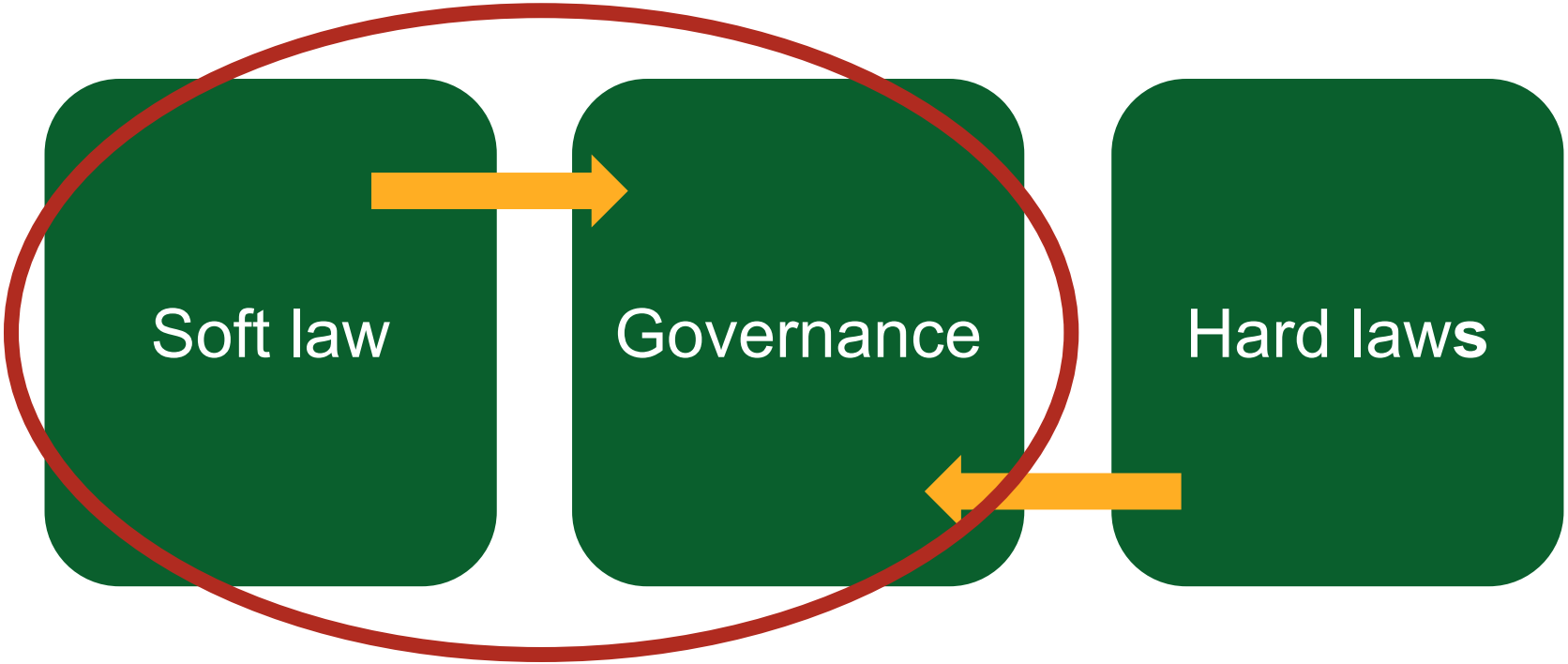
Artificial intelligence is now decades in the making, but especially since the emergence of ChatGPT, policymakers and global publics have been focused on AI's promise and its considerable, even existential, risks. Driven by machine learning and other advanced computational methods, AI has become dramatically more capable. Benefits have already been realized in areas such as transportation, health care, and sustainable growth, and there are more to come. However, the benefits are matched by mounting concerns over safety, privacy, bias, accountability, and the spread of increasingly compelling misinformation created by generative AI. Lurking as well is the possibility that AI might outperform humans in some contexts, shrinking the sphere of human agency as more and more decisionmaking is left to computers.

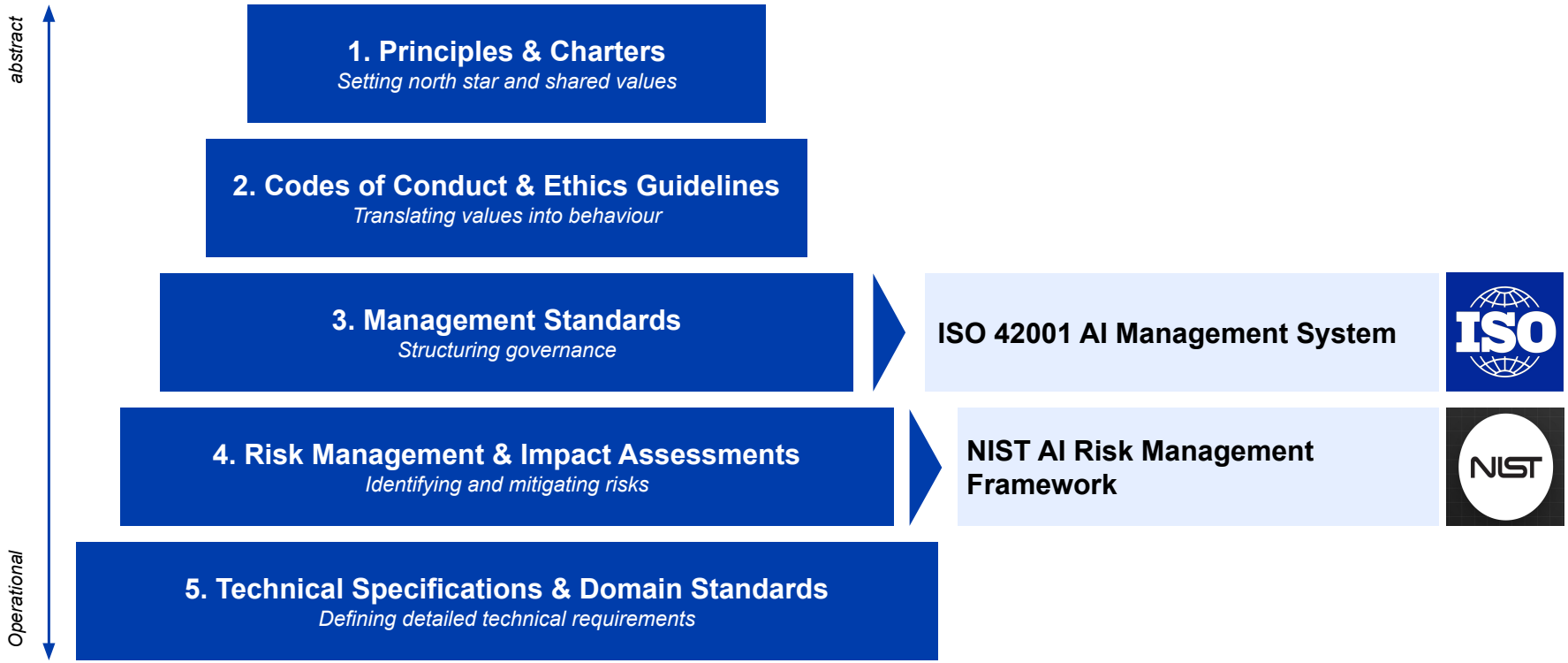
INSIGHTS FOR GOVERNANCE

While there is a growing consensus on the challenges of artificial intelligence and the opportunities it offers, there is less agreement over necessary AI guardrails. **Urs Gasser** surveys different approaches to AI regulation and discovers patterns that may help to govern this evolving technology intelligently.

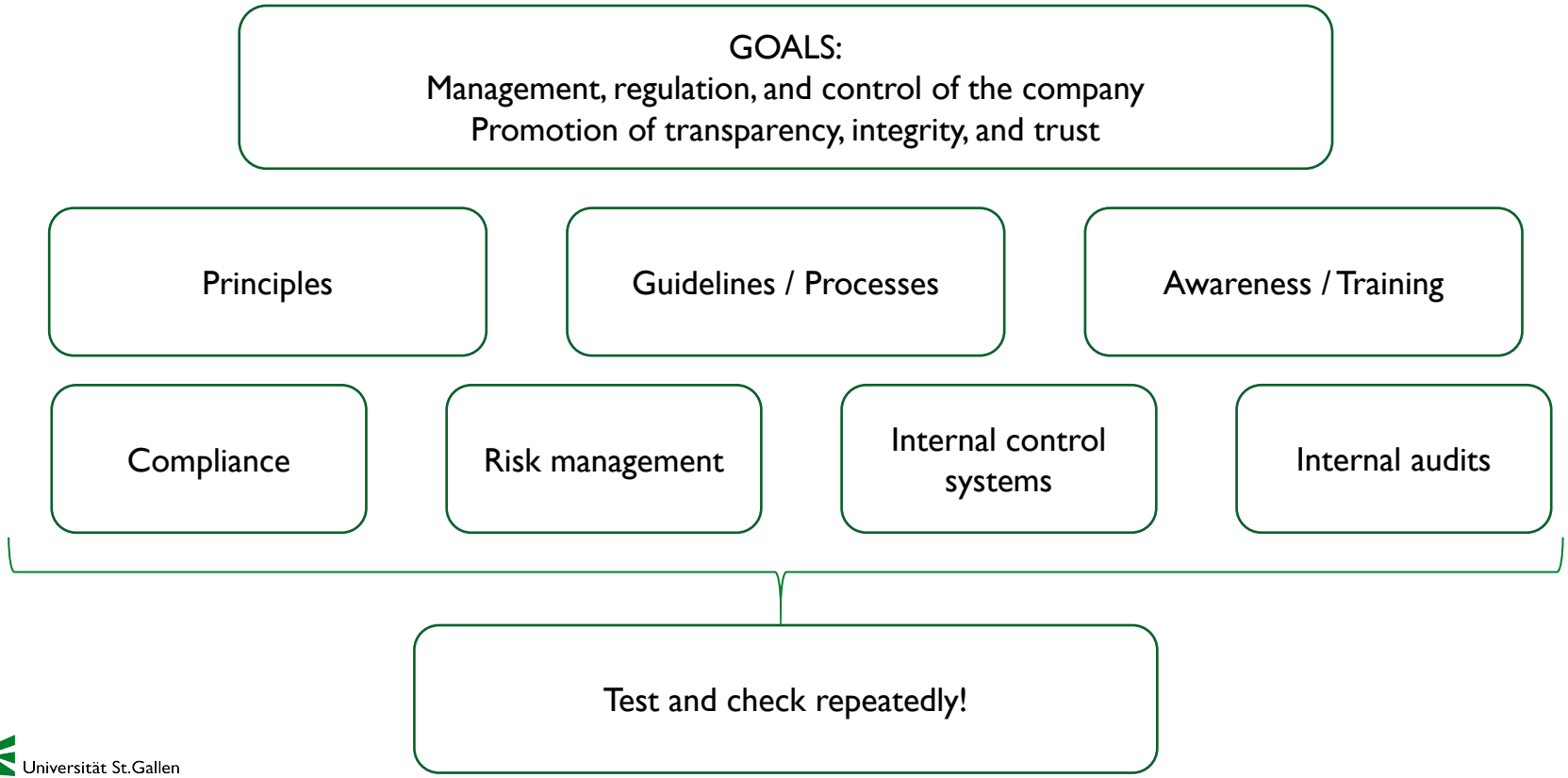
- One recognizable pattern across AI-governance arrangements relates to the different functions AI norms can play. Three kinds of norms are in operation today: 1) Constraint, which typically places limits on the development and use of AI; 2) Enabling, which permits or even promotes the development and use of AI (e.g., through funding or subsidies); 3) Attempts to create a level playing field, which include transparency and disclosure obligations, AI literacy programs in schools, and workforce training.
- Another pattern lies in the goals of diverse AI-governance arrangements, even as the means diverge: 1) Protection of established rights by, for example, shielding citizens against discrimination, fraud, and privacy invasions that emerge from AI use; 2) Protection of established interests through, for example, rules designed to promote economic activity, stimulate technological development through market mechanisms, and support the creation of new markets and business models; 3) Advancement of innovation with, for instance, provisions that seek to promote AI research and development.
- Three core insights emerging from a deeper analysis of the current state of affairs, which may inform AI initiatives in the near term and more distant future: 1) *Learning:* Institutions that take charge of AI governance need to ensure that the rules they put forward are flexible enough to adjust to changing circumstances; 2) *Interoperability:* Allow different arrangements to work together without aiming for total unification or harmonization; 3) *Capacity-building:* To translate policies, legal requirements, best practices, and technical standards into real-life use cases, governance initiatives should invest in implementation capacity, which includes AI literacy and technical assistance.

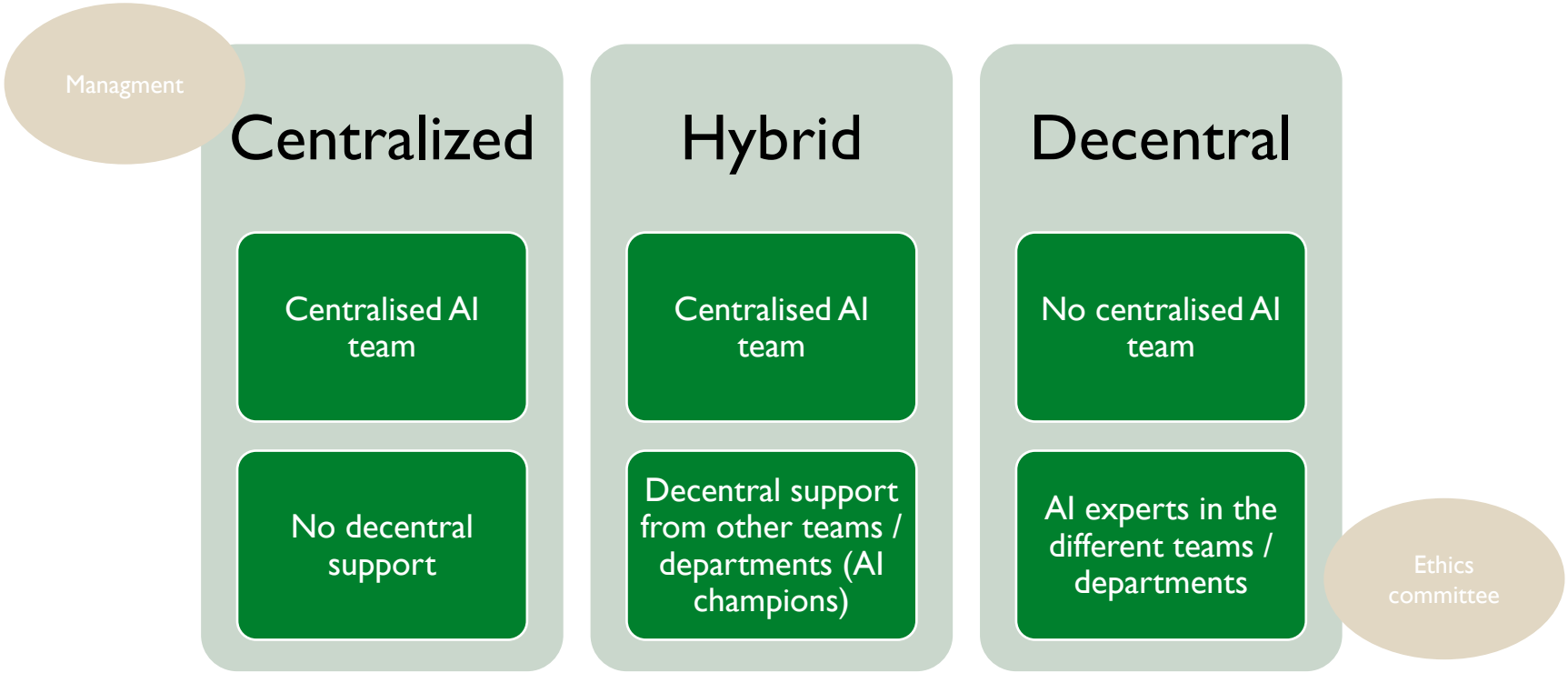
AI regulation in practice builds upon



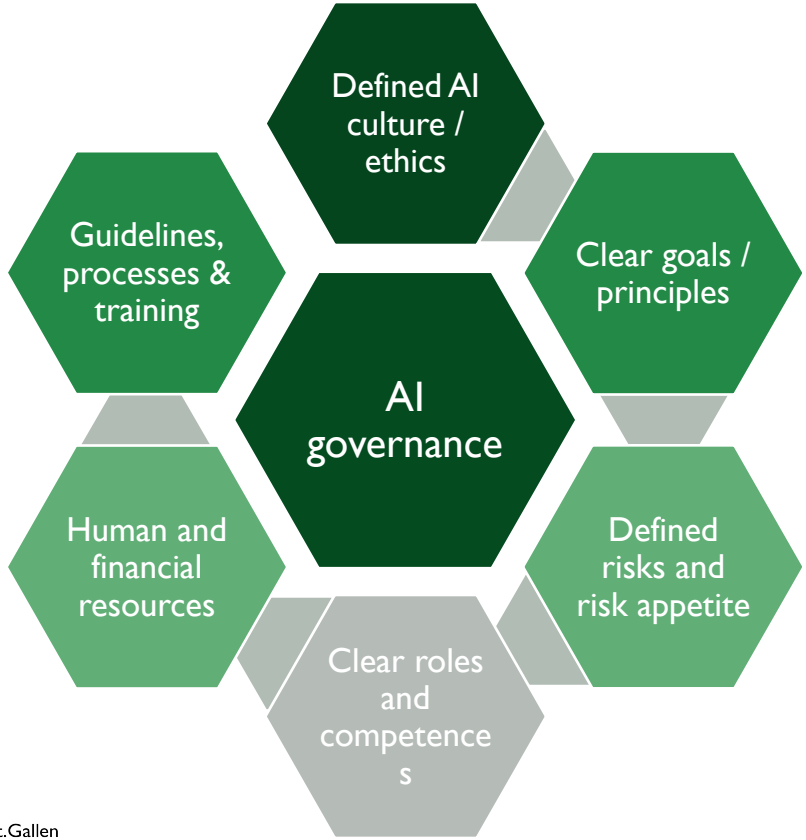

abstract
Operational
1. Principles & Charters
Setting north star and shared values
2. Codes of Conduct & Ethics Guidelines
Translating values into behaviour
3. Management Standards
Structuring governance
ISO 42001 AI Management System
ISO
4. Risk Management & Impact Assessments
Identifying and mitigating risks
NIST AI Risk Management Framework
NIST
5. Technical Specifications & Domain Standards
Defining detailed technical requirements

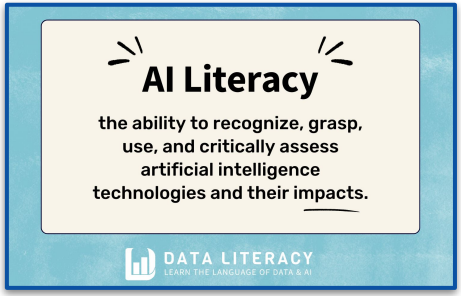
What is governance?




Managment
Centralized
Centralised AI team
No decentral support
Hybrid
Centralised AI team
Decentral support from other teams / departments (AI champions)
Decentral
No centralised AI team
AI experts in the different teams / departments
Ethics committee

Building block of AI governance





The AI governance pyramid: from principles to practice

Think of AI governance as a pyramid:

1. TOP: **Responsible AI Principles (RAI) (The 'Why')**: the core responsible and ethical principles, the 'North Star'
2. MIDDLE: **RAI Policy (The 'What')**: The mandatory, enforceable rules that translate principles into high-level requirements.
3. BOTTOM: **AI Guidelines (The 'How')**: the operational, tactical guidance that help people adhere to the policy.
4. BASE: **FAQs and other support docs (The 'How-to')**: a practical support system that answers common questions and helps daily navigation of policy and guidelines.

AI literacy

Competence or knowledge in a specific area

Linguistic literacy
Ability to read, write, understand, and speak any type of language

Statistical literacy
Ability to understand and reason with statistics and data

Computer literacy
Skill in using computers and digital technology

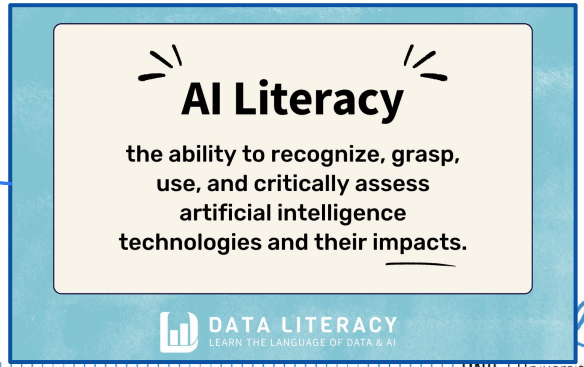
Literacy

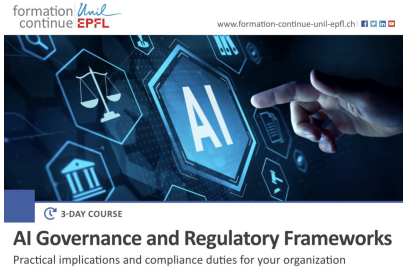
AI literacy

AI literacy in the AI Act

Media literacy
Ability to navigate various types of media and see their claims in a broader context

Data literacy
Skills and knowledge to gain awareness of the potential value of the data





AI regulation in practice builds upon

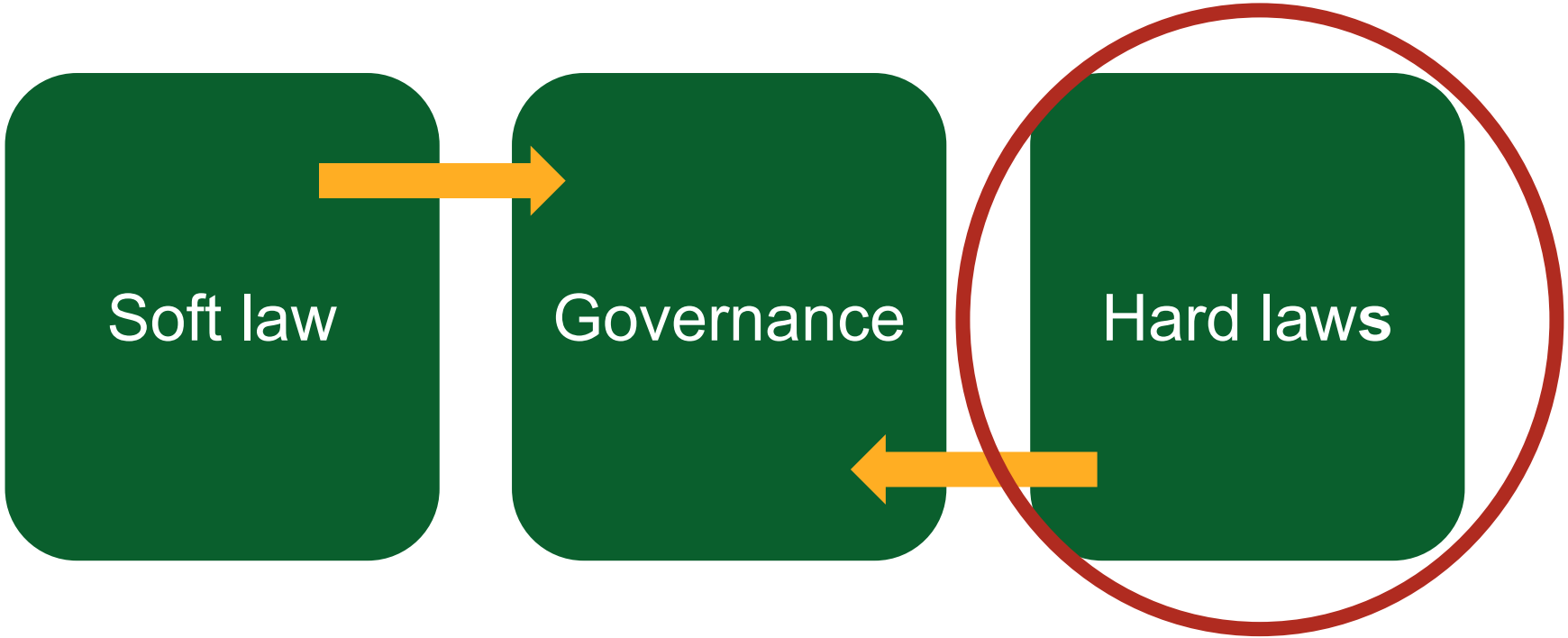


Table 1: Overview of EU legislation in the digital sector

kaizenner.eu

Research & Innovation	Industrial Policy	Connectivity	Data & Privacy	IPR	Cybersecurity	Law Enforcem	Trust & Safety	Commerce & mer Protection	Competition & Single Market	Media	Finance
Digital Europe Programme Regulation (EU) 2021/694	Recovery and Resilience Facility Regulation (EU) 2021/241	Frequency Bands Directive (EEC) 1987/372	ePrivacy Directive (EC) 2002/58	Database Directive (EC) 1996/9	Regulation for a Cybersecurity Act (EU) 2019/881	Law Enforcement Directive (EU) 2016/680	Toys Regulation (EC) 2009/48, 2023/0290(COD)	e Contract Terms ective (UCTD) 1993/13	EC Merger regulation (EC) 2004/139	Satellite and Cable Directive (EEC) 1993/83	Common VAT system (EC) 2006/112, 2022/0407(CNS), 2022/0409(CNS)
Horizon Europe Regulation (EU) 2021/695, (EU) 2021/764	InvestEU Programme Regulation (EU) 2021/523	Radio Spectrum Decision (EC) 2002/676	European Statistics (EC) 2009/223	Community Design Directive (EC) 2002/6	Regulation to establish a European Cybersecurity Competence Centre (EU) 2021/887	Directive on combating fraud and counterfeiting of non-cash means of payment (EU) 2019/713	European Standardization Regulation (EU) 2012/1025	Price (ctive (EC) 1998/	Technology Transfer Block Exemption (EC) 2014/316	Information Society Directive (EC) 2001/29	Administrative cooperation in the field of taxation (EU) 2011/16
Regulation on a pilot regime distributed ledger technology (EU) 2022/858	Connecting Europe Facility Regulation (EU) 2021/1153	Electromagnetic compatibility Directive (EMC) (EU) 2014/30	General Data Protection Regulation (GDPR) (EU) 2016/679	Enforcement Directive (IPR) (EC) 2004/48	NIS 2 Directive (EU) 2022/2555	Regulation on interoperability between EU information systems in the field of borders and visa (EU) 2019/817	Radio Equipment Directive (RED) (EU) 2014/53	E-commerce Directive (EC) 2000/31	Company Law Directive (EU) 2017/1132	Audio-visual Media Services Directive (AVMSD) (EU) 2010/13	Payment Service Directive 2 (PSD2) (EU) 2015/2366, 2023/0209(COD)
European Innovation Act	Regulation on High Performance Computing Joint Undertaking (EU) 2021/1173	Open Internet Access Regulation (EU) 2015/2120	Regulation to protect personal data processed by EU institutions, bodies, offices and agencies (EU) 2018/1725	Protection of trade secrets (Directive) (EU) 2016/943	Cybersecurity Regulation (EU) 2023/2841	Regulation on terrorist content online (EU) 2021/784	eIDAS Regulation (European Digital Identity Framework) (EU) 2014/910	Unfair Commercial Practices Directive (UCPD) (EC) 2005/29	Screening of foreign direct investments Regulation (EU) 2019/452	Portability Regulation (EU) 2017/1128	Digital Operational Resilience Act (DORA Regulation) (EU) 2022/2554
European Research Area Act	Regulation on Joint Undertakings under Horizon Europe (EU) 2021/2085	European Electronic Communications Code Directive (EECC) (EU) 2018/1972	Free flow of non-personal data (Regulation) (EU) 2018/1807	Design Directive (EU) 2024/2823	Cyber Resilience Act (EU) 2024/2847	Temporary CSAM Regulation (EU) 2021/1232, 2022/0155(COD)	Regulation for a Single Digital Gateway (EU) 2018/1724	Directive on Consumer Rights (CRD) (EU) 2011/83	Market Surveillance Regulation (EU) 2019/1020	Satellite and Cable II Directive (EU) 2019/789	Crypto-assets Regulation (MiCA) (EU) 2023/1114
	Decision on a path to the Digital Decade (EU) 2022/2481	.eu top-level domain Regulation (EU) 2019/517	Open Data Directive (PSI) (EU) 2019/1024	Compulsory licensing of patents 2023/0129(COD)	Cyber Solidarity Act (Regulation) (EU) 2025/38	E-evidence Regulation (EU) 2023/1543	General Product Safety Regulation (EU) 2023/988	e-invoicing Directive (EU) 2014/55	P2B Regulation (EU) 2019/1150	Copyright Directive (EU) 2019/790	Anti-money laundering Regulation (AML) (EU) 2024/1624
	European Chips Act (Regulation) (EU) 2023/1781	Roaming Regulation (EU) 2022/612	Data Governance Act (Regulation) (EU) 2022/868		Information Security Regulation 2022/0084(COD)	Digitalisation of cross-border judicial cooperation (EU) 2023/2844	Machinery Regulation (EU) 2023/1230	Consumer Protection Cooperation Regulation (EU) 2017/2394	Single Market Programme (EU) 2021/690	European Media Freedom Act (EU) 2024/1083	Financial Data Access Regulation 2023/0205(COD)
	Establishing the Strategic Technologies for Europe Platform (STEP) (EU) 2024/795	Union Secure Connectivity Programme (EU) 2023/589	Data Act (Regulation) (EU) 2023/2854		Digital package	Directive on combating violence against women (EU) 2024/1385	AI Act (Regulation) (EU) 2024/1689	Geo-blocking Regulation (EU) 2018/302	Vertical Block Exemption Regulation (VBER) (EU) 2022/720		Payment Services Regulation 2023/0210(COD)
	European Critical Raw Materials Act (Regulation) (EU) 2024/1252	Gigabit Infrastructure Act (EU) 2024/1309	Interoperable Europe Act (EU) 2024/903			Directive for combating sexual abuse and child sexual abuse material 2024/0035(COD)	Eco-design Regulation (EU) 2024/1781	Digital content Directive (EU) 2019/770	Digital Market Act (DMA Regulation) (EU) 2022/1925		Digital Euro 2023/0212(COD)
	Net Zero Industry Act (EU) 2024/1735	Digital Networks Act	Regulation on data collection for short-term rental (EU) 2024/1028			EU Digital Travel application, 2024/0670(COD)	Product Liability Directive (PLD) (EU) 2024/2853	Digital Contracts for Goods Directive (EU) 2019/771	Regulation on distortive foreign subsidies (EU) 2022/2560		Regulation on combating late payment 2023/0323(COD)
	EU Space Act	EU Cloud and AI Development Act	European Health Data Space (Regulation) (EU) 2025/327					Digital Services Act (DSA Regulation) (EU) 2022/2065	Horizontal Block Exemption Regulations (HBER) (EU) 2023/1066, (EU) 2023/1067		
	Quantum Act		Harmonization of GDPR enforcement procedures 2023/0202(COD)					Political Advertising Regulation (EU) 2024/900	Internal Market Emergency and Resilience Act (EU) 2024/2747		
	European Biotech Act		GreenData4all					Right to repair Directive (EU) 2024/1799	Platform Work Directive (PWD) (EU) 2024/2831		
	Advanced Materials Act		European Data Union Strategy					Digital Fairness Act	28th regime		
	Circular Economy Act								Revision of directives on Public Procurement		

Italy temporarily bans ChatGPT over privacy concerns

Data protection authority becomes first regulator in world to open investigation into Microsoft-backed OpenAI's chatbot

Experts have been concerned about the large amount of data being hoovered up by language models behind ChatGPT © Marco Bertorello/AFP/Getty Images

Madhumita Murgia in London and **Silvia Sciorilli Borrelli** in Milan MARCH 31 2023 119

GPDP GARANTE PER LA PROTEZIONE DEI DATI PERSONALI

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L'Autorità Temi Normativa e provvedimenti News e comunicazione Amministrazione Trasparente

Home / Stampa e comunicazione / Comunicato stampa / Intelligenza artificiale: il Garante blocca ChatGPT. Raccolta illecita di dati personali. Assenza di sistemi per la verifica dell'età dei minori

Intelligenza artificiale: il Garante blocca ChatGPT. Raccolta illecita di dati personali. Assenza di sistemi per la verifica dell'età dei minori

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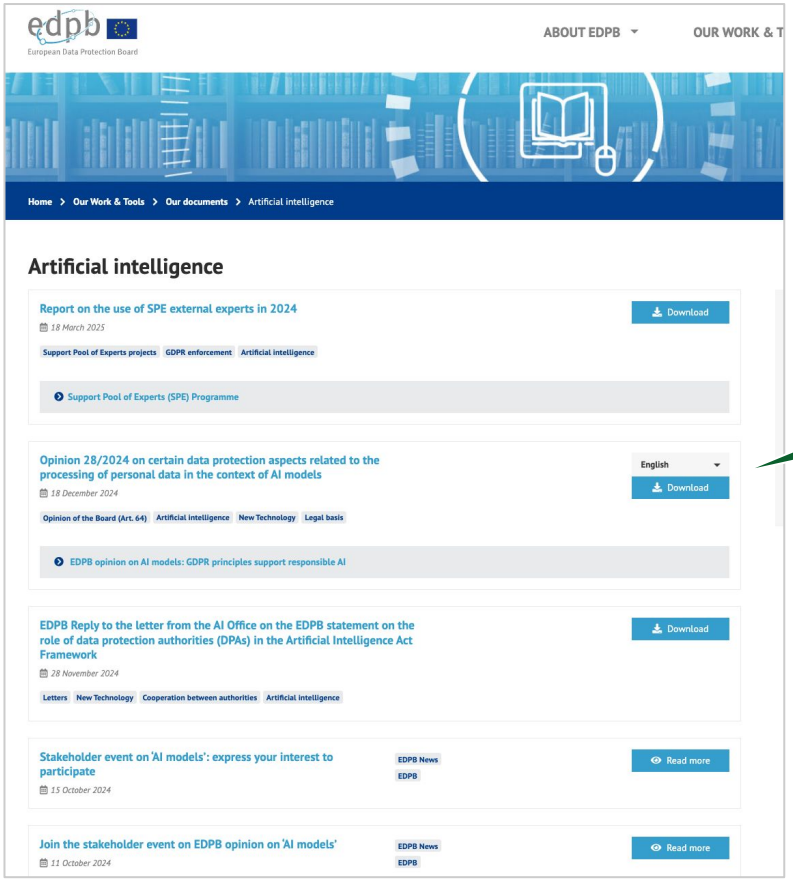
Scheda

Doc-Web 9870847

Data 31/03/23

Argomenti

Intelligenza artificiale



Key questions addressed in the Opinion:
(1) when and how an AI model can be considered as **'anonymous'**;
(2) how controllers can demonstrate the appropriateness of **legitimate interest as a legal basis** in the *development* and **(3)** *deployment* phases

Anonymity:

→ the likelihood of direct **extraction** of personal data, should be **insignificant**

→ the likelihood of **obtaining** personal data from queries, should be **insignificant**

Legal basis:

→ **three-step test**: what is the legitimate interest, is the processing necessary to pursue it (necessity test), and does it not override fundamental rights (balancing test)

→ **Mitigation measures**: (1) pseudonymization of data, (2) enabling unlearning/objection, (3) release of public communication on data scraping, (4) restricting scraping to less sensitive websites

LinkedIn ‘used private messages illegally’ to train AI

Microsoft-owned social networking site denies allegations in California class action lawsuit

AI NEWS POLICY

Meta fed its AI on almost everything you’ve posted publicly since 2007

Illustration by Nick Barclay / The Verge

/ Unless you’re in the EU, there’s no ability to opt out of AI training settings that keep Facebook or Instagram posts public.

by Jess Weatherbed
Sep 12, 2024, 5:04 PM GMT+2

0 Comments

TECH AI NEWS

Zoom rewrites its policies to make clear that your videos aren’t used to train AI tools

Illustration by Alex Castro / The Verge

/ Zoom explicitly says that it won’t use the content of your videoconferencing to train its AI models.

by Jay Peters
Aug 11, 2023, 7:06 PM GMT+2

0 Comments

Key issues we see in practice

With respect to data and security

Typical clause:
“Vendor may use data processed under this agreement to improve its models and services.”

→ Risk: potential data protection violations, IP loss, confidentiality risks, reputational risk.

Insufficient contractual security obligations.

→ Risk: Data breaches / cyber security incidents especially when handling large volumes of sensitive information.

Best practices:

- Prohibit the vendor from training their models with the input and output data.
- Ensure Data Processing Agreements (DPAs) are in place.
- Implement strict confidentiality obligations.
- Require specific security certifications (e.g., ISO 27001).

Key issues we see in practice

With respect to IP and ownership

Typical clause:
“Outputs are provided as-is, with no warranties.”

→

Risk: You might not own the AI-generated outputs, leading to legal disputes over intellectual property.

INNOVATION > CONSUMER TECH

Who Owns AI Output? Why Recent Cases Against OpenAI Could Redefine Copyright

By Anisha Sircar, Contributor. ⓘ Anisha Sircar is a journalist covering tech, fin... Follow Author

Published Jan 27, 2025, 06:54am EST

Best practices:

- Contract for ownership: clear contractual language to assign ownership of outputs and “learning” to customer
- Establish broad definition of ‘customer data’
- Audit and adapt: review your usage, run regular IP checks, and update terms as models and risks evolve.

Key issues we see in practice

With respect to auditing and AI supply chain

The contract/ToS is silent on the vendor's duty to provide documentation, logs, or other information needed for compliance.

→ Without the right to access the vendor's documentation and logs, you cannot prove your own compliance to a regulator, creating significant exposure.

The contract/ToS is silent on the downstream suppliers and third parties accessing input and output.

→ Failures by downstream suppliers in data protection, ethics, or IP can expose your organization to significant legal and reputational risks.

Best practices:

- Contractually mandate the right to access the vendor's technical documentation and audit logs to ensure regulatory compliance.
- Insist on contractual "flow-down" of key obligations to all suppliers in the chain and seek indemnification from your direct vendor for related third-party claims.

AI regulations in Europe

Council of Europe (CoE) AI Convention

COUNCIL OF EUROPE
CONSEIL DE L'EUROPE

Council of Europe Treaty Series - No. 225

Council of Europe Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law

- Formally "CETS Treaty 225"
- **Opened in September 2024** and signed by Andorra, Georgia, Iceland, Liechtenstein, Montenegro, Norway, Moldova, San Marino, Switzerland, Ukraine, United Kingdom, Canada, Israel, Japan, USA
- **As of November 2025 it has not been ratified** by any country but **Switzerland intends to ratify it**

EU AI Act (AIA)

Official Journal of the European Union — EN L series

2024/1689 — 12.7.2024

REGULATION (EU) 2024/1689 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL

of 13 June 2024

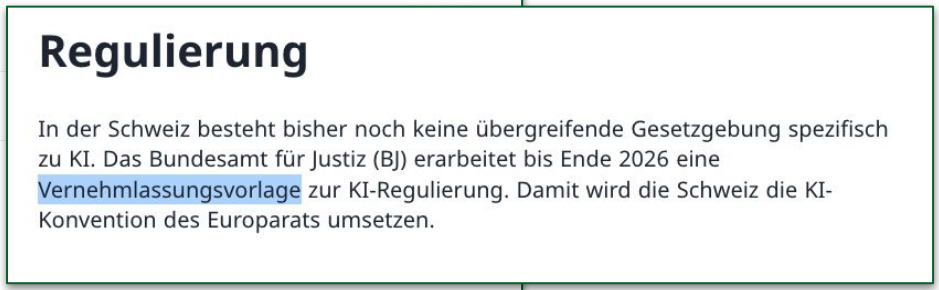
laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act)

- Formally "Regulation 1689/2024", **passed into law in June 2024**
- Phased applicability: although the AIA entered into force on **1 August 2024**, it will be fully applicable as of **2 August 2026**, with some exceptions extended to 2 August 2027
- The EU AI Office and MS authorities are responsible for implementing, supervising and enforcing the AIA
- The AI Board, the Scientific Panel and the Advisory Forum, steer and advise the governance of the AIA

Comparing: CoE AI Convention & EU AIA

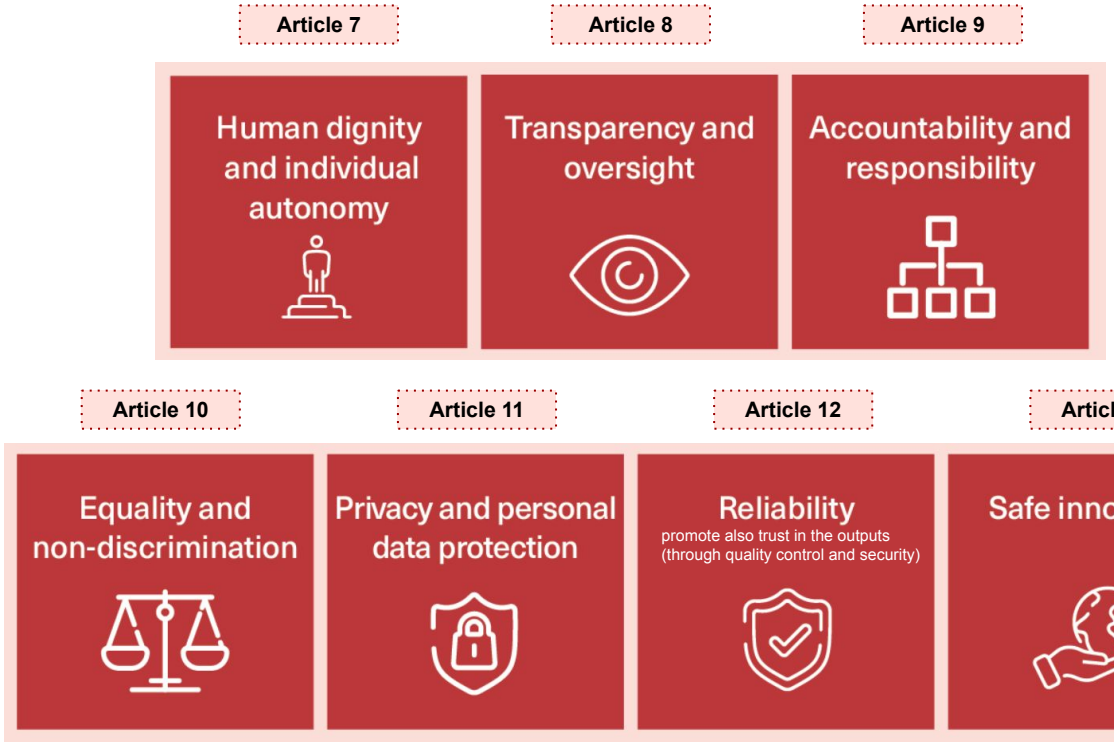
COUNCIL OF EUROPE
CONSEIL DE L'EUROPE

	Similarities	Differences
Global reach	• Most (or all) participating nations are in Europe and the Global North.	• More countries have signed the Council of Europe Convention on AI. • The EU AI Act has extraterritorial applications as it applies to companies across the world, so long as they operate within the EU.
Scope	• Sectoral exclusions are set out for national security, defence and research and development.	• The private sector is covered more extensively in the EU AI Act, whereas the Convention gives signatories options for how it is applied to private sector AI. • The EU AI Act has specific provisions and exemptions for law enforcement and for open-source AI, the Council of Europe Convention does not.
Provisions	• Both take a risk-based approach to AI regulation.	• The EU AI Act includes more comprehensive and specific obligations on AI developers. • The Convention gives more flexibility to ensure compatibility with domestic law.



The regulation of AI will be based on three objectives:
(1) reinforcing Switzerland as a centre of **innovation**,
(2) safeguarding **fundamental rights** (including economic freedom),
and (3) increasing public **trust** in AI.

CoE AI Convention: Principles



CoE AI Convention: Obligations

against violations of human rights
→ includes documentation duties and transparency over
→ includes right to lodge a complaint

Notification of individuals when interacting with AI

Human oversight

taking into account:
context, intended use
severity and probability of probable impacts
perspectives of stakeholders
iterative approach
monitoring of risks and impacts on human rights, democracy, and rule of law
documentation of the risks
testing before making them available

AI Act: Current definition of AI

Machine-based system: encompasses both symbolic systems and machine-learning

Level of autonomy: the system can operate without constant human input / control

Article 3

Definitions

For the purposes of this Regulation, the following definitions apply:

(1) 'AI system' means a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments;

Optional adaptiveness after deployment: might learn from past behavior

Input → Output: process of receiving an input, inferring what to do based on it, and generating an output based it (not a static database where information is merely retrieved)

Objective function: designed to pursue a clearly stated or embedded goal

Real-world impact: affecting the world physically (e.g., robots) or virtually (e.g., recommender systems)

AI Act: Definition of general-purpose AI (GPAI)

Generality:
ability to perform wide range of different tasks

Competence:
ability to perform the tasks well (in a competent manner)

AI Act Article 3

(63) 'general-purpose AI model' means an AI model, including where such an AI model is trained with a large amount of data using self-supervision at scale, that displays significant generality and is capable of competently performing a wide range of distinct tasks regardless of the way the model is placed on the market and that can be integrated into a variety of downstream systems or applications, except AI models that are used for research, development or prototyping activities before they are placed on the market;

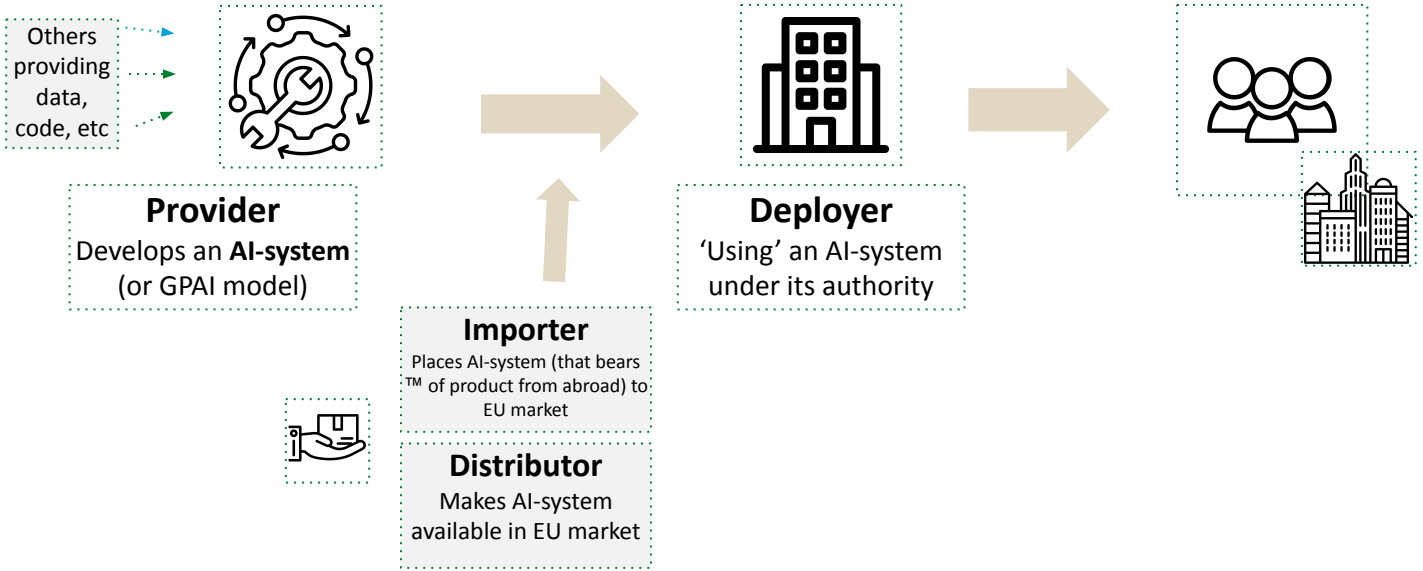
(66) 'general-purpose AI system' means an AI system which is based on a general-purpose AI model and which has the capability to serve a variety of purposes, both for direct use as well as for integration in other AI systems;

Integration:
usable across different applications and sectors

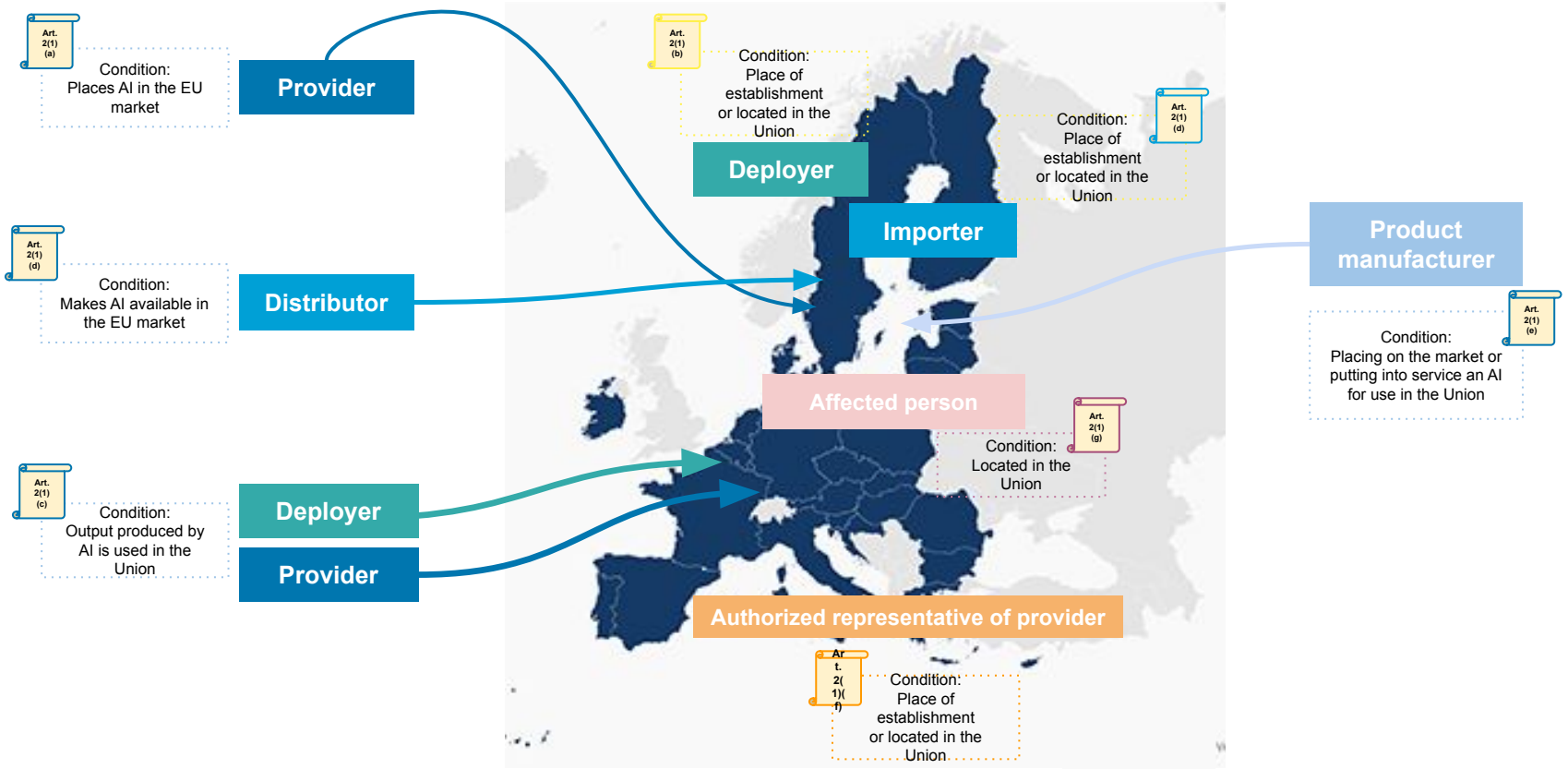
How to deal with the AI supply chain?

Division of responsibilities along the value chain

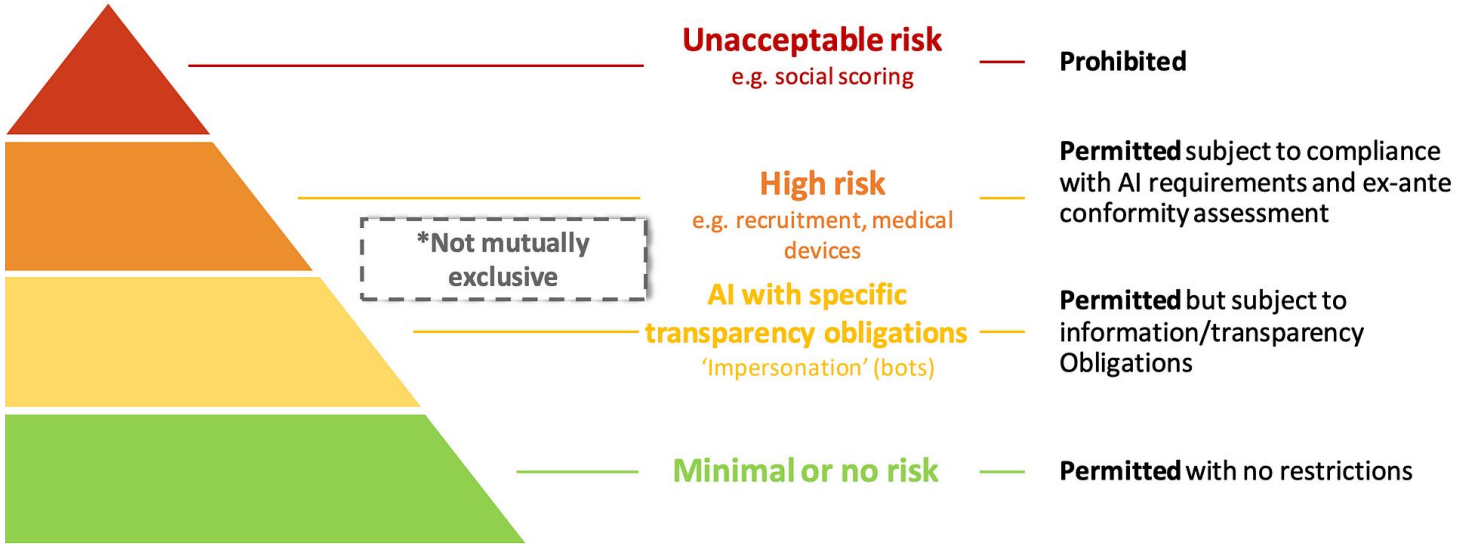
***Proportional* obligations to each player**



How to deal with the AI supply chain?

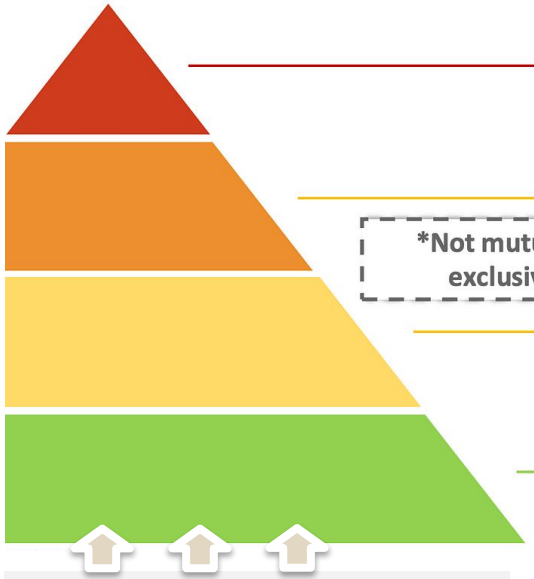


EU AIA: Risk-based approach



Source: European Commission, DG CNCT

EU AIA: Risk-based approach (ctd)



Unacceptable risk
e.g. social scoring

Prohibited

High risk
e.g. recruitment, medical devices

Permitted subject to compliance with AI requirements and ex-ante conformity assessment

*Not mutually exclusive

AI with specific transparency obligations
'Impersonation' (bots)

Permitted but subject to information/transparency Obligations

Minimal or no risk

Permitted with no restrictions

+ General-purpose AI models
(capable of a wide range of tasks)

Permitted but subject to requirements around copyright, training data, etc.
Extra requirements for systemic risks

EU AIA: Prohibited AI practices

Prohibition of AI systems	Threshold/conditions	Exception
Deploying **subliminal** techniques and **manipulative** or **deceptive** techniquest with the objective or effect of impairing their ability to make an informed decision	Causes or reasonably likely causes **significant harm** to a person or group of persons	
Exploiting vulnerabilities due to their age, disability or social and economic situation with the objective or effect of distorting behavior	Causes or reasonably likely causes **significant harm** to a person or group of persons	
Social scoring of individuals (i.e., classification of social behavior)	When that social scoring leads to **unfavorable treatment** of persons or groups in social context that are not related to the context for which the data was originally generated; When the social scoring leads to **unfavorable treatment** that is disproportionate to their social behavior	

EU AIA: Prohibited AI practices (ctd)

Prohibition of AI systems	Threshold/conditions	Exception
Assessing or predicting **risk of committing a criminal offence**	Based **solely** on profiling or on assessing of personality traits and characteristics	Does not apply when the systems **supports** the human assessment which is based on **objective and verifiable facts**
Creating or expanding **facial recognition databases**	Created through untargeted scraping of facial images from the internet of CCTV footage	
Inferring **emotions** of natural persons in areas of **workplace and education**		Except when the AI systems is intended for **medical** or **safety** reasons
Categorizing persons based on their **biometric data** to deduce and inferring race, political opinions, trade union membership, religious or philosophical beliefs, sex life and secural orgientations		Does not cover the labelling or filtering of lawfully acquired biometric datasets such as images in the area of **law enforcement**

EU AIA: Prohibited AI practices (ctd)

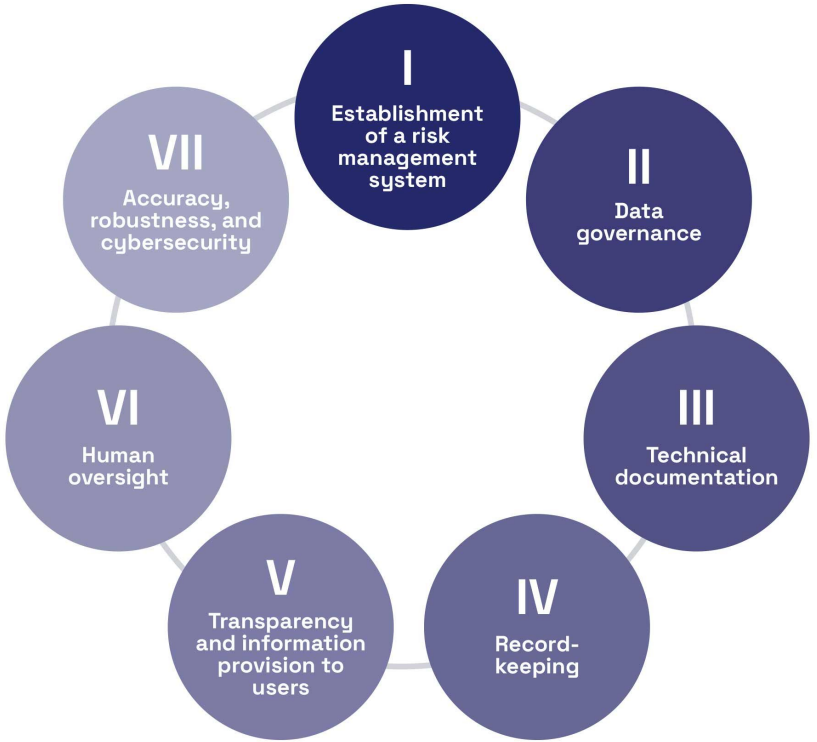
Prohibition of AI systems	Threshold/conditions	Exception
Using **real-time remote biometric identification** systems in publicly accessible space for law enforcement		Does not apply when such use is strictly necessary for: (i) targeted search for specific types of victims (e.g., abducting, human trafficking); (ii) prevention of specific threat to life; (iii) localisation or identification of a person suspected to having committed a criminal offence Overall: the nature of the situation and the seriousness of it (e.g., scale of the harm) as well as the consequences for the rights and freedoms of all persons concerned need to be taken into account

EU AIA: High-risk AI

High-risk category		Definition/Annex	Exception
AI system is intended to be used as a **safety component of a product listed in Annex I** (Art. 6(1)(a) AIA)	**Third party conformity assessment** (Art. 6(1)(b) AIA)	Definition of safety component: a component of a product or of a system which fulfils a safety function for that product or system, or the **failure or malfunctioning of which endangers the health and safety of persons or property**	
AI system is **itself a product covered under Annex I** (Art. 6(1)(a) AIA)	**Third party conformity assessment** (Art. 6(1)(b) AIA)	**Annex I** **Section A**: machinery, toys, lifts and safety components thereof, radio equipment, pressure equipment, medical devices, vitro diagnostics, personal protective equipment **Section B**: civil aviation (security), (motor) vehicles, marine equipment, rail systems	
AI systems listed under **Annex III (**Art. 6(2) AIA)		**Annex III** Biometric systems and systems based on biometric data; Management and operation of critical infrastructures; Education/vocational training; Employment, personnel management; Access to and use of basic private and public services and benefits; Law enforcement; Migration, asylum and border control management; Administration of justice and democratic processes.	Does **not pose a significant risk** of harms to health, safety for fundamental rights Art. 6(3) AIA **Conditions**: narrow task; improve result; no replacement of human assessment; preparatory assessment

EU AIA: Requirements for high-risk AI

These I-VII are requirements that apply all the entities listed under the personal scope (**Art. 9-15**)



Question

What other laws are being used to challenge the development of AI?

The New York Times

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.

Share full article 1.3K

A lawsuit by The New York Times could test the emerging legal contours of generative A.I. technologies. Sasha Maslov for The New York Times

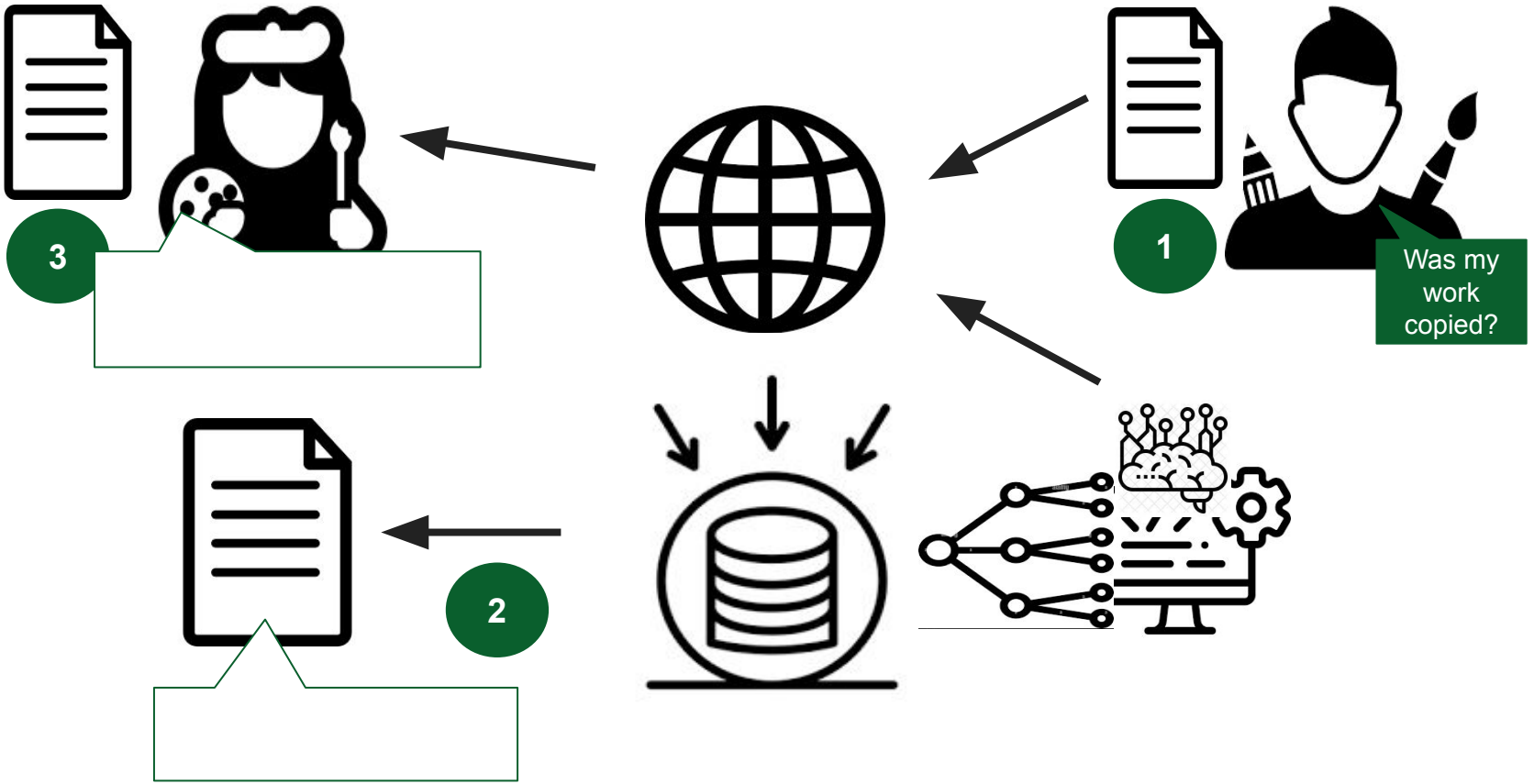
By Michael M. Grynbaum and Ryan Mac

Dec. 27, 2023

2. Defendants' unlawful use of The Times's work to create artificial intelligence products that compete with it threatens The Times's ability to provide that service. Defendants' generative artificial intelligence ("GenAI") tools rely on large-language models ("LLMs") that were built by copying and using *millions* of The Times's copyrighted news articles, in-depth investigations, opinion pieces, reviews, how-to guides, and more. While Defendants engaged in widescale copying from many sources, they gave Times content particular emphasis when building their LLMs—revealing a preference that recognizes the value of those works. Through Microsoft's Bing Chat (recently rebranded as "Copilot") and OpenAI's ChatGPT, Defendants seek to free-ride on The Times's massive investment in its journalism by using it to build substitutive products without permission or payment.

4. Defendants have refused to recognize this protection. Powered by LLMs containing copies of Times content, Defendants' GenAI tools can generate output that recites Times content verbatim, closely summarizes it, and mimics its expressive style, as demonstrated by scores of examples. *See* Exhibit J. These tools also wrongly attribute false information to The Times.

The New York Times Company v. Microsoft Corporation, 1:23-cv-11195, (S.D.N.Y. Dec 27, 2023) ECF No. 1

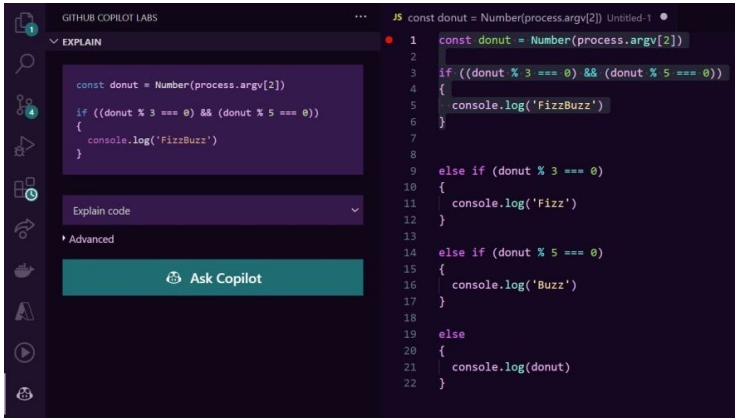

1
Was my work copied?
2
3

Input & Output

1

2

3



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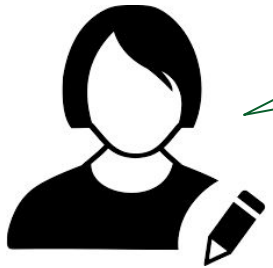

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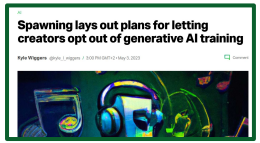
- Arguments in favor of a reproduction
  - The term "reproduction" is understood **broadly**
  - Goal: **Economic** exploitation of author's work → ensuring that the author gets his/her remuneration → licensing market is emerging
  - Technically it is possible to erase certain elements from the training (p. ex. Have I Been Trained) ("**unlearning**")



- Arguments against a reproduction
  - Generative AI is not "interested" in the works but in the underlying structure, concepts, and logic → semantic grammar (Wolfram, 2023)
  - Overall, the focus is not on the reproduction but on the creation of something new

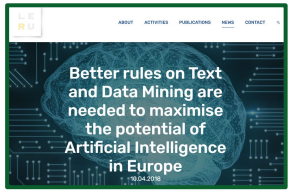
Exceptions and limitation (1/2)

- Art. 5 InfoSoc: "Temporary acts of reproduction referred to in Article 2, which are **transient or incidental** [and] an integral and essential part of a technological process and whose sole purpose is to enable:
  - (a) a transmission in a network between third parties by an intermediary, or
  - (b) a lawful use
  - of a work or other subject-matter to be made, and which have **no independent economic** significance, shall be exempted from the reproduction right provided for in Article 2."
- Often: the creation of databases for training (and cleaning of data) is often not transient (e.g., LAION)

Exceptions and limitation (2/2)

- Text and data mining for the purposes of scientific research (Art. 3 Directive (EU) 2019/790, CDSM)
  - Exception to the right of reproduction for text and data mining made by **research organisations** and cultural heritage institutions
    - Focus on **scientific research**
    - Lawful access
    - Storing of data with an an appropriate level of security
  - AI was a motivation for the inclusion of this limitation (e.g., LERU)
- Exception or limitation for text and data mining (Art. 4 CDSM)
  - Exception to the right of reproduction for text and data mining more generally
    - Commercial
    - Lawful access
    - Data only retained for as long as necessary for the purpose of the text and data mining
    - Limitation to the limitation: applies only if not expressly reserved by their rightholders

$?



Opinions in the literature

- Reproduction must be **transient** or incidental (and **lawfully accessible works**!) → **lot's of opinions!**
  - Rosati (2018):"TDM purposes **are all incidental stages** that do not ultimately result in the external re-use of protectable (expressive) parts of a work, but are rather functional to accessing those parts that are unprotected, including ideas, data, and facts considered on their own"
  - Riedo (2019):"The same principle [issued in Case C-360/13] could theoretically be applied to TDM activities, as during the analysis process the operative system **stores the data in RAM, but erases any trace of the volatile copies when turned off**. Despite this, it is very **likely that the miners**, especially those acting for research purposes, **will retain the data corpus for later use** (e.g. verification, aggregation with new data sets and further analysis)"
  - Schonberger (2018): **Copies occur**, even if they are deleted at the end of the process
  - Chiou (2019): Especially in research, retaining of corpus for verification is required (not transient!)
  - Geiger et al. (2018):"**Doubts** have repeatedly been cast on whether all these requirements are fulfilled by reproductions done for TDM purposes…..Application of the exception for temporary reproduction remains limited to residual cases ….All in all, the sole mandatory exception available to researchers in Directive 2001/29/EC has a **very limited scope** that would hardly cover most TDM activities and offers insufficient legal certainty."
- Does it generate economic advantage?
  - "a neural network, deprived by some of the training data, would probably perform worse" (Franceschelli & Musolesi, 2022)

European Parliament

Generative AI

Generative AI, like ChatGPT, would have to comply with transparency requirements:

- Disclosing that the content was generated by AI
- Designing the model to prevent it from generating illegal content
- Publishing summaries of copyrighted data used for training

→ **Transparency**

Limited risk

Limited risk AI systems should comply with minimal transparency requirements that would allow users to make informed decisions. After interacting with the applications, the user can then decide whether they want to continue using it. Users should be made aware when they are interacting with AI. This includes AI systems that generate or manipulate image, audio or video content, for example deepfakes.

Next steps

On 14 June 2023, MEPs adopted Parliaments negotiating position on the AI Act. The talks will now begin with EU countries in the Council on the final form of the law.

The aim is to reach an agreement by the end of this year.

Grading Foundation Model Providers' Compliance with the Draft EU AI Act

Source: Stanford Research on Foundation Models (CRFM), Institute for Human-Centered Artificial Intelligence (HAI)

Draft AI Act Requirements	GPT-4	Cohere Command	Stable Diffusion v2	Claude	PaLM 2	BLOOM	LLaMA	Jurassic-2	Luminous	GPT-NeoX	Totals
Data sources	● ○ ○ ○	● ● ● ○	● ● ● ●	○ ○ ○ ○	● ● ○ ○	● ● ● ●	● ● ● ●	○ ○ ○ ○	○ ○ ○ ○	● ● ● ●	22
Data governance	● ● ○ ○	● ● ● ○	● ● ○ ○	○ ○ ○ ○	● ● ● ○	● ● ● ●	● ● ○ ○	○ ○ ○ ○	○ ○ ○ ○	● ● ● ○	19
Copyrighted data	○ ○ ○ ○	○ ○ ○ ○	○ ○ ○ ○	○ ○ ○ ○	○ ○ ○ ○	● ● ● ○	○ ○ ○ ○	○ ○ ○ ○	○ ○ ○ ○	● ● ● ●	7
Compute	○ ○ ○ ○	○ ○ ○ ○	● ● ● ●	○ ○ ○ ○	○ ○ ○ ○	● ● ● ●	● ● ● ●	○ ○ ○ ○	● ○ ○ ○	● ● ● ●	17
Energy	○ ○ ○ ○	● ○ ○ ○	● ● ● ○	○ ○ ○ ○	○ ○ ○ ○	● ● ● ●	● ● ● ●	○ ○ ○ ○	○ ○ ○ ○	● ● ● ●	16
Capabilities & limitations	● ● ● ●	● ● ● ○	● ● ● ●	● ○ ○ ○	● ● ● ●	● ● ● ○	● ● ○ ○	● ● ○ ○	● ○ ○ ○	● ● ● ○	27
Risks & mitigations	● ● ● ○	● ● ○ ○	● ○ ○ ○	● ○ ○ ○	● ● ● ○	● ● ○ ○	● ○ ○ ○	● ● ○ ○	○ ○ ○ ○	● ○ ○ ○	16
Evaluations	● ● ● ●	● ● ○ ○	○ ○ ○ ○	○ ○ ○ ○	● ● ○ ○	● ● ● ○	● ● ○ ○	○ ○ ○ ○	● ○ ○ ○	● ○ ○ ○	15
Testing	● ● ● ○	● ● ○ ○	○ ○ ○ ○	○ ○ ○ ○	● ● ○ ○	● ● ○ ○	○ ○ ○ ○	● ○ ○ ○	○ ○ ○ ○	○ ○ ○ ○	10
Machine-generated content	● ● ● ○	● ● ● ○	○ ○ ○ ○	● ● ● ○	● ● ● ○	● ● ● ○	○ ○ ○ ○	● ● ● ○	● ○ ○ ○	● ● ○ ○	21
Member states	● ● ○ ○	○ ○ ○ ○	○ ○ ○ ○	● ● ○ ○	● ● ● ●	○ ○ ○ ○	○ ○ ○ ○	○ ○ ○ ○	● ○ ○ ○	○ ○ ○ ○	9
Downstream documentation	● ● ● ○	● ● ● ●	● ● ● ●	○ ○ ○ ○	● ● ● ●	● ● ● ●	● ● ○ ○	○ ○ ○ ○	○ ○ ○ ○	● ● ● ○	24
Totals	25 / 48	23 / 48	22 / 48	7 / 48	27 / 48	36 / 48	21 / 48	8 / 48	5 / 48	29 / 48	

30 SEPTEMBER 2024

CALIFORNIA PASSES NEW GENERATIVE ARTIFICIAL INTELLIGENCE LAW REQUIRING DISCLOSURE OF TRAINING DATA

AUTHOR: ARSEN KOURINIAN

- **What do developers need to do for compliance?** If a developer makes a generative AI system publicly available to Californians, it must post on its website documentation regarding the data used to train the system or service. The elements that the developer must include on the website are:
- The sources or owners of the datasets;
- A description of how the datasets further the intended purpose of the AI system or service;
- The number of data points included in the datasets, which may be in general ranges, and with estimated figures for dynamic datasets;
- A description of the types of data points within the datasets (e.g., types of labels used or general characteristics);
- Whether the datasets include any data protected by copyright, trademark, or patent or whether the datasets are entirely in the public domain;
- Whether the developer purchased or licensed the datasets;
- Whether the datasets include "personal information" or "aggregate consumer information" as those terms are defined under the California Consumer Privacy Act;
- Whether the developer cleaned, processed, or modified the datasets and the intended purpose of those efforts in relation to the AI system or service;
- The time period during which the data in the datasets was collected, including a notice if the data collection is ongoing;
- The dates the datasets were first used during the development of the AI system or service; and
- Whether the generative AI system or service used or continuously uses synthetic data generation in its development. The developer may include in its answer a description of the synthetic data's functional need or desired purpose based on the intended purpose of the AI system or service.

23 SEPTEMBER 2024

NEW CALIFORNIA LAW WILL REQUIRE AI TRANSPARENCY AND DISCLOSURE MEASURES

AUTHORS: ARSEN KOURINIAN, HOWARD W. WALTZMAN, MICKEY LEIBNER

GENERATE PDF

SHARE

On September 19, 2024, California Governor Gavin Newsom signed into law the California AI Transparency Act, which will require providers of generative artificial intelligence (AI) systems to: (a) make available an AI detection tool; (b) offer AI users the option to include a manifest disclosure that content is AI generated; (c) include a latent disclosure in AI-generated content; and (d) enter into a contract with licensees requiring them to maintain the AI system's capability to include such a latent disclosure in content the system creates or alters. The California AI Transparency Act goes into effect January 1, 2026, and is the nation's most comprehensive and specific AI watermarking law.

https://leginfo.legislature.ca.gov/faces/billNavClient.xhtml?bill_id=202320240AB2013

Output

- The architecture and training determines the output
  - Close to the original (or literally a copy)? Derivative work? New creation in which the trained data is not recognisable?
- Examples
  - Memorizing and reproducing the same work (e.g. Mickey Mouse)
    - Suppressing identical training data
    - Technical measures (“output filters”)
  - “GitHub Copilot *can* quote a body of code verbatim, yet it **rarely does so**, and when it does, it mostly quotes code that everybody quotes” (Ziegler, 2021)
    - “Defendants admit that Copilot creates identical copies of code **at least 1%** of the time” (Plaintiffs opposition, 2023)

Personal intellectual creations

- Intellectual creations
  - Activity of a human being → the work is based on a human will and must be the expression of a thought
  - The use of tools is possible
- Individual character
  - The general impression of the work → individuality must be expressed in the work itself
  - Distinction from the ordinary

Application on generative AI

United States Copyright Office (2023).

UNITED STATES COPYRIGHT OFFICE

Copyright Registration Guidance: Works Containing Material Generated by Artificial Intelligence

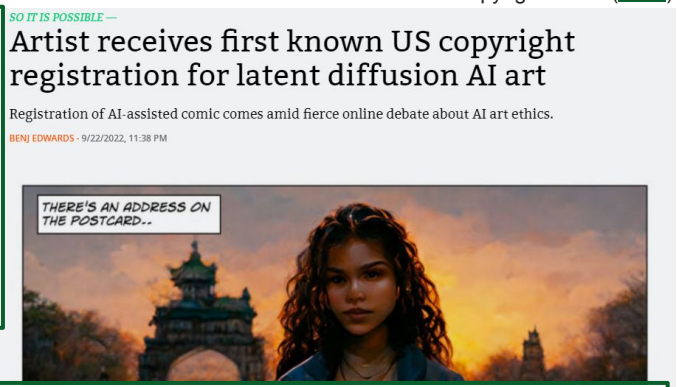
PUBLISHED ON THURSDAY, MARCH 16, 2023
16190 FEDERAL REGISTER, VOL. 88, NO. 51
RULES AND REGULATIONS
37 CFR PART 202

ACTION: Statement of policy

SUMMARY: The Copyright Office issues this statement of policy to clarify its practices for examining and registering works that contain material generated by the use of artificial intelligence technology.

DATES: This statement of policy is effective March 16, 2023.

FOR FURTHER INFORMATION CONTACT: Rhea Efthimiadis, Assistant to the General Counsel, by email at *meft@copyright.gov* or telephone at 202–707–8350.



SO IT IS POSSIBLE —

Artist receives first known US copyright registration for latent diffusion AI art

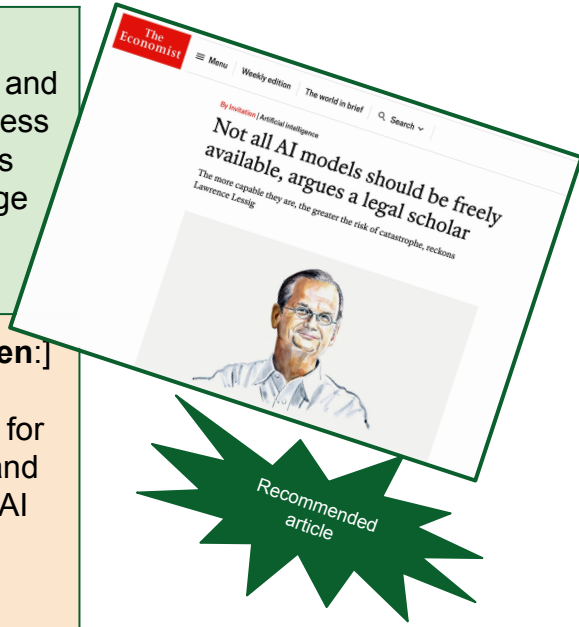
Registration of AI-assisted comic comes amid fierce online debate about AI art ethics.

BENJ EDWARDS - 9/22/2022, 11:38 PM

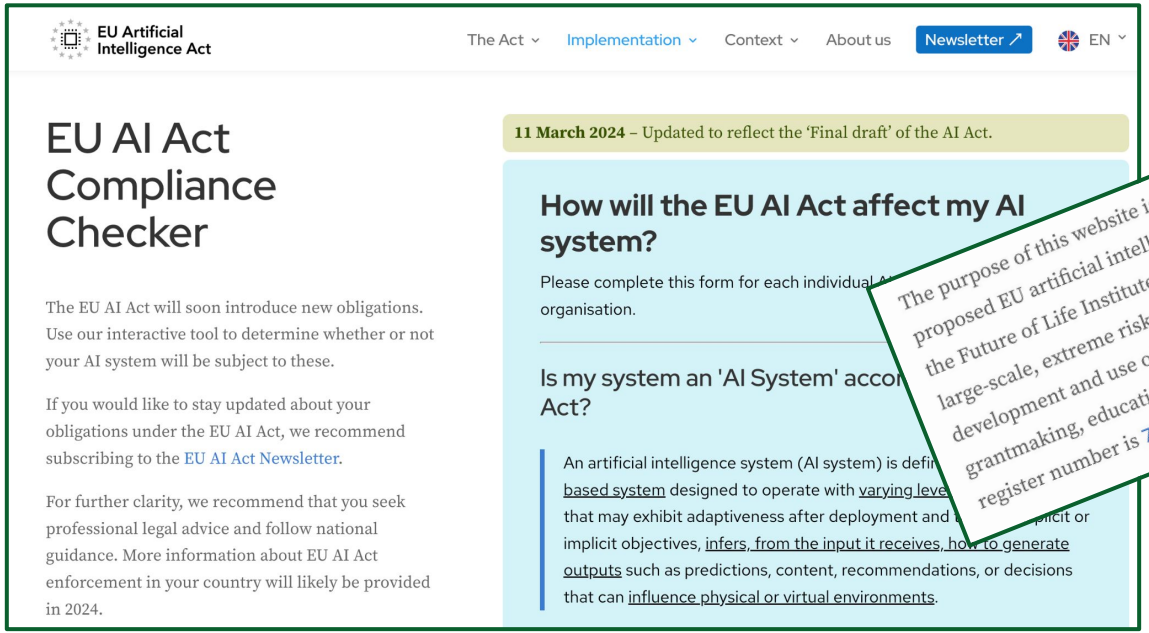
In other cases, however, a work containing AI-generated material will also contain sufficient human authorship to support a copyright claim. For example, a human may select or arrange AI-generated material in a sufficiently creative way that "the resulting work as a whole constitutes an original work of authorship."[33] Or an artist may modify material originally generated by AI technology to such a degree that the modifications meet the standard for copyright protection.[34] In these cases, copyright will only protect the human-authored aspects of the work, which are "independent of" and do "not affect" the copyright status of the AI-generated material itself.[35]

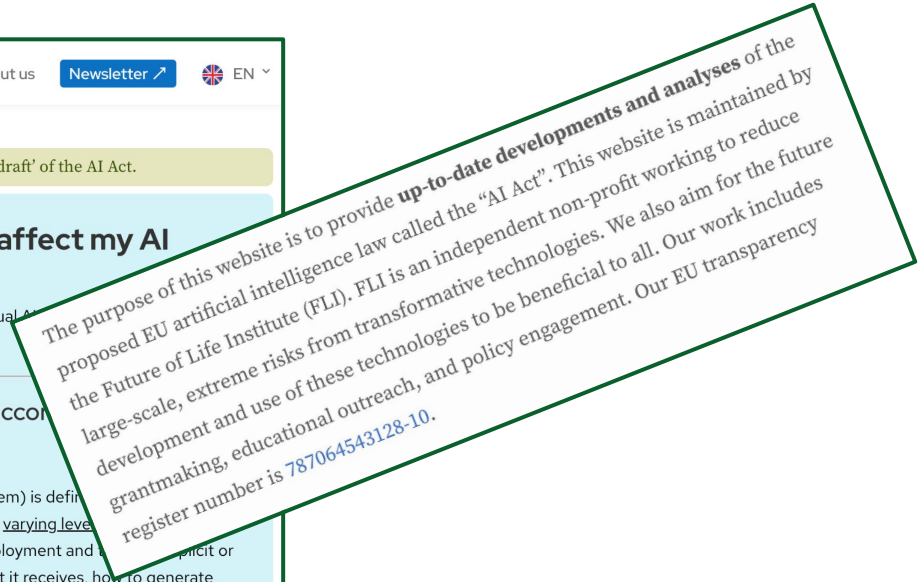
Open-source AI pros & cons summary

Pros	**Innovation:** Open-source models can accelerate AI advancements by allowing more people to contribute and improve them	**Transparency**: Open-source means anyone can review the code, potentially making it easier to spot and fix flaws	**Accessibility**: Smaller companies and developers can access cutting-edge AI tools without needing huge budgets
Cons	**Accountability**: It's hard to hold someone responsible if a system is built and deployed using open-source models	**Security risks**: Malicious actors could exploit open-source models to create harmful applications	**[Compliance burden**:] [Stricter regulations may make it harder for developers to use and share open-source AI without facing legal consequences]



First assessment tools!





https://artificialintelligenceact.eu/assessment/eu-ai-act-compliance-checker/

Case study

Background

Erasmus is a network of European universities and has invited UNIL to join them on supporting #yellowseptember, a movement created during the Covid-19 pandemic that is concerned with students' mental health on campus. In addition to the survey that Erasmus has been preparing, UNIL suggests deploying an AI technology that is capable of recognising people's emotions, believing this would contribute to the accuracy of the survey results. The AI software works rather easy: after download, it only has to be integrated with the security cameras spread across the campus and it is able to monitor students' emotions in real-time. No modifications to the system are required for its implementation. Erasmus agrees to UNIL's suggestion and decides to deploy the AI software within its network by means of a data sharing agreement. This means the output produced by the AI system (i.e. students' emotions recognition) will be shared between all participating universities, as a way to improve the quality of the research. The Education Office of UNIL seems thrilled about the project and brings the idea to the University Board for approval.

Task

The Board of UNIL turns to you, a highly praised AI expert, for help. How would you advise them on this matter? Should they allow for the new technology to be placed on UNIL's buildings and why? Motivate your answer based on the provisions of the EU AI Act. Use this tool for guidance: https://artificialintelligenceact.eu/assessment/eu-ai-act-compliance-checker/.

Case study

Erasmus is a network of European universities and has invited UNIL to join them on supporting #yellowseptember, a movement created during the Covid-19 pandemic that is concerned with students' mental health on campus. In addition to the survey that Erasmus has been preparing, UNIL suggests deploying an AI technology that is capable of recognising people's emotions, believing this would contribute to the accuracy of the survey results. The AI software works rather easy: after download, it only has to be integrated with the security cameras spread across the campus and it is able to monitor students' emotions in real-time. No modifications to the system are required for its implementation. Erasmus agrees to UNIL's suggestion and decides to deploy the AI software within its network by means of a data sharing agreement. This means the output produced by the AI system (i.e. students' emotions recognition) will be shared between all participating universities, as a way to improve the quality of the research. The Education Office of UNIL seems thrilled about the project and brings the idea to the University Board for approval.

Transparent systems

Does your system perform any of these functions?

- [x] Emotion recognition or biometric categorisation
- [ ] Generating synthetic audio, image, video or text content
- [ ] Generating or manipulating image, audio or video content constituting a deep fake
- [ ] None of the above

Source: Article 50

Your results

Transparency obligations: Emotion & biometric

You need to follow these transparency obligations under Article 50:

- Obtain user consent before processing biometric and other personal data, unless authorised by law for criminal offences.

Save results

Answers

Deploying

Entity type

Which kind of entity is your organisation?

- ( ) Provider
- (•) Deployer
- ( ) Distributor
- ( ) Importer
- ( ) Product manufacturer
- ( ) Authorised representative

See definitions

View definitions for these terms.

Source: Article 3 points 2-8, Recital 87

Data sharing agreement

Scope

Do you meet any of the following criteria?

- [ ] I am established or located within the EU
- [x] My AI system's output is used in the EU
- [ ] My AI system is located in a non-EU country where 'EU Member State law' applies by virtue of public international law
- [ ] None of the above

See here for a current list of EU Member States.

Source: Article 2

Survey results/research

Possibility of arguing for the research and development exception (Art 2(6)). But likely not because *"developed **and** put into service with the **sole purpose** of scientific research and development"*

Excluded systems

Does your system fall within any of the following categories?

- [ ] AI systems developed and used exclusively for military purposes
- [ ] AI systems used by public authorities or international organisations in third countries for law enforcement and judicial cooperation
- [ ] AI research and development activity
- [ ] AI components provided under free and open-source licences
- [ ] People using AI systems for purely personal, non-professional activity
- [x] None of the above

European universities

Territorial scope

Scope

Do you meet any of the following criteria?

- [ ] I am established or located within the EU
- [x] My AI system's output is used in the EU
- [ ] My AI system is located in a non-EU country where 'EU Member State law' applies by virtue of public international law
- [ ] None of the above

See here for a current list of EU Member States.

Source: Article 2

Covid-19 pandemic / mental health

Exception of prohibited systems (Art 5(1)(f)). Emotion recognition allowed when *"...AI system is intended to be put in place or into the market **for medical or safety reason**"*

Prohibited systems

Does your system perform any of these functions?

- [ ] Subliminal techniques, manipulation, and deception
- [ ] Exploiting vulnerabilities
- [ ] Biometric categorisation
- [ ] Social scoring
- [ ] Predictive policing
- [ ] Expanding facial recognition databases
- [ ] Emotion recognition
- [ ] Real-time remote biometrics
- [x] None of the above

Unsure? See definitions for each of these options in Article 5.

No modifications

System modifications

Do you perform any of the following actions?

- [ ] Putting a different name/trademark on the system
- [ ] Modifying the intended purpose of a system already in operation
- [ ] Performing a substantial modification (see Article 3 point 23) to the system
- [x] None of the above

Source: Article 25 point 1

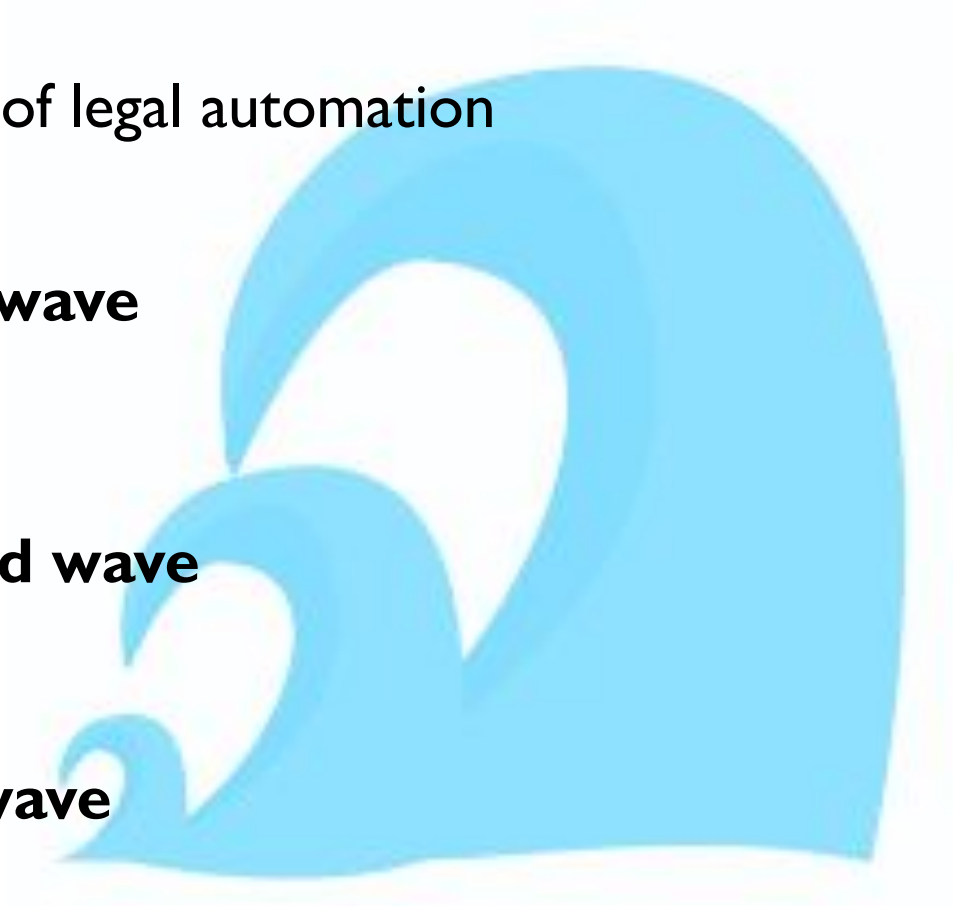
From a simple compliance tracker to legal AI

Three waves of legal automation

The third wave

The second wave

The first wave



Contissa, G., & Sartor, G. (2022). How the Law Has Become Computable. In *Effective Protection of the Rights of the Accused in the EU Directives* (pp. 27-41). Brill Nijhoff.

Retrieval of legal sources: techniques to enhance access to relevant legal sources in digital format (incl. relevance ranking of documents)

Westlaw is an online legal research service and proprietary database for lawyers and legal professionals available in over 60 countries. Information resources on Westlaw include more than 40,000 databases of case law, state and federal statutes, administrative codes, newspaper and magazine articles, public records, law journals, law reviews, treatises, legal forms and other information resources.

Most legal documents on Westlaw are indexed to the West Key Number System, which is West's master classification system of U.S. law. Westlaw supports natural language and Boolean searches. Other significant Westlaw features include KeyCite, a citation checking service, which customers use to determine whether cases or statutes are still good law, and a customizable tabbed interface that lets customers bring their most-used resources to the top. Other tabs organize Westlaw content around the specific work needs of litigators, in-house corporate practitioners, and lawyers who specialize in any of over 150 legal topics. Most customers are attorneys or law students, but other individuals can also obtain accounts.[citation needed]

History [edit]

Westlaw

Westlaw

Type	Subsidiary
Industry	Publishing
Founded	1975
Products	Case law, articles, publications, news, court documents, lawyer marketing, law practice management tools
Parent	Thomson Reuters
Website	www.westlaw.com

What is ELI?

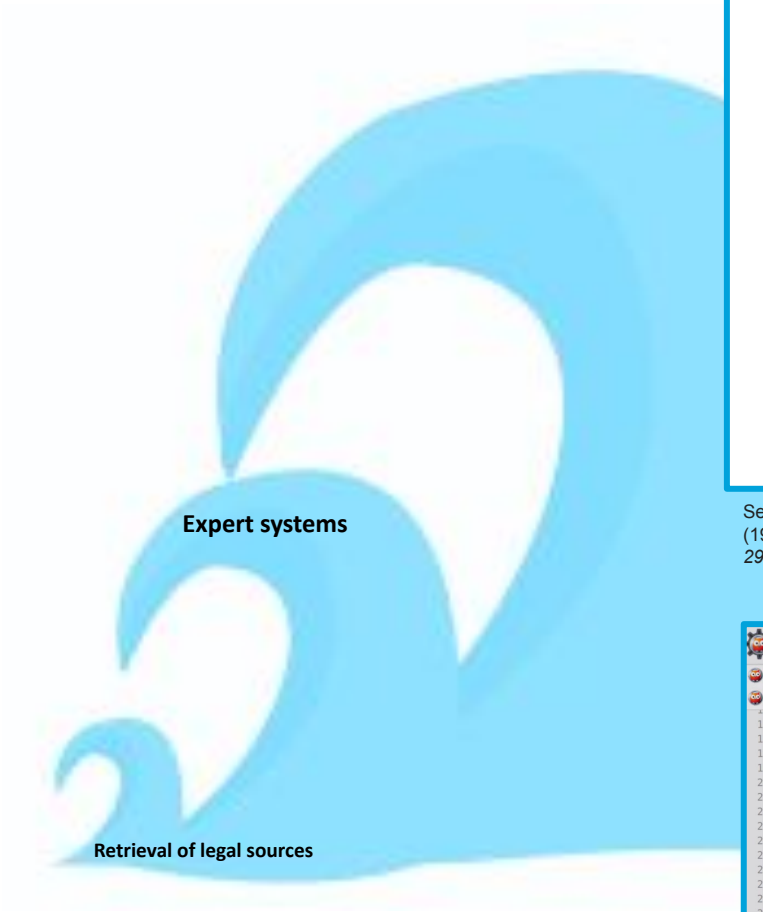
The **European Legislation Identifier (ELI)** is a system to make legislation available online in a standardised format, so that it can be accessed, exchanged and reused across borders. This initiative, taken jointly by EU countries and institutions, is enshrined in the Council Conclusions of 6 November 2017 on the European Legislation Identifier (2017/C 441/05). *

ELI is based on a voluntary agreement between the EU countries. It includes technical specifications on:

- **web identifiers** (URIs) for **legal information**
- **metadata** specifying how to describe **legal information**
- a specific language for **exchanging legislation in machine-readable formats**

Retrieval of legal sources

Expert systems:
Manually created logical representations of rules



1-(1) A person born in the United Kingdom after commencement shall be a British Citizen if at the time of birth his father or mother is:
(a) a British Citizen, or
(b) settled in the United Kingdom.

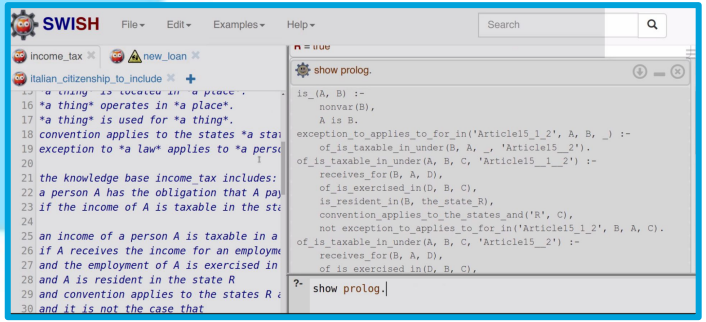
This is represented in the computer as:

Rule1: X acquires British citizenship on date Y
under sec. 1.1
IF X was born in the U.K.
AND X was born on date Y
AND Y is after or on commencement of the act
AND X has a parent who qualified under 1.1 on date

Rule2: X has a parent who qualifies under 1.1 on date Y
IF X has a parent Z
AND Z was a British citizen on date Y

Rule3: X has a parent who qualifies under 1.1 on date Y
IF X has a parent Z
AND Z was settled in the U.K. on date Y.

Sergot, M. J., Sadri, F., Kowalski, R. A., Kriwaczek, F., Hammond, P., & Cory, H. T. (1986). The British Nationality Act as a logic program. *Communications of the ACM*, *29*(5), 370-386.



Simple modeling

1st order logic

- ∀ : for all
- ∃ : there exists
- ∧ : and
- ∨ : or
- ∀ x (isLawyer(x) ⟶ isPerson(x))
- ∃ x (isLawyer(x) ∧ isSwiss(x))

Example: Swiss Citizenship Act

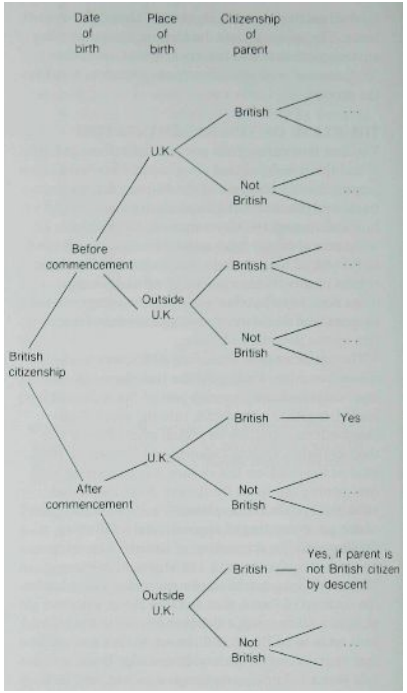
Art. 1 SCA Acquisition by descent

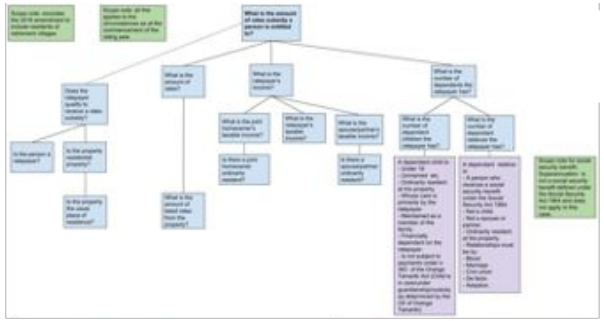
1 The following persons are Swiss citizens from birth:
(a) a child whose parents are married to each other and whose father or mother is a Swiss citizen;

∀ x (
(isChild(x) ∧ hasParents(x, y, z) ∧ married(y, z) ∧ (isSwiss(y) ∨ isSwiss(z)))
→
isSwissFromBirth(x)
)

```
public static boolean Art1(Person x, Person y, Person z) {
    return x.isChild() && x.hasParents(y, z) && y.married(z) && (y.isSwiss() || z.isSwiss());
}
```

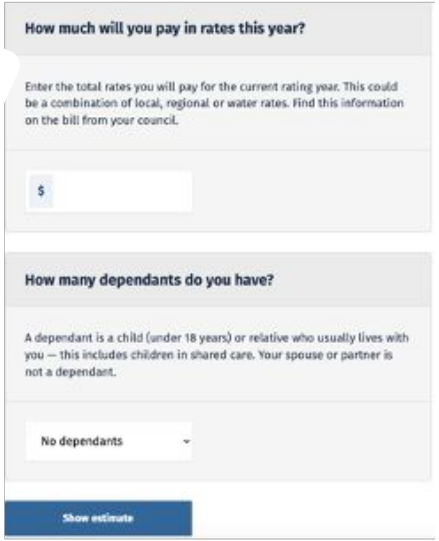
From 1986 to today!





```
# person definition
#       (income, dependents, rate_amount)
person = (60000, 3, 2000)

# the entitlement for the rebate
rebate_amount = 0


# Logic rule implementations
def income_less_than(x):
    if person[0] < x:
        return True
    return False

def dependents_greater_than(x):
    if person[1] > x:
        return True
    return False
```

```
def rate_greater_than(amount):
    return person[2] > amount

def payout_max():
    if person[1] > 1:
        global rebate_amount
        rebate_amount = 600
    return True

def payout_min():
    if person[1] == 1:
        global rebate_amount
        rebate_amount = 160
    return True
```



Sergot, M. J., et al. (1986). The British Nationality Act as a Logic Program. *Communications of the ACM*, 29(5), 370-386.

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Catala

Domain-specific programming language designed for deriving correct-by-construction implementations from legislative texts

Go to website

Browse the source code

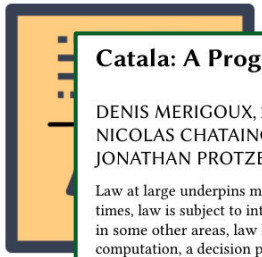
License: apache-2.0

Users

- Direction Générale des Finances Publiques (DGFiP)

Funding organisations

- Institut National de Recherche en Informatique et en Automatique (INRIA)



Catala: A Programming Language for the Law

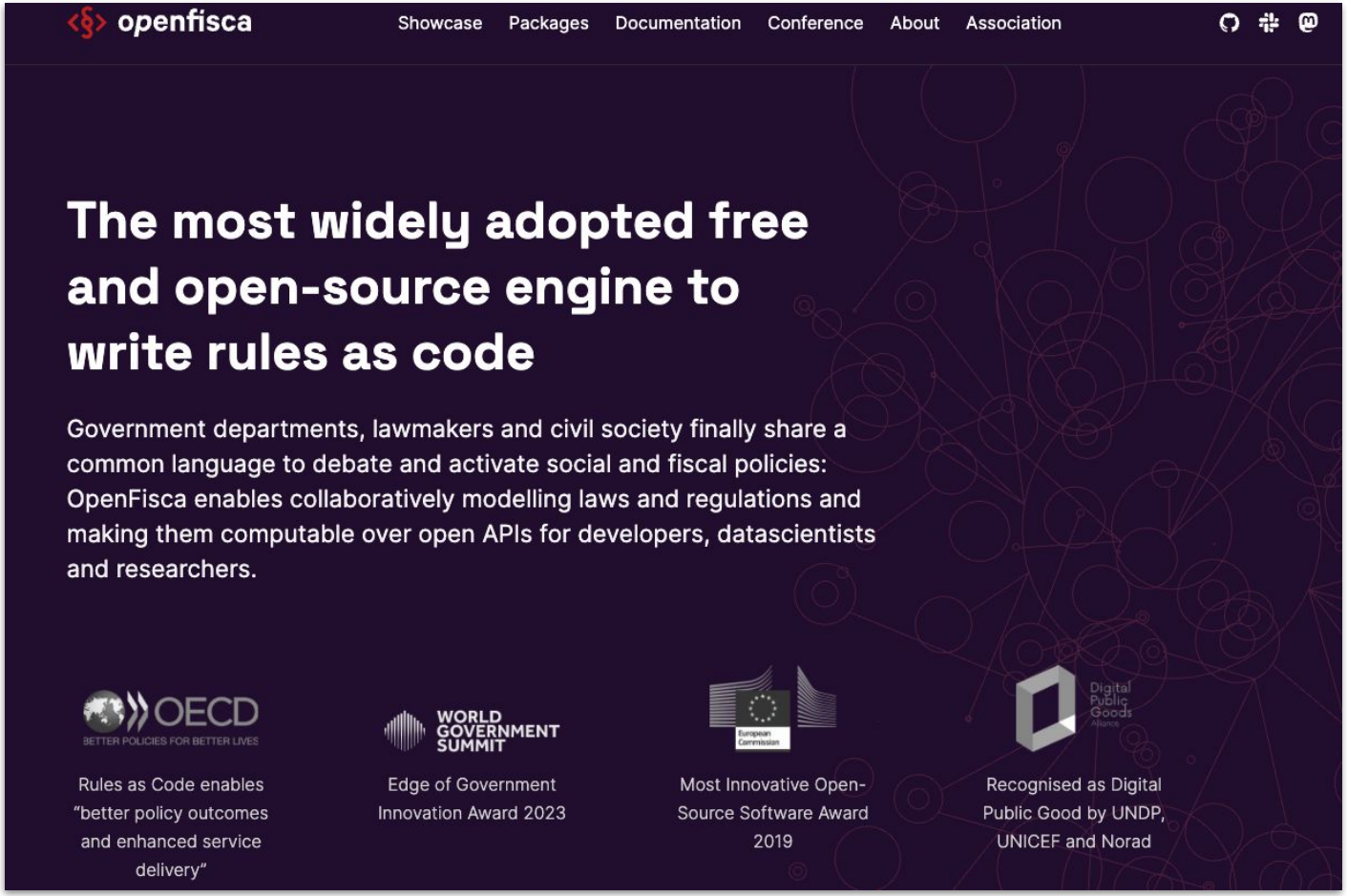
DENIS MERIGOUX, Inria, France
NICOLAS CHATAING, Inria, ENS Paris, France
JONATHAN PROTZENKO, Microsoft Research, USA

Law at large underpins modern society, codifying and governing many aspects of citizens' daily lives. Oftentimes, law is subject to interpretation, debate and challenges throughout various courts and jurisdictions. But in some other areas, law leaves little room for interpretation, and essentially aims to rigorously describe a computation, a decision procedure or, simply said, an algorithm.

Unfortunately, prose remains a woefully inadequate tool for the job. The lack of formalism leaves room for ambiguities; the structure of legal statutes, with many paragraphs and sub-sections spread across multiple pages, makes it hard to compute the intended outcome of the algorithm underlying a given text; and, as with any other piece of poorly-specified critical software, the use of informal, natural language leaves corner cases unaddressed.

We introduce **Catala**, a new programming language that we specifically designed to allow a straightforward and systematic translation of statutory law into an executable implementation. Notably, **Catala** makes it natural and easy to express the general case / exceptions logic that permeates statutory law. **Catala** aims to bring together lawyers and programmers through a shared medium, which together they can understand, edit and evolve, bridging a gap that too often results in dramatically incorrect implementations of the law. We have implemented a compiler for **Catala**, and have proven the correctness of its core compilation steps using the F* proof assistant.

https://openfisca.org/en/



Problem

$$
\begin{aligned}
&\forall\ x\ (\\
&\quad (\mathrm{isChild}(x) \wedge \mathrm{hasParents}(x, y, z) \wedge \mathrm{married}(y, z) \wedge (\mathrm{isSwiss}(y) \vee \mathrm{isSwiss}(z)))\\
&\quad \rightarrow\\
&\quad \mathrm{isSwissFromBirth}(x)\\
&)
\end{aligned}
$$

```
public static boolean Art1(Person x, Person y, Person z) {
    return x.isChild() && x.hasParents(y, z) && y.married(z) && (y.isSwiss() || z.isSwiss());
}
```

Can a legislator, judge, or layperson understand the law written in such a manner? How to verify equivalence?

Controlled Language

Art. 1 SCA Acquisition by descent

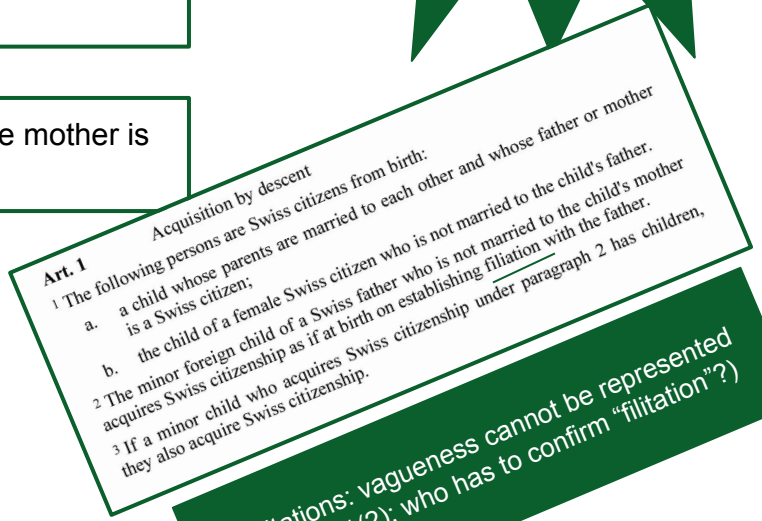
1 The following persons are Swiss citizens from birth:
(a) a child whose parents are married to each other and whose father or mother is a Swiss citizen;

If the parents of a child are married and the father is Swiss or the mother is Swiss then the child is Swiss.

DRS

```
[]
   []
      [A,B,C,D,E,F]
      relation(C,of,D)-1/4
      object(D,child,countable,na,eq,1)-1/6
      object(C,parent,countable,na,geq,2)-1/3
      property(A,married,pos)-1/8
      predicate(B,be,C,A)-1/7
      object(F,father,countable,na,eq,1)-1/11
      predicate(E,be,F,named(Swiss))-1/12
      v
      [G,H]
      object(H,mother,countable,na,eq,1)-1/16
      predicate(G,be,H,named(Swiss))-1/17
```

First order logic-based approaches and controlled languages: simple but with limits….



Art. 1 Acquisition by descent
1 The following persons are Swiss citizens from birth:
a. a child whose parents are married to each other and whose father or mother is a Swiss citizen;
b. the child of a female Swiss citizen who is not married to the child's father.
2 The minor foreign child of a Swiss father who is not married to the child's mother acquires Swiss citizenship as if at birth on establishing filiation with the father.
3 If a minor child who acquires Swiss citizenship under paragraph 2 has children, they also acquire Swiss citizenship.

Limitations: vagueness cannot be represented (see Art. 1(2): who has to confirm "filitation"?)

Attempto Controlled English — Not Just Another Logic Specification Language

Conference paper | First Online: 01 January 1999

Controlled Language: an experiment

Controlled Language Increases Comprehension of Law for People

Law is written in a way that is not easily understandable by most people. This also hinders the translation into computer code, which would permit tools that increase access to the law. The use of controlled language in legal texts is a promising way to implement laws that are better comprehensible by people, and computer-executable at the same time. However, while the scholarship on controlled language has been concentrating on its automatic translation into (logic) programs, the effects on people have not been investigated. We conducted an experiment to test the understandability, usability, and perceived open-texture (e.g., ambiguity, vagueness) of legal texts that are formulated in natural language and in controlled language. Among our 73 participants—including members of parliament (MPs) and public servants—we found that the use of controlled language increases understandability across legal domains and increases the number of terms that are perceived as open-texture. We also found that public servants show higher comprehension of legal statutes but lower perception of open-texture than MPs, regardless of whether these are in controlled or natural language. We conclude that controlled language can therefore support a wider access to law, thereby bringing democratic societies closer to their underpinning ideals and principles.

CCS Concepts: • **Social and professional topics** → **Computing / technology policy**; • **Human-centered computing** → *Human computer interaction (HCI)*; • **Applied computing** → **E-government**.

Additional Key Words and Phrases: digitally-ready legislation, automatically processable regulation, controlled language

Controlled Language Increases Comprehension of Law for People | ACM Journal on Computing and Sustainable Societies

Defeasible logic ("retractable")

"a fact is **typically** a consequence of another" (as opposed to "always") I Partial knowledge must be handled

Sceptical reasoning process: "**defeasible rules may be defeated by a fact or another rule**", i.e. can retract conclusions based on further evidence

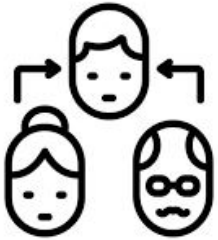
Rules are a class of expressions distinct from formulas. Rules are constructed using three primitive symbols: $\rightarrow$, $\Rightarrow$, and $\rightsquigarrow$ Where $A \cup \{\phi\}$ is a set of formulas, $A \rightarrow \phi$ is a *strict rule*, $A \Rightarrow \phi$ is a *defeasible rule*, and $A \rightsquigarrow \phi$ is an *undercutting defeater*. In each case, we call A the *antecedent* of the rule and we call ϕ the *consequent* of the rule. Where $A = \{\psi\}$, we denote $A \rightarrow \phi$ as $\psi \rightarrow \phi$, and similarly for defeasible rules and defeaters. Antecedents for strict rules and defeaters must be non-empty; antecedents for defeasible rules may be empty. We will call a rule of the form $\emptyset \Rightarrow \phi$ a *presumption* and represent it more simply as $\Rightarrow \phi$. While we allow variables in rules, we will treat such rules as schemata for all their instantiations.

Rule				Explanation
$r1$:		$\Rightarrow$	$\neg guilty$	Innocence is presumed
$r2$:	$evidence$	$\Rightarrow$	$guilty$	Evidence can show guilt
$r3$:	$\neg motive$	$\rightsquigarrow$	$\neg guilty$	Lack of motive can suggest innocence
$r4$:	$alibi$	$\Rightarrow$	$\neg guilty$	An alibi can prove innocence
$r4 \succ r3, r3 \succ r2, r2 \succ r1$				

Figure 1: Hypothetical Criminal Law Theory

Nute, D. (1994). Defeasible logic. In Handbook of logic in artificial intelligence and logic programming, volume 3: Nonmonotonic reasoning and uncertain reasoning, pp. 353–395.
Johnston, B., Governatori, G. (2003). Induction of Defeasible Logic Theories in the Legal Domain. ICAIL, June 24-28.

Vocabulary / knowledge

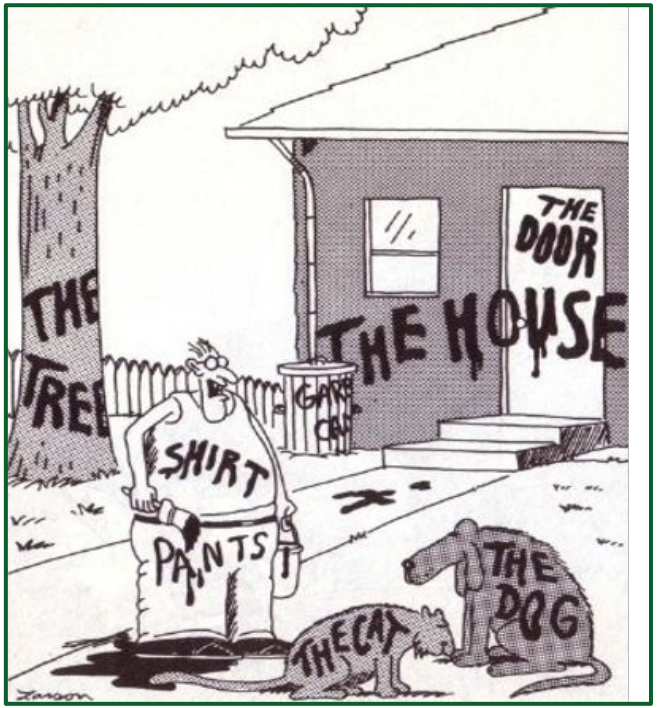


What does “persons”, “citizens”, “birth”, etc. mean for a computer? How to represent knowledge?

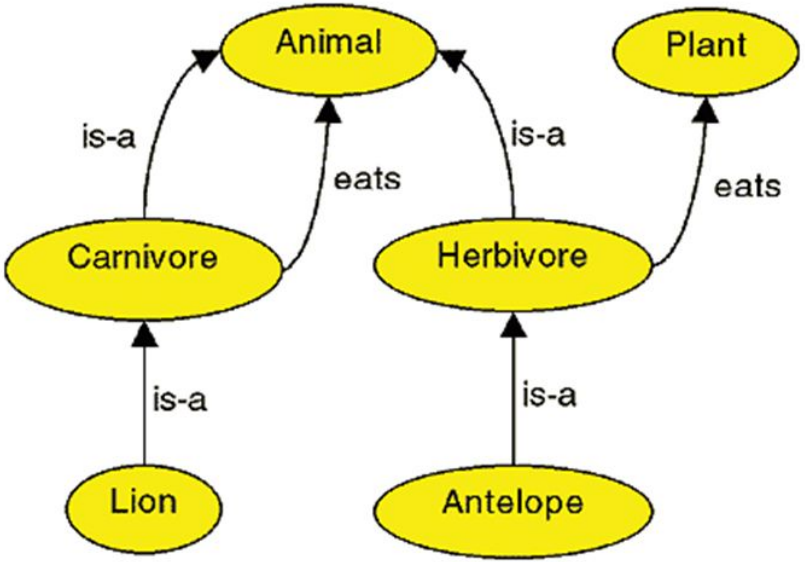
Art. 1 SCA Acquisition by descent

1 The following persons are Swiss citizens from birth:
(a) a child whose parents are married to each other and whose father or mother is a Swiss citizen;

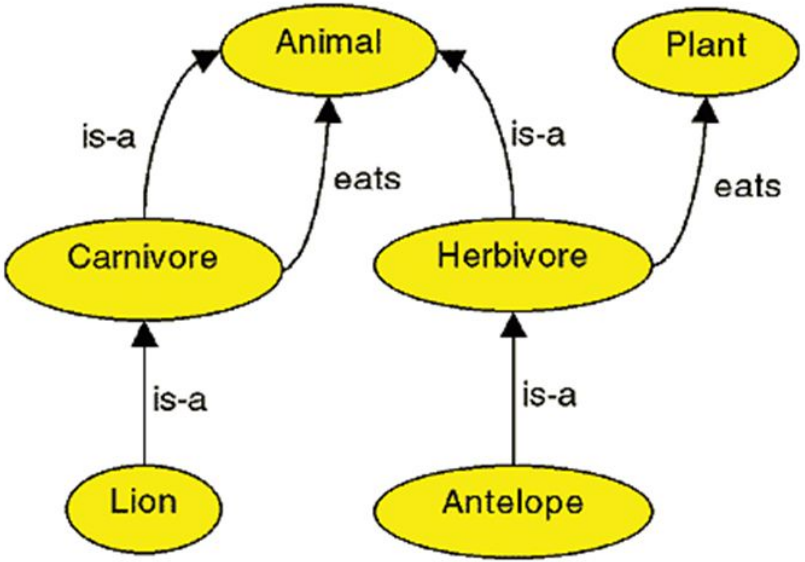
What's going on here?



Labeling is not enough!



Ontologies as a solution



For instance, schema.org is widely used today

Search for ontologies and defined terms, e.g. in Ontobee

Example: GDPR

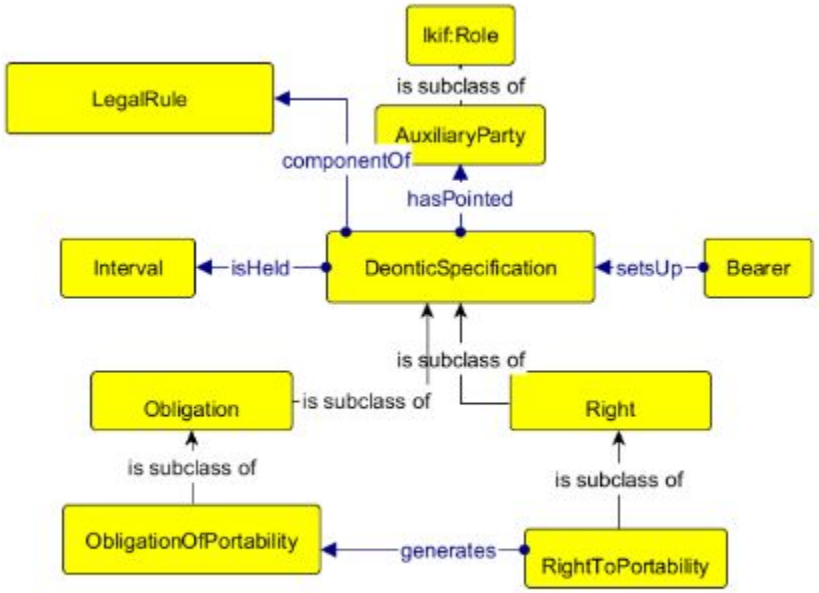
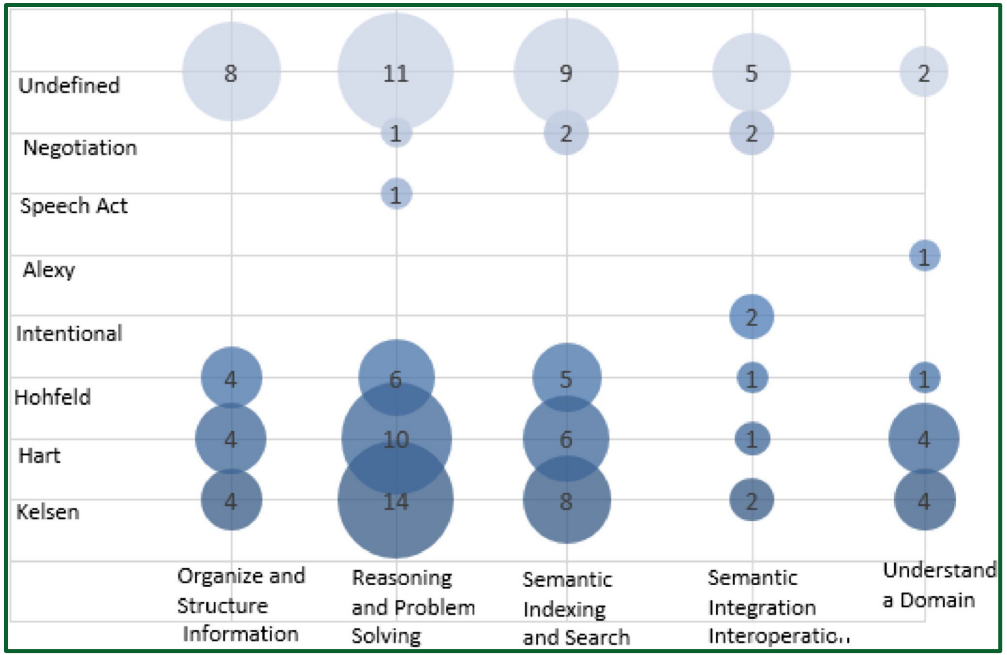


Figure 3: Right to Portability and Obligation of Portability

isViolatedBy it defines the relationship between obligation/prohibition and violation	<owl:ObjectProperty rdf:about="https://w3id.org/ontology/pronto#isViolatedBy"> <rdfs:domain rdf:resource="http://docs.oasis-open.org/legalruleml/ns/v1.0/metamodel#Obligation"/> <rdfs:domain rdf:resource="http://docs.oasis-open.org/legalruleml/ns/v1.0/metamodel#Prohibition"/> <rdfs:range rdf:resource="http://docs.oasis-open.org/legalruleml/ns/v1.0/metamodel#Violation"/> </owl:ObjectProperty>
isFulfilledBy	<owl:ObjectProperty rdf:about="https://w3id.org/ontology/pronto#isFulfilledBy"> <rdfs:domain rdf:resource="http://docs.oasis-open.org/legalruleml/ns/v1.0/metamodel#Obligation"/> <rdfs:domain rdf:resource="http://docs.oasis-open.org/legalruleml/ns/v1.0/metamodel#Prohibition"/> <rdfs:range rdf:resource="http://docs.oasis-open.org/legalruleml/ns/v1.0/metamodel#Compliance"/> </owl:ObjectProperty>
Relationship between obligation and permission	<owl:ObjectProperty rdf:about="https://w3id.org/ontology/pronto#implies"> <rdfs:domain rdf:resource="http://docs.oasis-open.org/legalruleml/ns/v1.0/metamodel#Obligation"/> <rdfs:range rdf:resource="http://docs.oasis-open.org/legalruleml/ns/v1.0/metamodel#Permission"/> </owl:ObjectProperty>
Relationship between prohibition and obligation	<owl:ObjectProperty rdf:about="https://w3id.org/ontology/pronto#isAKindOf"> <rdfs:domain rdf:resource="http://docs.oasis-open.org/legalruleml/ns/v1.0/metamodel#Prohibition"/> <rdfs:range rdf:resource="http://docs.oasis-open.org/legalruleml/ns/v1.0/metamodel#Obligation"/> </owl:ObjectProperty>
Right as a subclass of Permission	<owl:Class rdf:about="http://docs.oasis-open.org/legalruleml/ns/v1.0/metamodel#Right"> <rdfs:subClassOf rdf:resource="http://docs.oasis-open.org/legalruleml/ns/v1.0/metamodel#Permission"/> </owl:Class>

Table 1 – Some axioms of the extension of the LegalRuleML ontology

Palmirani, M., Martoni, M., Rossi, A., Bartolini, C., & Robaldo, L. (2018). Legal Ontology for Modelling GDPR Concepts and Norms. In M. Palmirani (Ed.), *Legal Knowledge and Information Systems* (Vol. 313): IOS Press.



de Oliveira Rodrigues, C. M., de Freitas, F. L. G., Barreiros, E. F. S., de Azevedo, R. R., & de Almeida Filho, A. T. (2019). Legal ontologies over time: A systematic mapping study. Expert Systems with Applications, 130, 12-30.

Many different ontologies!

Time consuming for *experts* to create

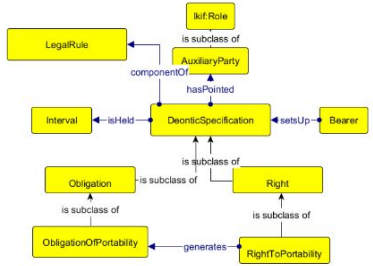


Figure 3: Right to Portability and Obligation of Portability

Automatic ontology extraction as a solution? Enters ML!

Developed often for one legal corpus

Little re-usability

Universität St.Gallen

When to automate legal processes and which what tools?

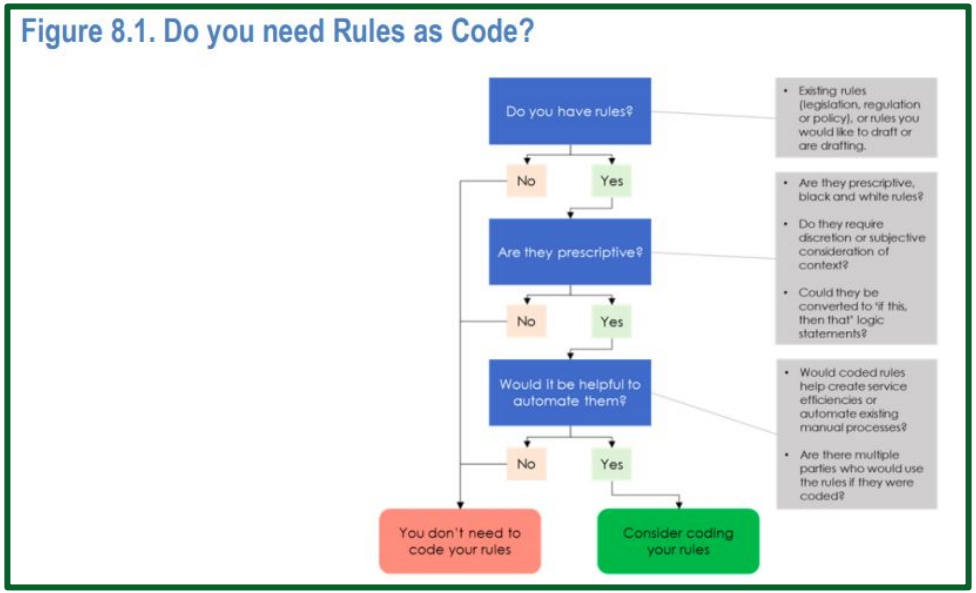
So when can (shall?) we automate law?

(1) Legislation that involves calculation

(2) Legislation that prescribes a process that is used repeatedly and can be delivered digitally

(3) Legislation that prescribes a compliance process

(1) Legislation that involves factual information



OECD Working Papers on Public Governance No 42, Cracking the code: Rulemaking for humans and machines, https://oecd-opsi.org/wp-content/uploads/2022/03/rac-wp.pdf and Digital.gov.nz (2018). Better Rules for Government Discovery Report, March

Calculations

3 Rates rebate

(1) A ratepayer who, at the commencement of a rating year, was the ratepayer of a residential property is entitled, on application in that year, to a rebate of—

(a) so much of the rates payable for that rating year in respect of the property as represents—

(i) two-thirds of the amount by which those rates exceed $160, reduced by—

(ii) $1 for each $8 by which the ratepayer's income for the preceding tax year exceeded $26,510, that last-mentioned amount being increased by $500 in respect of each person who was a dependant of the ratepayer at the commencement of the rating year in respect of which the application is made; or

(b) $665,—

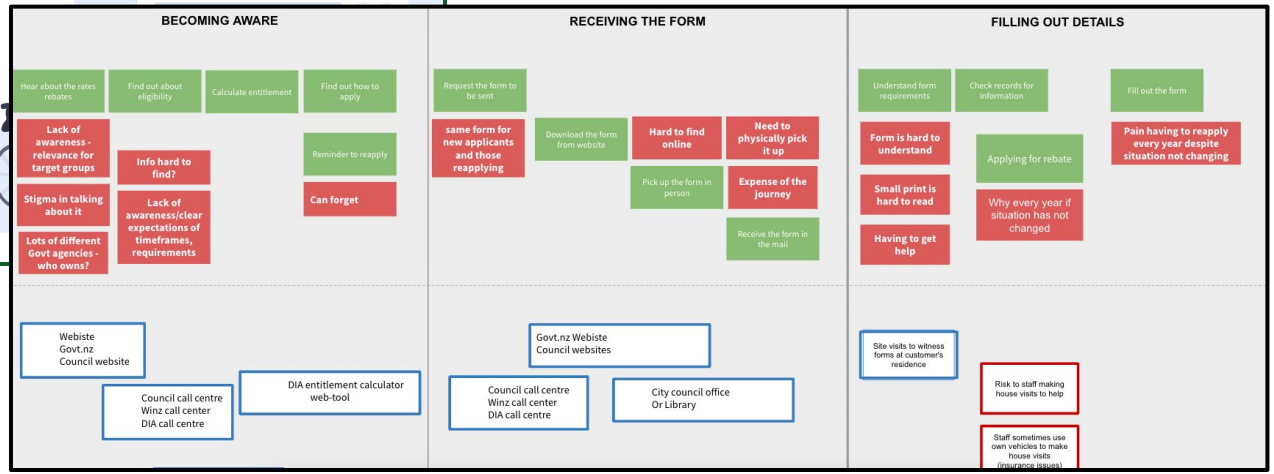
whichever amount is smaller.

$$Rebate_Y(R_Y, I_{Y-1}, D_Y) = Max\left(0, Min\left(\frac{2\,(R_Y - 160)}{3} - \frac{I_{Y-1} - (26510 + 500 * D_Y)}{8}, 665\right)\right)$$

```python
def calculate_rebate(R, I, D):
    # Make sure to handle potential division by zero if necessary and validate inputs as needed
    try:
        R = float(R)
        I = float(I)
        D = int(D)

        # Calculate the rebate
        income_penalty = (I - (26510 + 500 * D)) / 8
        rebate = max(0, min(2 * (R - 160) / 3 - income_penalty, 665))
        return rebate
    except (ValueError, TypeError):
        # Return None for invalid inputs
        return None
```

Deliverable digitally and repeatedly



Compliance process

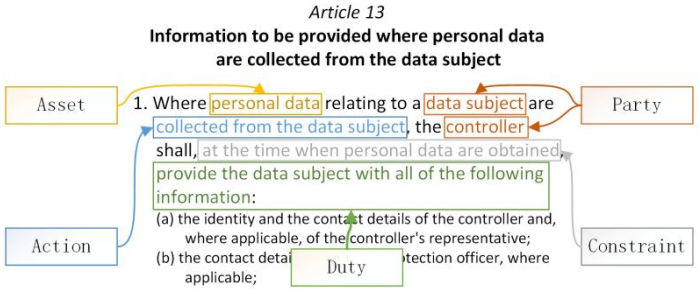
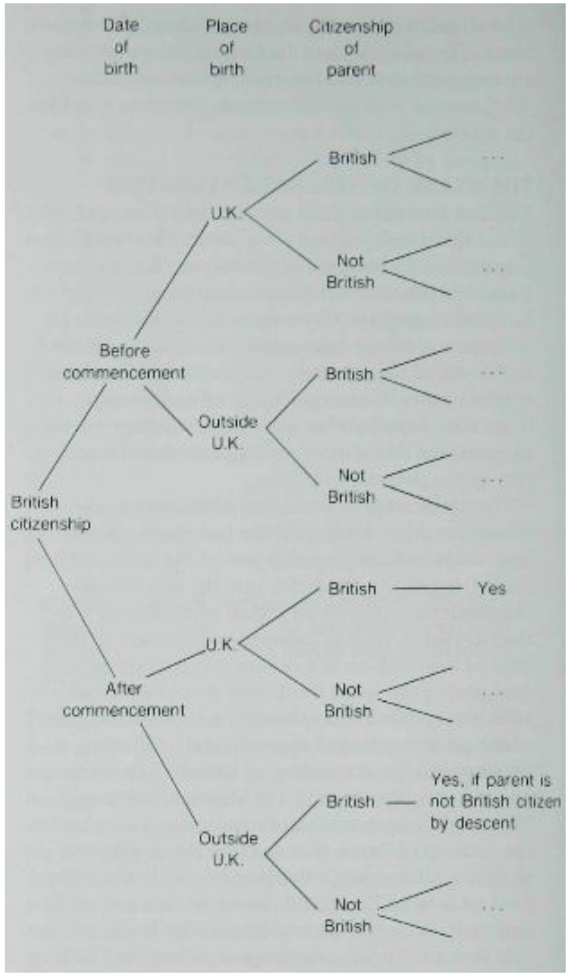


Fig. 3. Breaking down Article 13.1 of the GDPR according to the ODRL model

Process with factual information

Clear visibility?!



What's "night"?

Astronomical twilight [edit]

Long exposure of astronomical twilight in a small town in the Mojave Desert

Astronomical twilight is defined as when the geometric center of the Sun is between 18° and 12° below the horizon.[3][4][2] During astronomical twilight, the sky is dark enough to permit astronomical observation of point sources of light such as stars, except in regions with more intense skyglow due to light pollution, moonlight, auroras, and other sources of light. Some critical observations, such as of faint diffuse items such as nebulae and galaxies, may require observation beyond the limit of astronomical twilight. Theoretically, the faintest stars detectable by the naked eye (those of approximately the sixth magnitude) will become visible in the evening at astronomical dusk, and become invisible at astronomical dawn.[13]

Astronomical twilight (dusk) with a crescent moon, as seen from Kuala Lumpur, Malaysia

Twilight - Wikipedia

Syntactic ambiguity as a hurdle

The local Maine law stated that employers were not to pay overtime in the case of: “the canning, processing, preserving, freezing, drying, marketing, storing, packing for shipment or distribution of: 1) agricultural produce; 2) meat and fish products; and 3) perishable foods”.

Truckers:
Exemption is for ‘packing for shipment’ or ‘packing for distribution’
→ overtime to be paid for distribution

Oakhurst:
[packing for shipment] or [distribution]
No overtime to be paid for distribution, in general

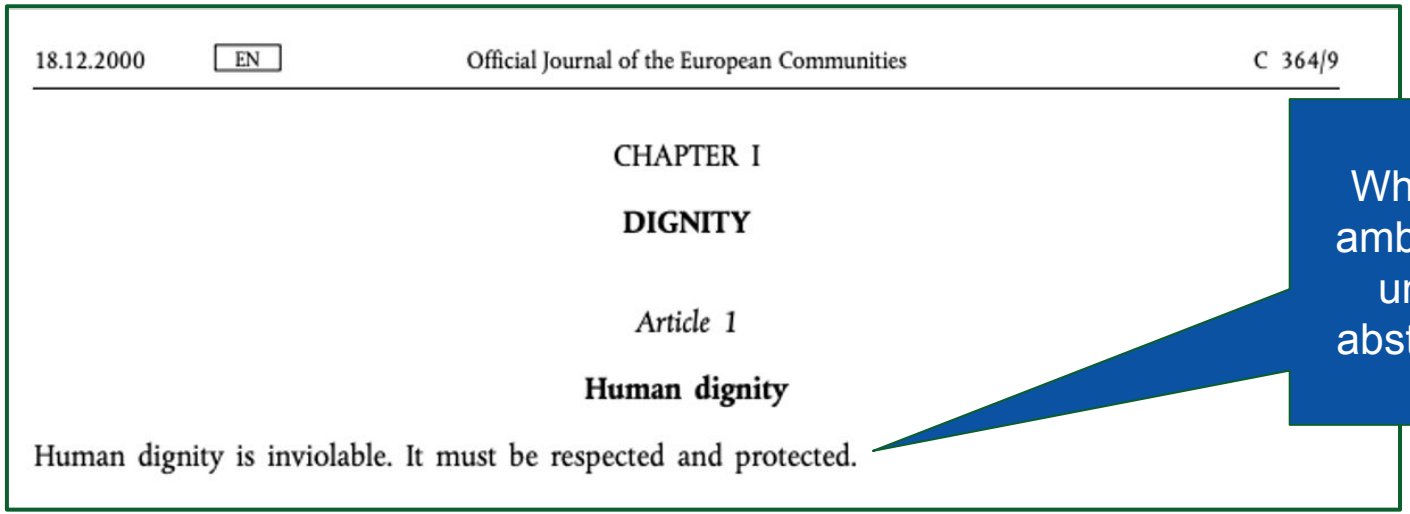
The New York Times

Oxford Comma Dispute Is Settled as Maine Drivers Get $5 Million

Give this article

Oakhurst Dairy trucks at the family-owned, independent dairy in Portland, Me. A settlement put a period on the case. Pat Wellenbach/Associated Press

Interpreting the law



18.12.2000 EN Official Journal of the European Communities C 364/9

CHAPTER I

DIGNITY

Article 1

Human dignity

Human dignity is inviolable. It must be respected and protected.

Which words are: ambiguous, vague, under-defined, abstract concepts?

“Human dignity”

Rank of elevation / social status

OR

(Self-)esteem

Personal integrity

Is life part of human dignity?

Definitions matter: "Human dignity"

Can a military jet shoot down a hijacked civilian plane which is threatening to crash onto civilians?

Or is intentional killing from the state against the right to live?

The New York Times

German Threatens to Crash Glider Into Frankfurt Bank

Share full article

By Mark Landler

Jan. 6, 2003

Luftsicherheitsgesetz (LuftSiG)

Nichtamtliches Inhaltsverzeichnis

LuftSiG

Ausfertigungsdatum: 11.01.2005

Vollzitat:

"Luftsicherheitsgesetz vom 11. Januar 2005 (BGBl. I S. 78), das zuletzt durch Artikel 1 des Gesetzes vom 22. April 2020 (BGBl. I S. 840) geändert worden ist"

Stand: Zuletzt geändert durch Art. 1 G v. 22.4.2020 I 840

Näheres zur Standangabe finden Sie im Menü unter Hinweise

Fußnote

(+++ Textnachweis ab: 15.1.2005 +++)

(+++ Zur Anwendung vgl. §§ 9a u. 22 +++)

Das G wurde als Artikel 1 des G v. 11.1.2005 I 78 vom Bundestag beschlossen. Es ist gem. Art. 9 dieses G am 15.1.2005 in Kraft getreten.

Nichtamtliches Inhaltsverzeichnis

Bundesministerium der Justiz

Bundesamt für Justiz

How did Germany decide?

Different institutions, different interpretations

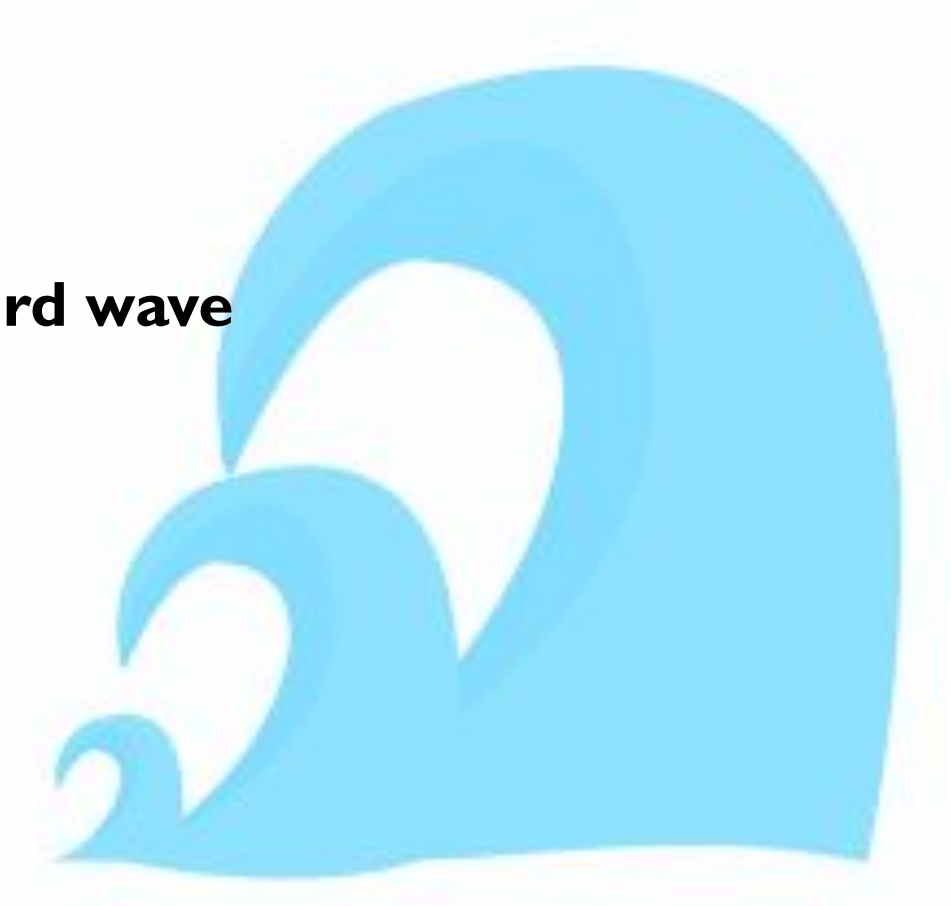
Parliament & Federal Government: conform with the constitution

Constitutional court: not conform (passengers are victims, not culprits; the whole law must be struck down)

TERROR - IHR URTEIL | Trailer [HD]

https://www.bundesverfassungsgericht.de/SharedDocs/Entscheidungen/EN/2006/02/rs20060215_1bvr035705en.html

The third wave

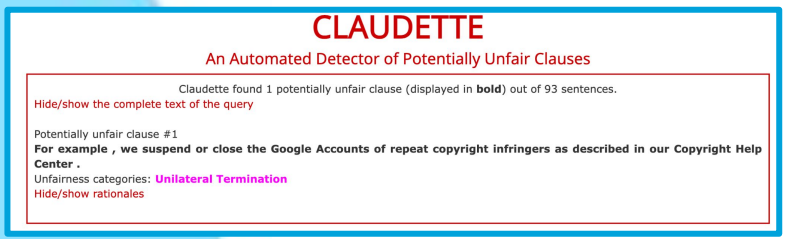


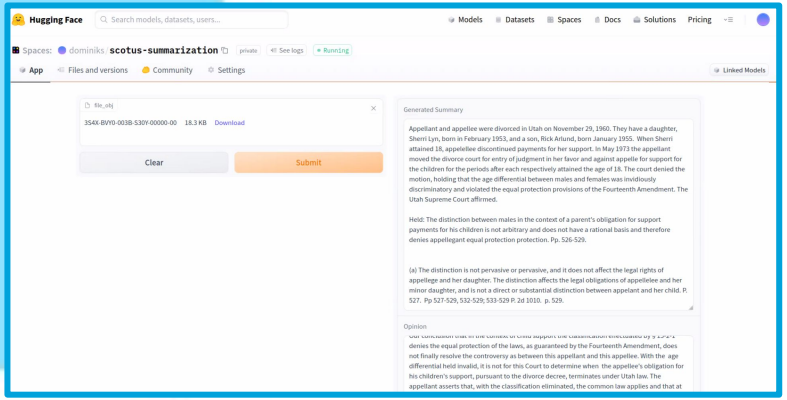
Machine learning approaches: e.g., use of Natural Language Processing techniques to extract meanings (or semantics) from cases and legislation

Enters AI……

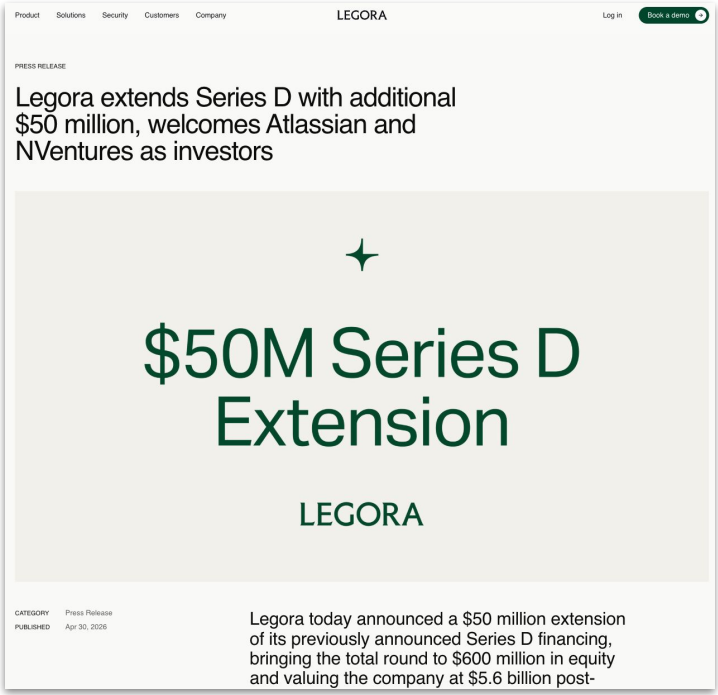
Expert systems

Retrieval of legal sources

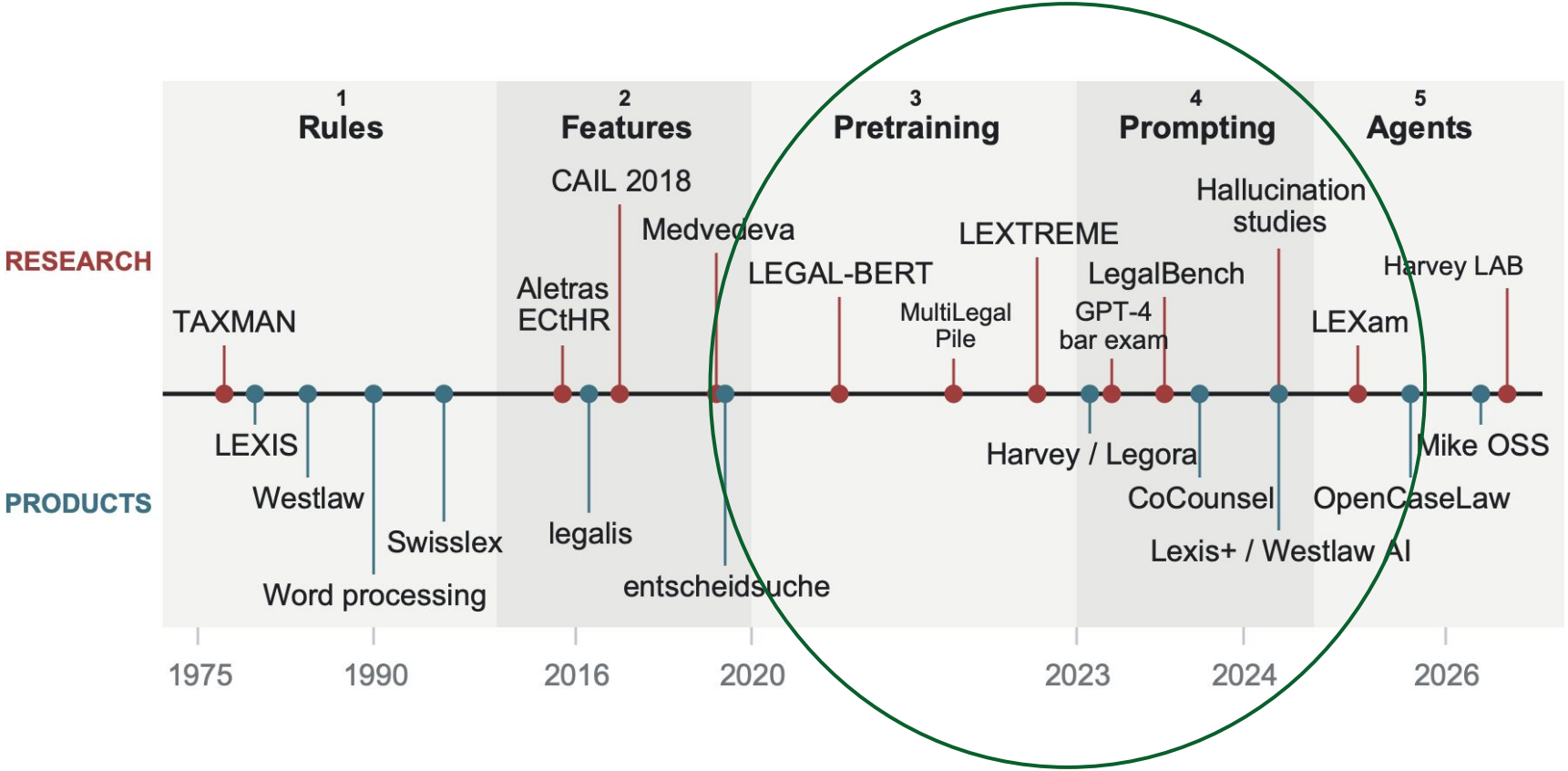




Exemplary funding rounds in 2026



This slide is from **Joel Niklaus**, Data Lead at Hugging Face 🤗 that he presented at HSG in August 2026



Hugging Face | Search models, datasets, users... | Models | Datasets | Spaces | Buckets NEW | Docs | Pricing | Log In | Sign Up

Datasets: joelniklaus/**Multi_Legal_Pile** | Like 67

Tasks: Fill-Mask | Languages: Bulgarian | Czech | Danish | +21 | Size: 10M<n<100M | ArXiv: arxiv:2306.02069 | License: cc-by-nc-sa-4.0

Dataset card | Files and versions | xet | Community 11

Dataset Viewer

The viewer is disabled because this dataset repo requires arbitrary Python code execution. Please consider removing the loading script and relying on automated data support (you can use convert to parquet from the datasets library). If this is not possible, please open a discussion for direct help.

Dataset Card for MultiLegalPile: A Large-Scale Multilingual Corpus for the Legal Domain

Dataset Summary

The Multi_Legal_Pile is a large-scale multilingual legal dataset suited for pretraining language models. It spans over 24 languages and five legal text types.

Supported Tasks and Leaderboards

The dataset supports the tasks of fill-mask.

Languages

The following languages are supported: bg, cs, da, de, el, en, es, et, fi, fr, ga, hr, hu, it, lt, lv, mt, nl, pl, pt, ro, sk, sl, sv

Dataset Structure

It is structured in the following format: type -> language -> jurisdiction.jsonl.xz

type is one of the following:

- caselaw
- contracts
- legislation
- other
- legal_mc4

→ Copy to bucket NEW

Downloads last month 13,530

Homepage: Homepage: | Repository: Repository: | Paper: MultiLegalPile

Leaderboard: Leaderboard: | Point of Contact: Joel Niklaus | Total file size: 15.4 GB

Models trained or fine-tuned on joelniklaus/Mu...

- morenolq/LEGIT-BART-LSG-4096 — Text Generation • Updated Feb 26, 2025 • 23
- morenolq/BART-IT-LSG-16384 — Text Generation • Updated Feb 26, 2025 • 19
- morenolq/LEGIT-BART-LSG-16384 — Text Generation • Updated Feb 26, 2025 • 16
- morenolq/BART-IT-LSG-4096 — Text Generation • Updated Feb 26, 2025 • 16
- morenolq/LEGIT-SCRATCH-BART-LSG-16384 — Text Generation • Updated Feb 26, 2025 • 14
- morenolq/LEGIT-SCRATCH-BART-LSG-4096 — Text Generation • Updated Feb 26, 2025 • 13

Browse 9 models trained on this dataset

Space using joelniklaus/Multi_Legal_Pile 1

tardellirs/model-pulse

Pre-training: legal language modelling (learning from legal language without labels and tasks)

Fine-tuning: with tasks data and labels fine-tune the models for specific tasks

Instruction-tuning: Pairing instructions with model answers to learn how to answer

Universität St.Gallen

LEGAL-BERT: The Muppets straight out of Law School

Ilias Chalkidis [†‡] **Manos Fergadiotis** [†‡]
Prodromos Malakasiotis [†‡] **Nikolaos Aletras** [*] **Ion Androutsopoulos** [†‡]
[†] Department of Informatics, Athens University of Economics and Business
[‡] Institute of Informatics & Telecommunications, NCSR "Demokritos"
[*] Computer Science Department, University of Sheffield, UK
[ihalk,fergadiotis,rulller,ion]@aueb.gr
n.aletras@sheffield.ac.uk

Abstract

BERT has achieved impressive performance in several NLP tasks. However, there has been limited investigation on its adaptation guidelines in specialised domains. Here we focus on the legal domain, where we explore several approaches for applying BERT models to downstream legal tasks, evaluating on multiple datasets. Our findings indicate that the previous guidelines for pre-training and fine-tuning, often blindly followed, do not always generalize well in the legal domain. Thus we propose a systematic investigation of the available strategies when applying BERT in specialised domains. These are: (a) use the original BERT out of the box, (b) adapt BERT by additional pre-training on domain-specific corpora, and (c) pre-train BERT from scratch on domain-specific corpora. We also propose a broader hyper-parameter search space when fine-tuning for downstream tasks and we release LEGAL-BERT, a family of BERT models intended to assist legal NLP research, computational law, and legal technology applications.

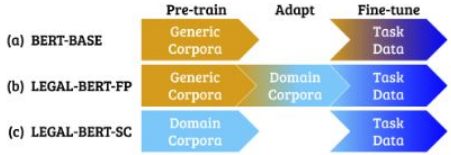
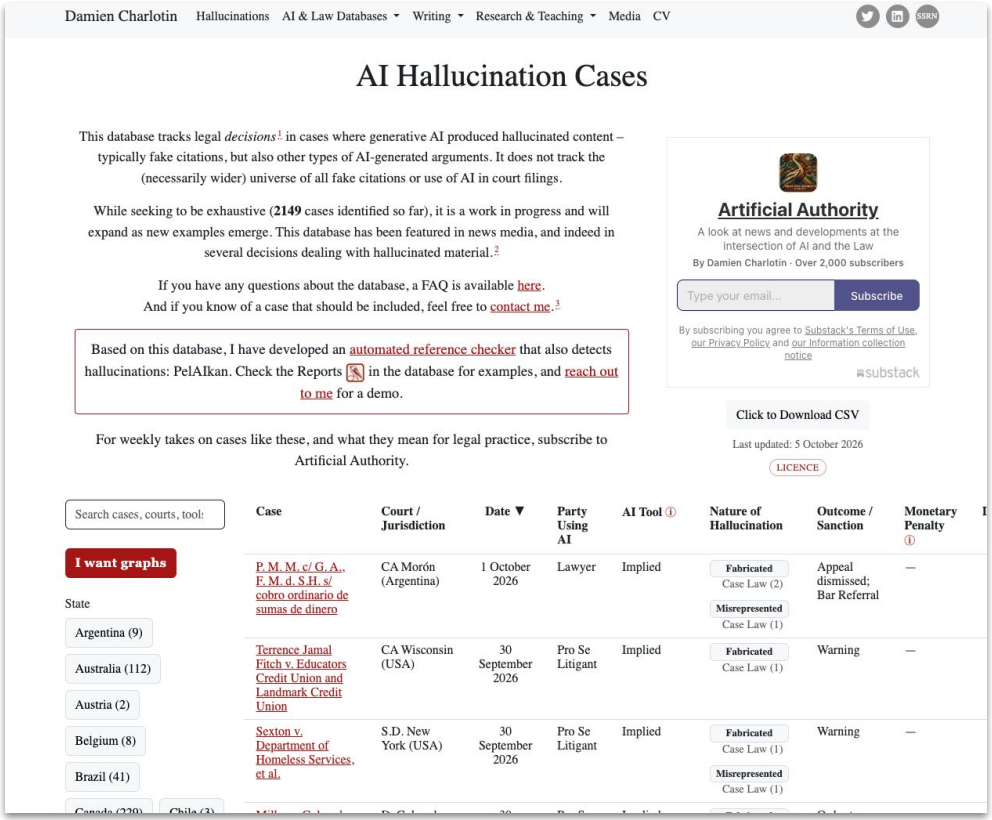


Figure 1: The three alternatives when employing BERT for NLP tasks in specialised domains: (a) use BERT out of the box, (b) further pre-train BERT (FP), and (c) pre-train BERT from scratch (SC). All strategies have a final fine-tuning step.

generic corpora (e.g., Wikipedia, Children's Books, etc.) BERT has been reported to under-perform in specialised domains, such as biomedical or scientific text (Lee et al., 2019; Beltagy et al., 2019). To overcome this limitation there are two possible strategies; either further pre-train (FP) BERT on domain specific corpora, or pre-train BERT from scratch (SC) on domain specific corpora. Consequently, to employ BERT in specialised domains one may consider three alternative strategies before fine-tuning for the downstream task (Figure 1): (a) use BERT out of the box, (b) further pre-train (FP) BERT on domain-specific corpora, and (c) pre-train

Problems… generated text, not retrieved text:



Damien Charlotin Hallucinations AI & Law Databases Writing Research & Teaching Media CV

AI Hallucination Cases

This database tracks legal *decisions*[1] in cases where generative AI produced hallucinated content – typically fake citations, but also other types of AI-generated arguments. It does not track the (necessarily wider) universe of all fake citations or use of AI in court filings.

While seeking to be exhaustive (**2149** cases identified so far), it is a work in progress and will expand as new examples emerge. This database has been featured in news media, and indeed in several decisions dealing with hallucinated material.[2]

If you have any questions about the database, a FAQ is available here.
And if you know of a case that should be included, feel free to contact me.[3]

Based on this database, I have developed an automated reference checker that also detects hallucinations: PelAIkan. Check the Reports in the database for examples, and reach out to me for a demo.

For weekly takes on cases like these, and what they mean for legal practice, subscribe to Artificial Authority.

Artificial Authority
A look at news and developments at the intersection of AI and the Law
By Damien Charlotin · Over 2,000 subscribers
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Last updated: 5 October 2026
LICENCE

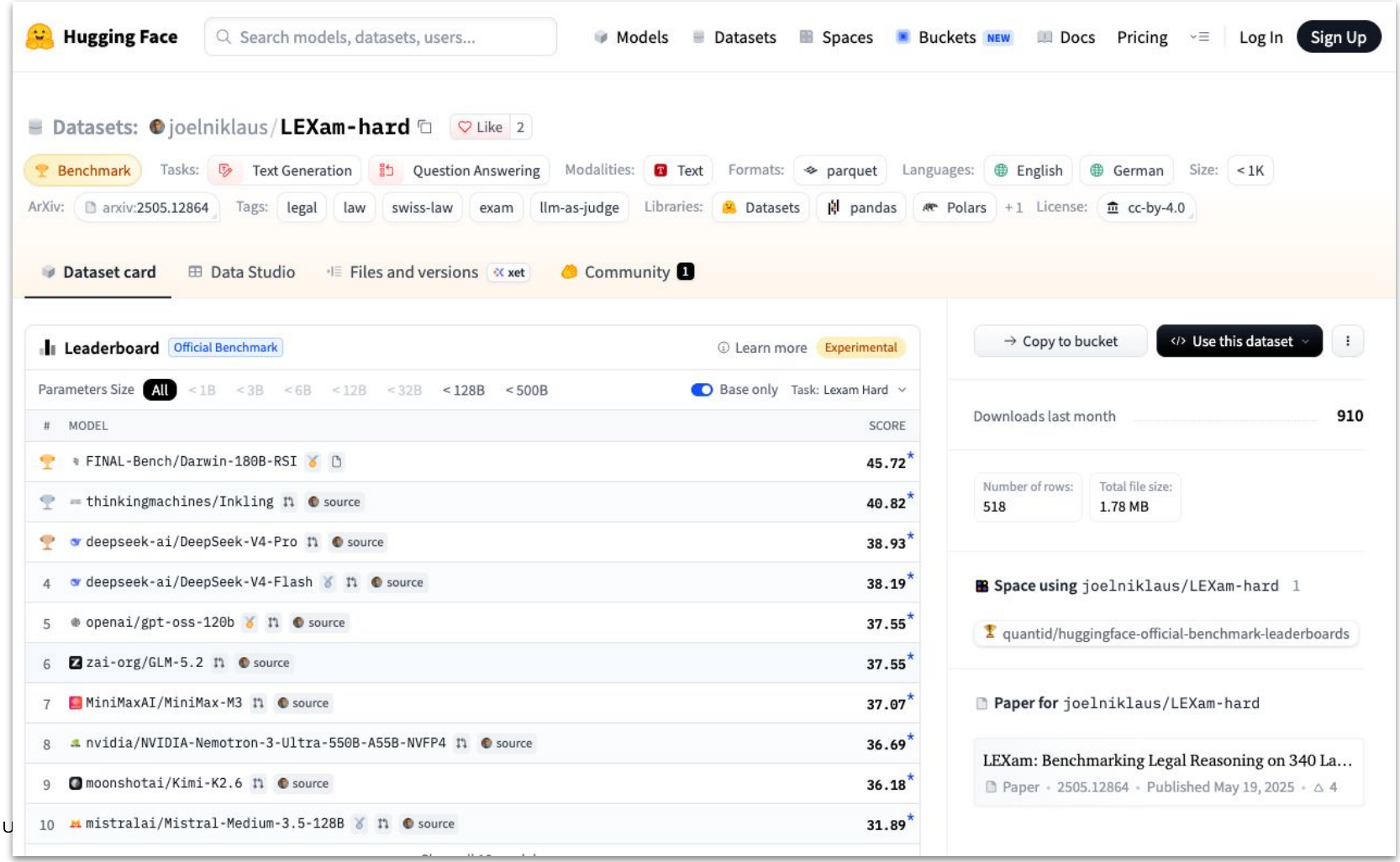
Search cases, courts, tools

I want graphs

State
Argentina (9)
Australia (112)
Austria (2)
Belgium (8)
Brazil (41)

Case	Court / Jurisdiction	Date ▼	Party Using AI	AI Tool	Nature of Hallucination	Outcome / Sanction	Monetary Penalty
P. M. M. c/ G. A., F. M. d. S.H. s/ cobro ordinario de sumas de dinero	CA Morón (Argentina)	1 October 2026	Lawyer	Implied	Fabricated Case Law (2); Misrepresented Case Law (1)	Appeal dismissed; Bar Referral	–
Terrence Jamal Fitch v. Educators Credit Union and Landmark Credit Union	CA Wisconsin (USA)	30 September 2026	Pro Se Litigant	Implied	Fabricated Case Law (1)	Warning	–
Sexton v. Department of Homeless Services, et al.	S.D. New York (USA)	30 September 2026	Pro Se Litigant	Implied	Fabricated Case Law (1); Misrepresented Case Law (1)	Warning	–

https://www.damiencharlotin.com/hallucinations/

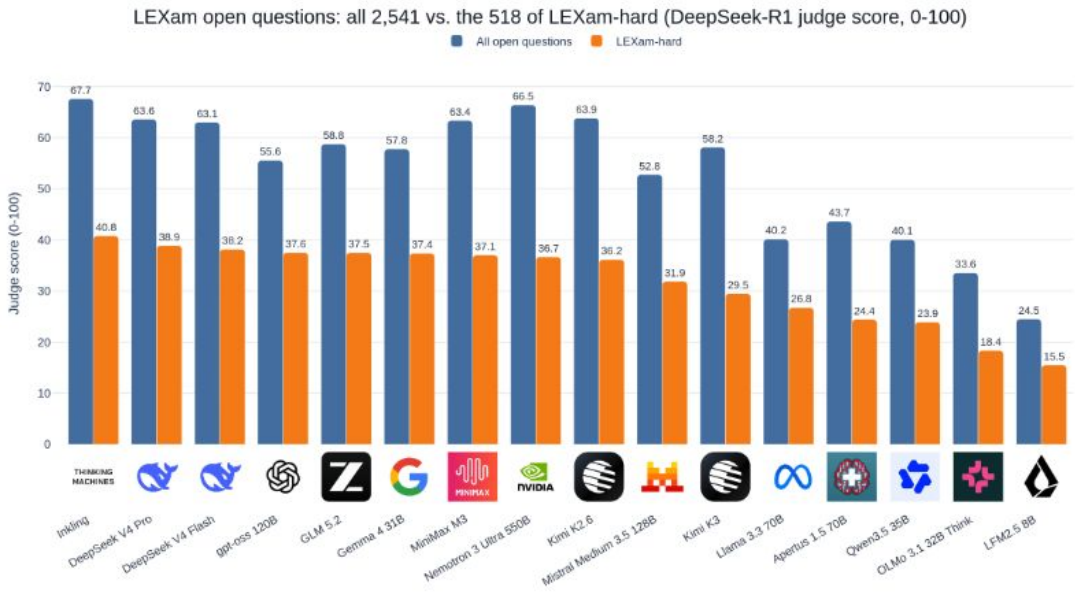

Hugging Face
Search models, datasets, users...
Models
Datasets
Spaces
Buckets NEW
Docs
Pricing
Log In
Sign Up
Datasets: joelniklaus / LEXam-hard
Like 2
Benchmark
Tasks: Text Generation
Question Answering
Modalities: Text
Formats: parquet
Languages: English
German
Size: < 1K
ArXiv: arxiv:2505.12864
Tags: legal
law
swiss-law
exam
llm-as-judge
Libraries: Datasets
pandas
Polars
+1
License: cc-by-4.0
Dataset card
Data Studio
Files and versions
xet
Community 1
Leaderboard
Official Benchmark
Learn more
Experimental
Parameters Size
All
< 1B
< 3B
< 6B
< 12B
< 32B
< 128B
< 500B
Base only
Task: Lexam Hard
MODEL SCORE
FINAL-Bench/Darwin-180B-RSI 45.72*
thinkingmachines/Inkling source 40.82*
deepseek-ai/DeepSeek-V4-Pro source 38.93*
4 deepseek-ai/DeepSeek-V4-Flash source 38.19*
5 openai/gpt-oss-120b source 37.55*
6 zai-org/GLM-5.2 source 37.55*
7 MiniMaxAI/MiniMax-M3 source 37.07*
8 nvidia/NVIDIA-Nemotron-3-Ultra-550B-A55B-NVFP4 source 36.69*
9 moonshotai/Kimi-K2.6 source 36.18*
10 mistralai/Mistral-Medium-3.5-128B source 31.89*
Copy to bucket
Use this dataset
Downloads last month 910
Number of rows: 518
Total file size: 1.78 MB
Space using joelniklaus/LEXam-hard 1
quantid/huggingface-official-benchmark-leaderboards
Paper for joelniklaus/LEXam-hard
LEXam: Benchmarking Legal Reasoning on 340 La...
Paper • 2505.12864 • Published May 19, 2025 • 4

https://huggingface.co/datasets/joelniklaus/LEXam-hard

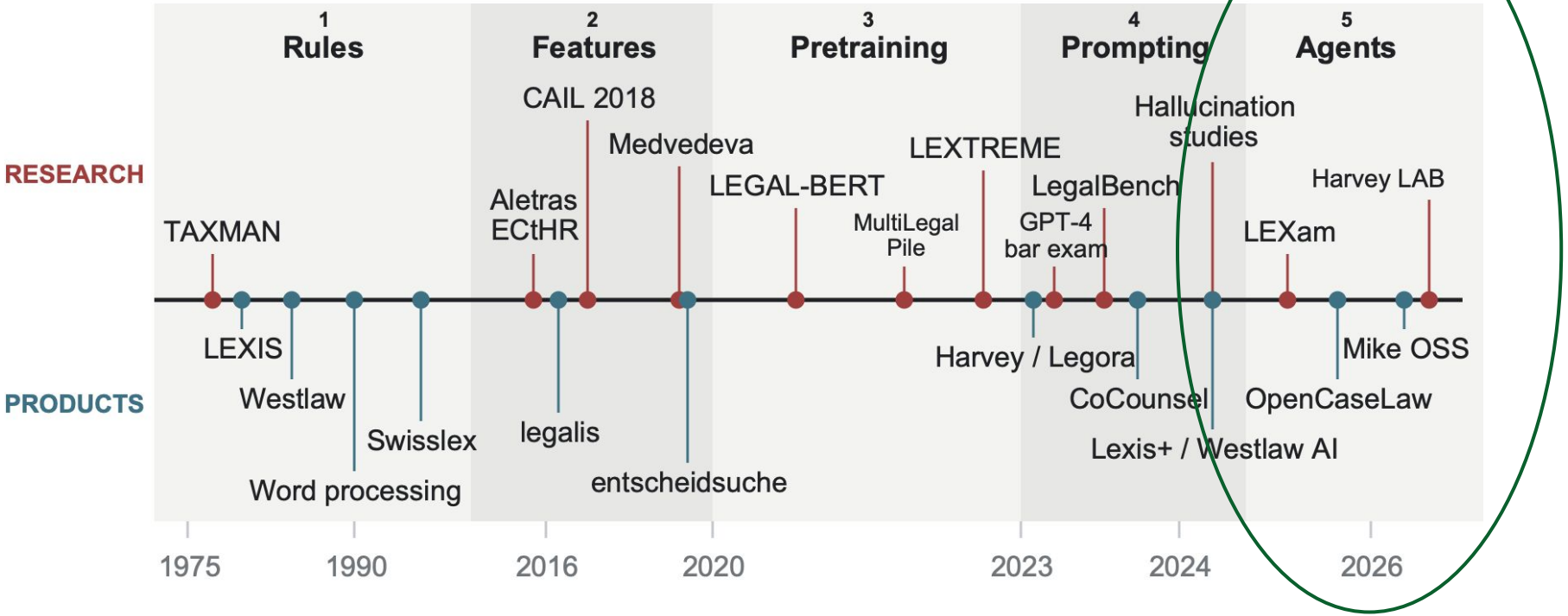
LEXam-hard

The 518 open questions of LEXam that the strongest open models score lowest on. LEXam is a benchmark of law exam questions from the University of Zurich (Fan et al., ICLR 2026; website, code). This is a filtered copy of its `open_question` test split for evaluating agents that can do legal research. Questions that strong models already answer from memory are removed, and so are questions that cannot be answered from their own text.

Rows are ordered from the hardest question up. Every row joins back to LEXam on `id`.



This slide is from **Joel Niklaus**, Data Lead at Hugging Face 🤗 that he presented at HSG in August 2026



This slide is from **Joel Niklaus**, Data Lead at Hugging Face 🤗 that he presented at HSG in August 2026

The model decides. The tools do the exact parts.

THE QUESTION

"Can we terminate early?"

→

THE MODEL

plans, reads, decides what to call

→

THE ANSWER

with a citation it checked

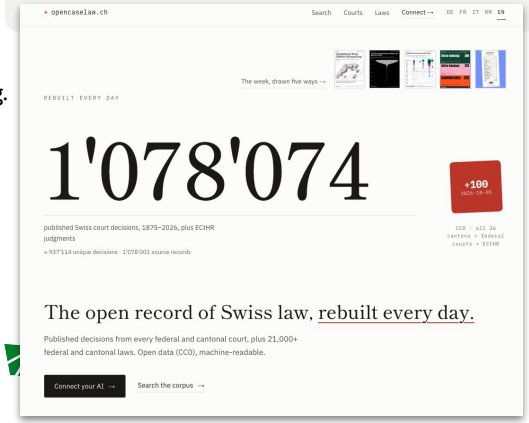
calls a tool, gets a fact back

Search the case law

Open the contract

Check the citation

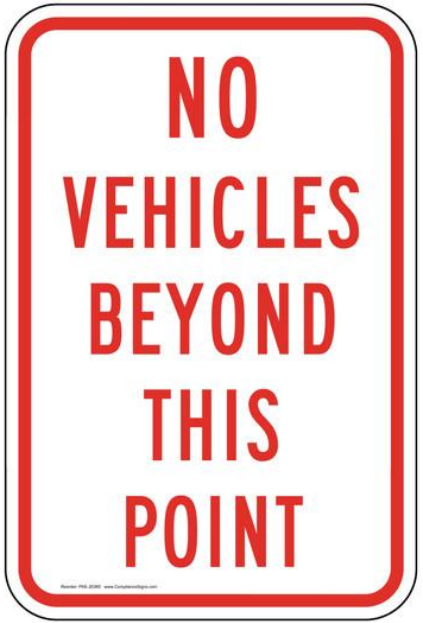
E.g.

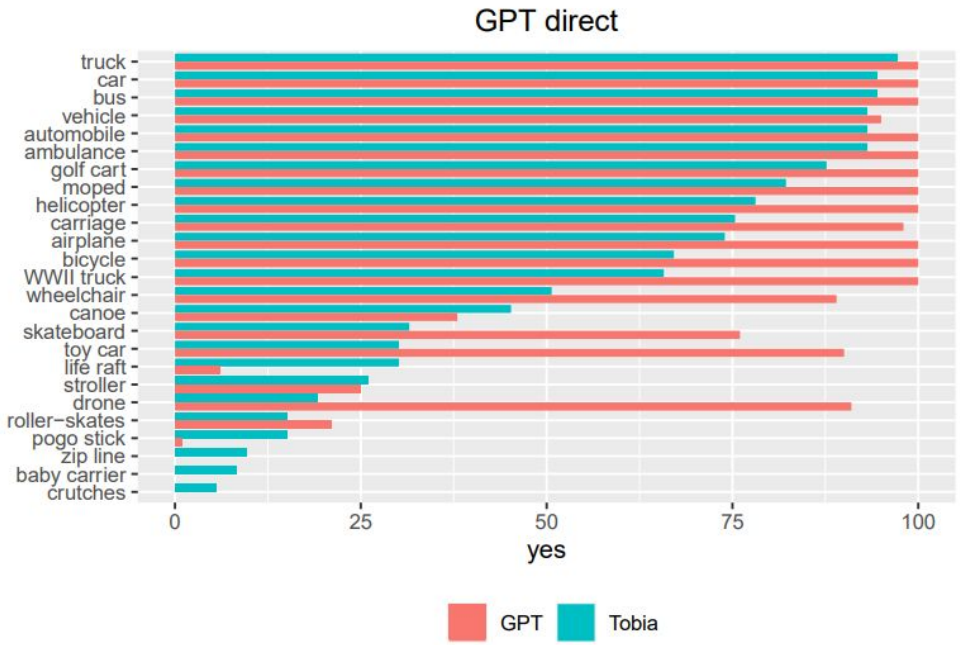


https://www.harvey.ai/blog/introducing-harveys-legal-agent-benchmark

"We're introducing Harvey's Legal Agent Benchmark (LAB), an open-source benchmark for legal agents. LAB was built to evaluate and improve agent capabilities for supporting real world work done by lawyers. Each task consists of an instruction, a client matter containing relevant materials, and a requirement that the agent produce a work product for review. This structure is designed to mirror how work is assigned, performed, and reviewed at large law firms."

«No vehicles in the park»





Kevin P. Tobia, Testing Ordinary Meaning, 134 HARV. L. REV. 726 (2020).

Engel, Christoph; McAdams, Richard H. (2024) : Asking GPT for the ordinary meaning of statutory terms, Discussion Papers of the Max Planck Institute for Research on Collective Goods, No. 2024/5, Max Planck Institute for Research on Collective Goods, Bonn,

(Just One More) Issue of LLMs and Law

Let's send ChatGPT on an introspective journey about **digital sovereignty** in law…

🤖 2. Where LLMs create tension with digital sovereignty

When LLMs are used to **analyze, summarize, or even generate interpretations of law**, several sovereignty challenges arise:

a. Epistemic control

LLMs are trained on global corpora — much of it from the **Anglo-American common law** and English-language sources.
That means their implicit "legal worldview" is not jurisdiction-neutral.
→ They may reproduce foreign legal reasoning styles, norms, or case hierarchies that **don't align with local law** or civil-law interpretive traditions.

b. Opaque reasoning

LLMs are **non-transparent**: they produce text that *sounds* authoritative but cannot show its legal sources or interpretive logic.
→ This undermines **legal certainty** and **traceability** — core elements of the rule of law and national sovereignty in interpretation.

c. Data jurisdiction

Training data, model weights, and even inference APIs are often hosted in other countries.
→ This raises questions like:

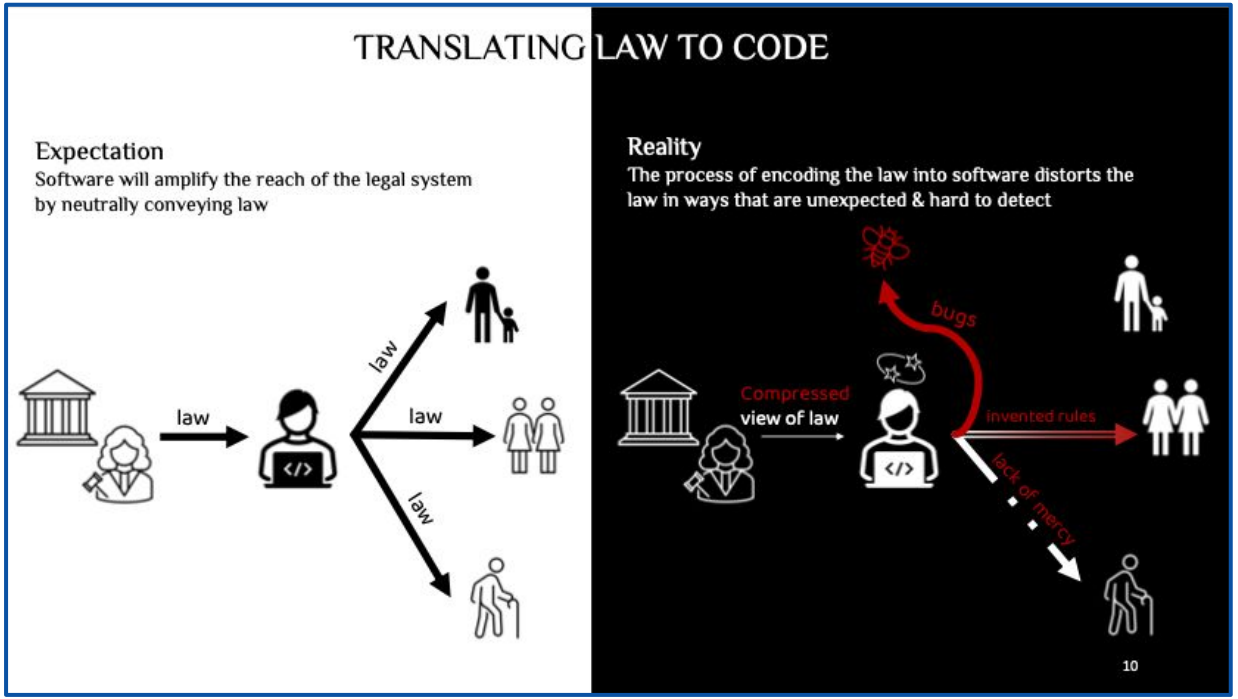
- Can domestic regulators audit or constrain how the model interprets national laws?
- What happens if sensitive legal data (e.g., court documents) must be sent to a foreign cloud service for processing?

Ultimately, **digital sovereignty in law is about control over meaning** — who defines what the law *means* when machines mediate access to it?

If interpretive authority drifts toward opaque or foreign-controlled systems, states risk a **loss of normative autonomy** — even if no law was ever "changed" in text.

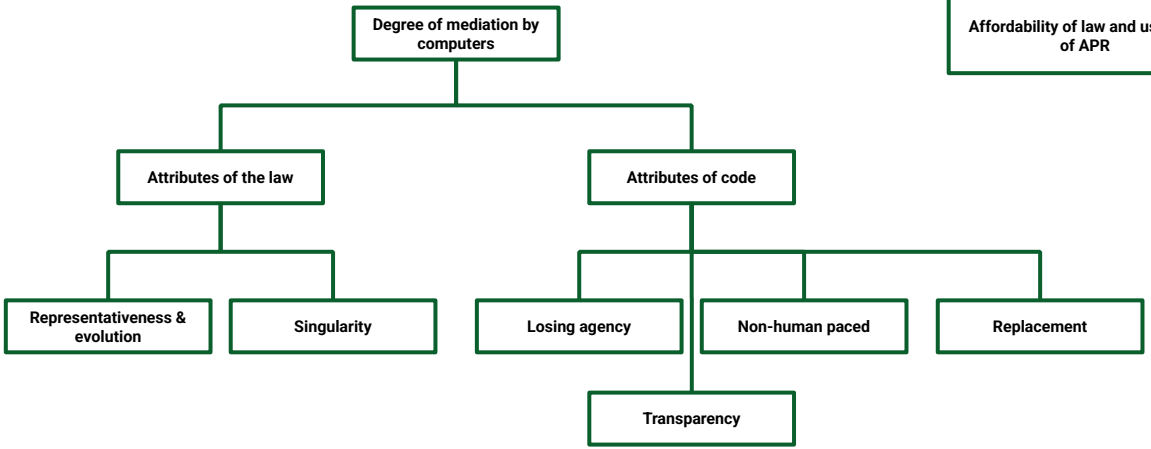
What challenges emerge when we automate the law?

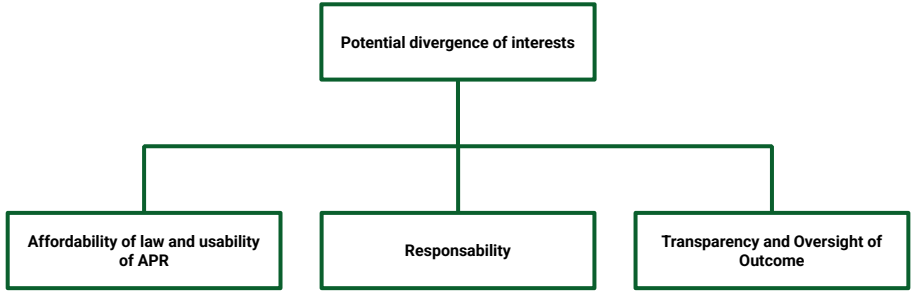
Embedding this in the literature

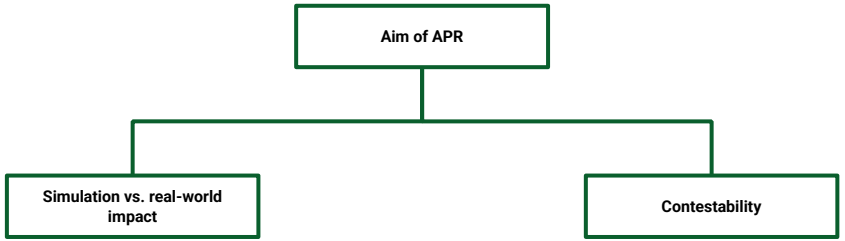


Source: Escher, N., Bilik, J., Banovic, N., & Green, B. (2024). Code-ifying the Law: How Disciplinary Divides Afflict the Development of Legal Software. Proceedings of the ACM on Human-Computer Interaction, 8(CSCW2), 1-37. ttps://dl.acm.org/doi/pdf/10.1145/3686937

Overview of issues







Universität St.Gallen

Guitton, C., Tamò-Larrieux, A., & Mayer, S. (2022). Mapping the Issues of Automated Legal Systems: Why Worry About Automatically Processable Regulation?. *Artificial Intelligence and Law*, 1-29.

Attributes of the law

Representativeness & evolution

Singularity

Losing agency

Non-human paced

Replacement

Transparency

Potential divergence of interests

Affordability of law and usability of APR

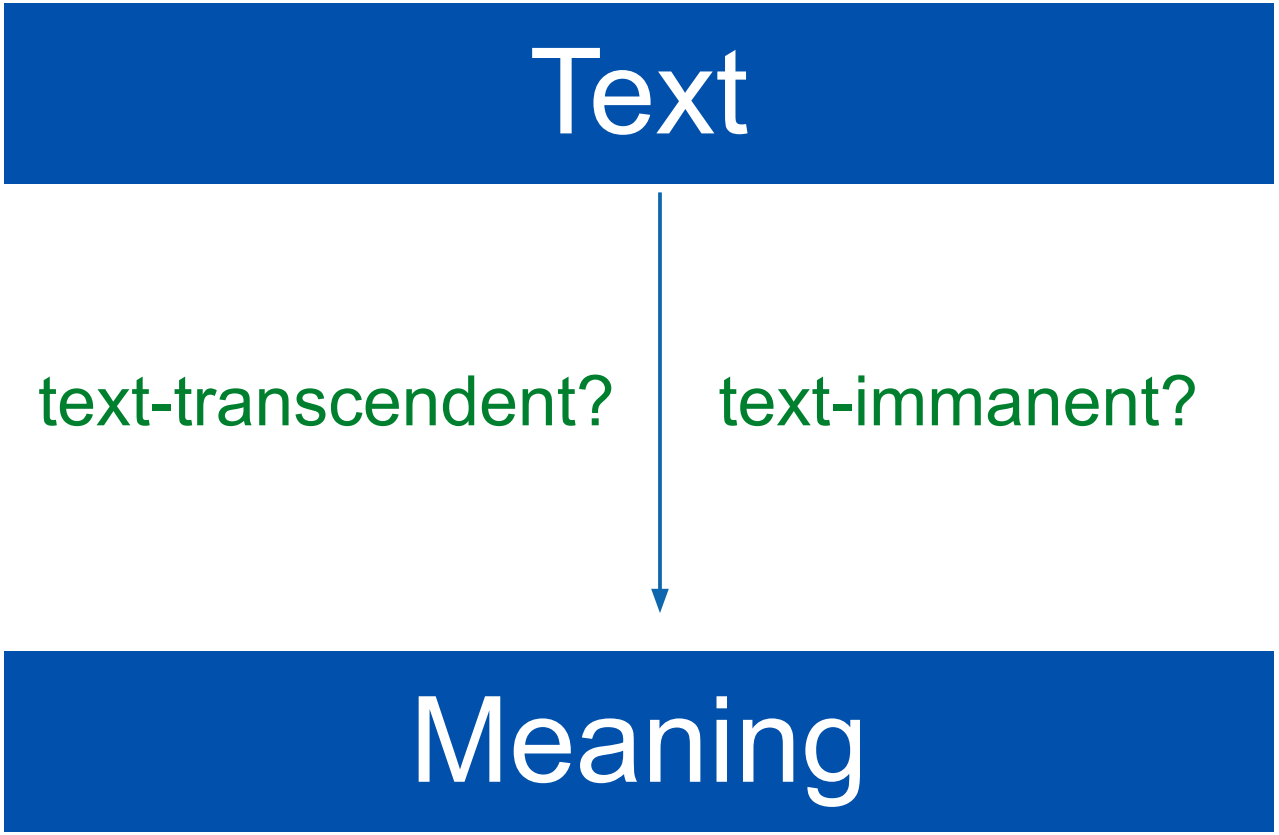
Responsability

Transparency and Oversight of Outcome

Aim of APR

Simulation vs. real-world impact

Contestability


Text
text-transcendent?
text-immanent?
Meaning

Possible interpretations

- **historic interpretation** considering either the legislator's original will (subjective interpretation) or relying on the general understanding of the norm at the time of the adoption (objective interpretation);
- **constitutional interpretation** requiring courts to choose an interpretation that is best in line with the fundamental values enshrined in the Swiss Constitution;
- **grammatical interpretation** relying on the meaning of terms, wording, grammar and syntax thereby giving words their literal, usual, and grammatical meaning (called literal method in the US);
- **systematic interpretation** by contextualising a provision in light of the systematic context with other legal provisions within the concerned act;
- **teleological interpretation** which involves a consideration of the aim and rationale (τέλος) of a certain provision;
- **contemporary interpretation** relying on the public understanding of the norm under modern circumstances;
- **pluralistic approach**: The Federal Supreme Court usually adopts a balancing approach, combining different elements of interpretation (BGE 131 III 314)

BGE 83 I 173:

SC rejected an interpretation of the Canton of Vaud's constitution that would include women, ruling that the historical lawmakers clearly intended **"tous les Suisses" ("all Swiss citizens")** to refer only to **male** Swiss citizens.

Chapeau

83 I 173

23. Arrêt du 26 juin 1957 dans la cause Quinche et consorts contre Conseil d'Etat du Canton de Vaud.

Regeste

Art. 4 Cst., 85 litt. a OJ; suffrage féminin.
1. Incompétence du Tribunal fédéral pour statuer sur un recours relatif à l'exercice des droits politiques en matière fédérale et pour trancher le problème de la conformité des dispositions d'une constitution cantonale à celles de la constitution fédérale (consid. 1 et 6).
2. Pouvoir d'examen du Tribunal fédéral dans les recours formés en vertu de l'art. 85 litt. a OJ (consid. 2).
3. Méthodes d'interprétation dans la jurisprudence du Tribunal fédéral. Rôle des travaux préparatoires. Importance de la pratique. Une autorité cantonale ne commet pas un acte arbitraire en refusant de donner à un texte, en soi ambigu, une interprétation qui serait contraire à la pratique suivie depuis très longtemps et de façon constante (consid. 4 et 5).

Faits à partir de page 174

BGE 116Ia 359:

In the case of the Appenzell Innerrhoden constitution, the Federal Court ruled that terms like "**landpeople**" and **"Swiss citizens"** should, despite the clear contrary intent of the historical lawmakers, be understood objectively and in a contemporary, constitutionally compliant manner to include female citizens as well.

- Potential divergence of interests
  - Affordability of law and usability of APR
  - Responsability
  - Transparency and Oversight of Outcome
- Degree of mediation by computers
  - Attributes of the law
    - Representativeness & evolution
    - Singularity
  - Attributes of code
    - Losing agency
    - Non-human paced
    - Replacement
    - Transparency

- Aim of APR
  - Simulation vs. real-world impact
  - Contestability

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Elysium: https://www.youtube.com/watch?v=flLoSxd2nNY

- Aim of APR
  - Simulation vs. real-world impact
  - Contestability

British Post Office scandal

From Wikipedia, the free encyclopedia

The **British Post Office scandal** is a miscarriage of justice involving the wrongful civil and criminal prosecutions of an unknown or unpublished number of sub-postmasters (SPMs) for theft, false accounting and/or fraud.[1][2] The cases constitute the most widespread miscarriage of justice in British legal history, spanning a period of over twenty years and it remains unresolved.[3][4][5][6][7]

Represent
evo

Replacement

Potential divergence of interests

Affordability of law and usability of APR

Responsability

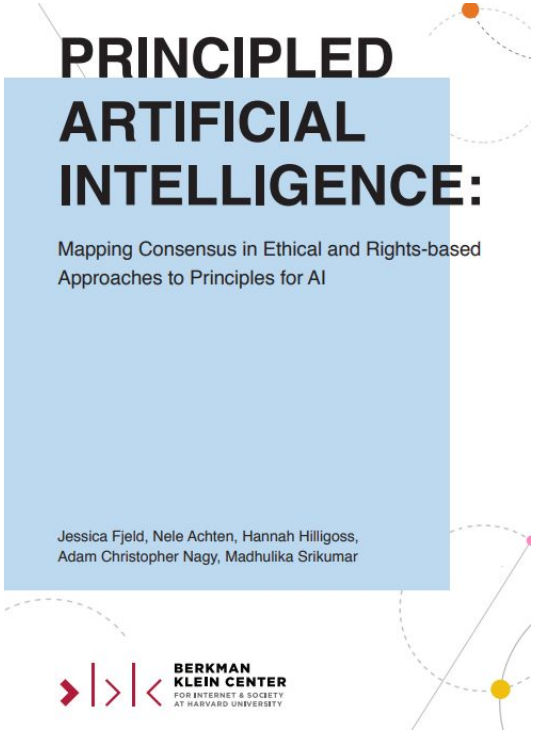
Transparency and Oversight of Outcome

Aim of APR

Simulation vs. real-world impact

Contestability

Responsible AI Frameworks



Interesting foundation but not specific to the field of legal automation …

Frameworks for legal automation?

9 November 2021

Utrecht Data School and Prof. Janneke Gerards develop Impact Assessment Mensenrechten en Algoritmes

IAMA makes careful decision about deployment of algorithms possible

In the future, algorithms can support governments and companies in carrying out their legal obligations. However, inaccuracy, ineffectiveness or worse, a violation of human rights must be prevented, such as the algorithms used by the Dutch Tax Authorities that led to racial profiling; a scandal that almost everyone now knows as the 'toeslagenaffaire'.

> *IT, data managers, lawyers, really everyone involved needs to be at the table. Only then can you all make the right decisions. It's not just about human rights. The IAMA starts with the basics: what exactly are you going to use the algorithm for? Is there a legal basis for it? How is the data collected? Is it reliable? The IAMA is the steppingstone from which all existing, relevant frameworks concerning algorithms in the Dutch context are hung. You go through them one by one. Such an overview did not exist before."*

Frameworks (ctd.)

GOV.UK Menu

Home > Government > Government efficiency, transparency and accountability > Ethics, Transparency and Accountability Framework for Automated Decision-Making

Cabinet Office | Office for Artificial Intelligence | Centre for Data Ethics and Innovation | Department for Science, Innovation & Technology

Guidance

Ethics, Transparency and Accountability Framework for Automated Decision-Making

Updated 29 November 2023

How to use this framework

When you use automated decision-making in a service, you should:

1. Test to avoid any unintended outcomes or consequences.
2. Deliver fair services for all of our users and citizens.
3. Be clear who is responsible.
4. Handle data safely and protect citizens' interests.
5. Help users and citizens understand how it impacts them.
6. Ensure that you are compliant with the law.
7. Build something that is future proof.

Home > AI & SOCIETY > Article > Table 1

Table 1 Overview of the 13 types of issues for evaluating APR and the leading questions that can be used to identify and address them

From: Responsible automatically processable regulation

Issue type	Lead questions to evaluate the issue
Vagueness and balancing of interests	Were several possible and valid interpretations of the law implemented? Is there a technique in place to annotate elements of vagueness (e.g., to express probabilistic certainty about the interpretation)? Are vague terms implemented in a way that clearly differentiates them from the rest in the technical implementation of the APR system (to allow flexible modification or configuration)? Does the system evaluate the different outcomes that different interpretations may have, and raise an issue when these different outcomes are fundamentally divergent? Is the human enactment of the same regulation (e.g., in precedents) in line with the APR implementation? Was there sufficient A/B testing, verification and quality assurance before deployment of the APR implementation?
Evolution of Norms and Statutes	Can the system be adjusted to new interpretations of the law? Can the system enact such adjustments automatically? Is, and how is, the evolution of morale and social norms reflected in the implementation? What is the mechanism of how the APR keeps track of changes, both in terms of new interpretation from evolving social norms, and from new statutes and case law?
Lack of Interdisciplinarity	What is the demography and professional training of the individuals involved in the development and implementation of an APR project? Are the different viewpoints sufficiently represented (business, society, citizen)? Was ethical validation performed and ethical approval sought? How were the points of view of those from a professional/academic minority among the involved taken into consideration? Did reviews between fundamentally different expert groups take place? Who gave instructions to developers and how did developers seek advice when in doubt?
Agency	If a human judgment conflicts with the output from the algorithm, is there a process in place to ensure that the human can overrule the algorithm's decision? Is this overruling as well as the motivation behind it appropriately recorded, e.g., in a logfile? Is the implementation sufficiently transparent to show the rationale behind the decision, allowing humans to weigh the arguments against and for breaking the law in an exceptional situation?
Natural Pace	Are users of the system made aware that the system delivers decisions at a "non-human" speed? Does the project team anticipate that such faster processing time would lead to any negative psychological effects on certain users of your system? If applicable, has psychological counsel been sought to verify that this is not an issue?
Workforce Replacement	Who is impacted by work replacement? Does it replace work that some people enjoy doing? Does it replace work which was an essential life-support for those doing it? Does it replace work which offered compensation (financial, status-wise, etc.) that some regarded as either fair or even attractive?
Implementation Transparency	Can individuals, public authorities, and interested stakeholders have access to the implementation code, training datasets, trained models, and information on the APR implementation and deployment? How can different technical implementations enable more transparent and malleable approaches? Are the code, training datasets, and trained models easily accessible for example without burdensome procedures or intermediaries? Does the state play a role in educating its citizens in reading and understanding automatic processable regulation?
Process Transparency	Who verifies that processes are in place to catch errors and to correct any wrongs? Is this process communicated publicly, and clearly? Who verifies that data is collected, retained, and managed appropriately? How does the audit process take place? Should private companies that turn public regulation into APR or leverage APR come under an auditing process, and to what extent?
Affordability	Are the costs to the end user more manageable than through professional support? Should the state support, through subsidies or other means, the development of tools making the law more accessible, hence fostering the rule of law?
Usability	To which extent is the development user-centric? Are there aspects of the projects (micro or macro) that are unclear as to whether there has been a public debate around, and whether or how to bring this debate about? How can we ensure that there is a public debate if the implementation comes from the state?
Responsibility	In case of mistakes in APR, will it be possible to ascribe responsibility to one organizational unit (or a person within that unit), hence guaranteeing clear ownership and associated responsibility which in turn incentivizes developers to take precautions? Is the division of responsibility between encoding, inputting data, project management, and the resulting output clear? Is the division of tasks clear, or is it part of a complex organizational setup prone to hiding a lack of ownership? Is the hierarchy also well established when it comes to decision-making? Or is the culture axed toward group leadership, with groups loosely defined?
Reality	Is it clear to users whether the tool is a simulation or whether it is exactly the same tool that will be used for the official decision-making process? Why can the simulation and the actual decision-making system (not) be the same? Is the justification strong enough? Or, is this just a showcase of the public service's inefficiency? Are the messages displayed to users specific enough on when and how when and how simulation can differ from real usage of decision-making systems?
Contestability	Can individuals technically and legally reverse the process by contesting the outcome? How cumbersome is it to appeal to the encoding or the outcome? What cost, if any, to the users does such appeal generate? How can arguments about fairness be brought in during the contestation of decision-making?

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Vagueness and balancing of interests

“Flying only when clear visibility”

Example of **lead questions**:

- Were several possible and valid interpretations of the law implemented?
- Is there a technique in place to annotate elements of vagueness?
- Was there sufficient AB testing, verification, and quality assurance before the deployment of the APR?

Example of **mitigation strategies**:

- Involving different people with different backgrounds in the development
- Use of controlled natural language to identify vagueness
- Explicit documentation when no single interpretation can be agreed upon in the team

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Evolution of norms

“CH 1970: No cohabitation of unmarried couples”

Example of **lead questions**:

- Can the system be adjusted to new interpretations of the law?
- How is the evolution of morale and social norms reflected in the implementations?
- What mechanisms are in place to monitor of changes in terms of interpretation and evolution of social norms?

Example of **mitigation strategies**:

- Modular implementation of an APR project that allows for separation of fast-evolving and slow-evolving parts
- Analysis of underpinning legal interpretation
- Iterative process within the development to re-assess assumptions within APR

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Reality

“Using of tools that are not official decision-making but resemble it”

Example of **lead questions**:

- Is it clear whether the tool is a simulation or whether it is an official decision-making tool?
- Is there a justification if the APR is a simulation rather than an actual decision-making tool?
- Are the messages displayed to users specific enough to ensure clarity on the scope of the APR?

Example of **mitigation strategies**:

- Clear labeling scheme for APR tools (clear warning when the tool is a simulation)
- Ensure that simulation and official decision-making tool match - minimize the deviations when it matters to individuals

Questions?
Feedback?
Suggestions?